

Algebraic Number Theory

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1 Motivations

One of the main goals of number theory is to solve *Diophantine equations*, which are polynomials in the ring of integer coefficients. A famous example is Fermat's Last theorem (proven by Wiles in 1994). Fermat's last theorem states that if there exists an integer solution to

$$x^n + y^n = z^n, \quad n \geq 3,$$

then $xyz = 0$. This won't be covered in this course, but let us consider the case when $n = 2$. Note that when we consider possible triplets (x, y, z) , we can assume that $x, y, z > 0$ and that x, y, z are pairwise coprime. Next, we deduce that one of x and y must be odd, and z must be odd. Now, let us write $x^2 + y^2 = z^2$ as

$$y^2 = (z - x)(z + x).$$

Note that first, we get that $(z - x, z + x) = 1$. This then implies that $z - x$ and $z + x$ needs to be squares (from the fact that $\mathbb{Z}$ is a UFD). From this, we can deduce that solutions are of the form

$$\left(\frac{\alpha^2 - \beta^2}{2}, \alpha\beta, \frac{\alpha^2 + \beta^2}{2} \right).$$

In the process above, we used the *separating powers trick*, for $n = 2$.

Lemma 1.0.1

Let $a, b, c \in \mathbb{Z} \setminus \{0\}$ and $n > 0$ such that $a^n = bc$. If $(b, c) = 1$, then $\exists b_1, c_1 \in \mathbb{Z}$ such that $b = \pm b_1^n$ and $c = \pm c_1^n$.

If $n = p$ a prime odd, then, we can write $x^p + y^p = z^p$ as

$$(x + p)(x + \zeta y) \dots (x + \zeta^{p-1} y) = z^p,$$

where ζ is the primitive p -th root of unity. Now, the factors are not integers, and they are in $\mathbb{Z}[\zeta]$. Some questions we ask about this ring is whether it has unique factorization, or whether there is an analogue of the separating powers which work. In this course, we will study arithmetic properties of rings similar to $\mathbb{Z}[\zeta]$ and use such properties to study Diophantine equations.

We discuss another examples to motivate our studies. This time, we analyze

$$y^2 = x^3 + 16.$$

If $x = 0$, we can easily see that $y = \pm 4$. Now suppose that $x \neq 0$. We can rewrite the above as

$$(y - 4)(y + 4) = x^3.$$

If y is odd, we attain that $(y - 4, y + 4) = 1$. Then, both $y - 4$ and $y + 4$ must be cubes. Then, we must get that

$$8 = \alpha^3 - \beta^3,$$

which is not possible, since no two odd cubes differ by 8. The case for when y is even follows similarly and by using the separating power trick. This is a specific example in the family of equations called **Mordell equations**, which are equations of the form

$$y^2 = x^3 + k, \quad k \in \mathbb{Z}.$$

These give rise to the ring $\mathbb{Z}[\sqrt{k}], \mathbb{Z}[i\sqrt{k}]$.

2 Classes of Rings

§2.1 Unique Factorization Domains

Remark 2.1.1 — Here, we assume all rings are commutative and have a unit.

Remark 2.1.2 — Since many definitions/ theorems have come up previously, I will only list important ones.

Lemma 2.1.3

In an integral domain R , if $a \in R \setminus \{0\}$ is prime, then a is irreducible.

Proof. Let $b, c \in R$ such that $a = bc$. Then, $a \mid bc \implies a \mid b$ or $a \mid c$. This implies that there exists $v \in R$ such that $b = av$, implying that $a = avc$. This implies that $a(1 - vc) = 0$, so $a = 0$ or $vc = 1$. Since we're in an integral domain and $a \neq 0$, it follows that $1 = vc$, so c must be a unit. $\square$

Lemma 2.1.4

If R is a UFD, then every irreducible element is prime.

Proof. Let $a \in R \setminus \{0\}$ be an irreducible element, and let $b, c \in R$ such that $a \mid bc$. In other words, we want to find a $d \in R$ such that $ad = bc$. If we write down the unique factorizations for b, c, d , then it must be that a divides an irreducible factor that arises in b or c . Therefore, $a \mid b$ or c . $\square$

Theorem 2.1.5

Every PID is a UFD.

Example 2.1.6

For $k \in \mathbb{Z}_{>0}$, $\mathbb{Z}[i\sqrt{k}]$ is a subring of $\mathbb{C}$.

Example 2.1.7

In $\mathbb{Z}[i\sqrt{3}]$, we have

$$2 \cdot 2 = 4 = (1 + i\sqrt{3})(1 - i\sqrt{3}).$$

We need to show that $2, 1 + i\sqrt{3}, 1 - i\sqrt{3}$ are all irreducible elements to show that the two factorizations are not equivalent.

To do this, we define the *norm map* $N: \mathbb{Z}[i\sqrt{3}] \rightarrow \mathbb{Z}$ that sends $a + i\sqrt{3}b \mapsto a^2 + 3b^2$. Note that $N(\alpha) = 2$ has no solution, so we must have that $N(\alpha) = 4$. This then shows that $2, 1 + i\sqrt{3}, 1 - i\sqrt{3}$ are irreducibles.

Remark 2.1.8 — This shows that $\mathbb{Z}[i\sqrt{3}]$ is not a UFD, as $2 \cdot 2 = (1 - i\sqrt{3})(1 + i\sqrt{3})$, but $2 \nmid 1 + i\sqrt{3}$.

Definition 2.1.9. For rings of the form $\mathbb{Z}[i\sqrt{k}]$, for $k \in \mathbb{Z}_{>0}$, we can define the **norm map** as follows

$$\begin{aligned} N: \mathbb{Z}[i\sqrt{k}] &\rightarrow \mathbb{Z}_{>0} \\ \alpha &\mapsto \alpha\bar{\alpha} \end{aligned}$$

More explicitly, if we write $\alpha = a + i\sqrt{k}b$, we have that

$$N(\alpha) = a^2 + kb^2.$$

Definition 2.1.10. For rings of the form $\mathbb{Z}[\sqrt{k}]$, for $k \in \mathbb{Z}_{>0}$, we can define the **norm map** as follows

$$\begin{aligned} N: \mathbb{Z}[\sqrt{k}] &\rightarrow \mathbb{Z}_{>0} \\ \alpha &\mapsto \alpha\bar{\alpha} \end{aligned}$$

In this case, if we write $\alpha = a + \sqrt{k}b$, we have that

$$N(\alpha) = a^2 - kb^2.$$

Proposition 2.1.11

Given $\mathbb{Z}[i\sqrt{k}]$, the norm N as defined above has the following properties:

1. $N(\alpha) = 0 \implies \alpha = 0$.
2. $N(\alpha\beta) = N(\alpha)N(\beta)$.
3. $N(\alpha) = 1 \iff \alpha \in \mathbb{Z}[i\sqrt{k}]^\times$.

§2.2 Euclidean Domains

Example 2.2.1

$\mathbb{Z}$, with $\phi: \mathbb{Z} \setminus \{0\} \rightarrow \mathbb{Z}_{>0}$, $a \mapsto |a|$ is a Euclidean domain.

Example 2.2.2

Let k be a field. The degree map $\deg: k[x] \setminus \{0\} \rightarrow \mathbb{Z}_{>0}$ is a Euclidean domain.

Theorem 2.2.3

The ring of Gaussian integers $\mathbb{Z}[i]$ is a Euclidean domain with respect to $N: \mathbb{Z}[i] \rightarrow \mathbb{Z}_{>0}$.

Proof. Let $x, y \in \mathbb{Z}[i] \setminus \{0\}$. Then, let $z := \frac{y}{x}$. There exists a $q \in \mathbb{Z}[i]$ such that $|z - q| < \frac{\sqrt{2}}{2}$. Letting $r = y - xq$, we have

$$N(r) = N(y - xq) = N(x)N(z - q) \leq \frac{1}{2}N(x) < N(x).$$

□

Theorem 2.2.4

Every ED is a PID; hence, a UFD.

Proof. Let $I \subseteq R$. If $I = (0)$, we're done. Otherwise, let $a \in I$ such that $\phi(a)$ is minimal. Assume that there is a $b \in I \setminus \{0\}$. Then, $a \nmid b$. Then, the Euclidean algorithm applies and so there exists $q, r \in R$ such that $b = qa + r$ with $\phi(r) < \phi(a)$ or $r = 0$. $r \neq 0$, since $a \nmid b$, and so $r = b - qa \in I$, which is a contradiction. □

Question: Can we mimic the proof for showing that the Gaussian integers is an ED for other domains $\mathbb{Z}[i\sqrt{k}]$?

Let $x, y \in \mathbb{Z}[i] \setminus \{0\}$ and take $z := \frac{y}{x}$. Then, there exists $q \in \mathbb{Z}[i\sqrt{k}]$ such that $|z - q|^2 \leq \left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{k}}{2}\right)^2$. Then,

$$N(y - xq) = N(x)N(z - q) = N(x)|z - q|^2 \leq \frac{1+k}{4}N(x) < N(y).$$

Thus, the same argument works for up to $k \leq 2$. Thus, $\mathbb{Z}[i\sqrt{2}]$ is also a Euclidean domain.

Remark 2.2.5 — For $k \geq 3$, $\mathbb{Z}[i\sqrt{k}]$ is not a Euclidean domain with respect to the usual norm

Here is a notion that lets us prove that there are certain classes of $\mathbb{Z}[i\sqrt{k}]$ aren't EDs.

Definition 2.2.6. Let R be an integral domain with fraction field K . We say that R is **integrally closed** if for every monic polynomial

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in R[x]$$

and every $\frac{a}{b} \in K$, the implications

$$f\left(\frac{a}{b}\right) = 0 \implies \frac{a}{b} \in R$$

holds.

Lemma 2.2.7

If R is a UFD, then R is integrally closed.

Proof. Take $\frac{a}{b} \in K$ such that there exists $a_0, a_{n-1} \in R$ such that

$$\left(\frac{a}{b}\right)^n + a_{n-1} \left(\frac{a}{b}\right)^{n-1} + \cdots + a_0 = 0.$$

WLOG, we can assume that (up to units), $(a, b) = 1$. Then, the above can be rewritten as

$$a^n = -b(a_{n-1}a^{n-1} + \cdots + a_0b^{n-1}).$$

This implies that $b \mid a^n$, so b must be a unit. Therefore, $\frac{a}{b} \in R$. $\square$

Theorem 2.2.8

$\mathbb{Z}[i\sqrt{k}]$ is not a Euclidean domain if $k \equiv 3 \pmod{4}$.

Proof. We show that $\mathbb{Z}[i\sqrt{k}]$ is not integrally closed, hence not a UFD, and hence not a ED. Assume that $k \equiv 3 \pmod{4}$. Take $\frac{1+i\sqrt{k}}{2} \notin \mathbb{Z}[i\sqrt{k}]$, but it is a root of the polynomial

$$T^2 - T + \frac{1+k}{4} \in \mathbb{Z}[T],$$

as $k \equiv 3 \pmod{4}$. $\square$

Example 2.2.9 (UFDs aren't EDs)

$\mathbb{Z}[x]$ is a UFD but not a PID. Thus, not an ED. It is a UFD by Gauss's lemma, but focusing on the ideal $(2, x)$ shows that it's not a PID.

§2.3 Algebraic Integers

Recall there is the Gauss lemma.

Lemma 2.3.1 (Gauss Lemma)

Let $g, h \in \mathbb{Q}[x]$ be monic polynomials such that $g \cdot h \in \mathbb{Z}[x]$. Then, $g, h \in \mathbb{Z}[x]$.

Definition 2.3.2. An element $\alpha \in \mathbb{C}$ is **algebraic** if there exists $f \in \mathbb{Q}[x]$ such that $f(\alpha) = 0$.

Definition 2.3.3. An element $\alpha \in \mathbb{C}$ is **algebraic integer** if there exists a monic polynomial $f \in \mathbb{Z}[x]$ such that $f(\alpha) = 0$.

Recall that given an algebraic element α , we can consider the ideal

$$I = \{f \in \mathbb{Q}[x] \mid f(\alpha) = 0\} \subseteq \mathbb{Q}[x].$$

Then, as $\mathbb{Q}[x]$ is a PID, there exists a monic polynomial m_α that generates I .

Proposition 2.3.4

Let $\alpha \in \mathbb{C}$ be an algebraic element with minimal polynomial $m_\alpha \in \mathbb{Q}[x]$. Then, α is an algebraic integer if and only if $m_\alpha \in \mathbb{Z}[x]$.

Proof. The $\Leftarrow$ direction is immediate, so the forward implication is what we'll show. Suppose that α is an algebraic integer, and let $f \in \mathbb{Z}[x]$ be a monic polynomial such that $f(\alpha) = 0$. By the definition of a monic polynomial, there exists a $g \in \mathbb{Q}[x]$ such that $f(x) = g(x) \cdot m_\alpha(x)$. Note that since f and m_α are monic, this implies that g is monic. Then, we can apply the Gauss lemma to say that g and m_α are in $\mathbb{Z}[x]$. $\square$

Example 2.3.5

Let $\alpha \in \mathbb{Q}$ be rational. Then, α is an algebraic integer if and only if $m_\alpha \in \mathbb{Z}[x]$, which is true if and only if $\alpha \in \mathbb{Z}$.

§2.3.i Quadratic Numbers

Definition 2.3.6. We say that $\alpha \in \mathbb{C}$ is a **quadratic integer** if $m_\alpha \in \mathbb{Z}[x]$ and degree 2.

Notation 2.3.7. A **quadratic field** is the field

$$\mathbb{Q}(\sqrt{d}) = \{a + b\sqrt{d} \mid a, b \in \mathbb{Z}\}.$$

Remark 2.3.8 — Note that any quadratic integer will lie in a quadratic field.

Remark 2.3.9 — Note that if $d_1, d_2 \in \mathbb{Z}$, then $d_1 \neq d_2 \implies \mathbb{Q}(\sqrt{d_1}) \neq \mathbb{Q}(\sqrt{d_2})$.

Definition 2.3.10. Conjugation on $\mathbb{Q}(\sqrt{d})$ is defined by

$$\overline{a + b\sqrt{d}} = a - b\sqrt{d}.$$

Now, note that for $\alpha \in \mathbb{Q}(\sqrt{d})$, we have

$$0 = \alpha^2 - \underbrace{(\alpha + \bar{\alpha})}_{\in \mathbb{Q}}\alpha + \underbrace{\alpha\bar{\alpha}}_{\in \mathbb{Q}}.$$

We can define the following functions.

- The trace: $\text{Tr}: \mathbb{Q}(\sqrt{d}) \rightarrow \mathbb{Q}$,

$$\alpha \mapsto \alpha + \bar{\alpha}$$

- The norm: $\mathcal{N}: \mathbb{Q}(\sqrt{d}) \rightarrow \mathbb{Q}$,

$$\alpha \mapsto \alpha\bar{\alpha}$$

Let $\alpha = a + b\sqrt{d}$. Then, we see that

$$\text{Tr}(\alpha) = 2a, \quad \mathcal{N}(\alpha) = a^2 - db^2.$$

Thus, note that α is a root of

$$p(x) = x^2 - \text{Tr}(\alpha)x + \mathcal{N}(\alpha) \in \mathbb{Q}[x].$$

If $\alpha \notin \mathbb{Q}$, then $p(x) = m_\alpha(x)$. $\alpha \notin \mathbb{Q}$ is an algebraic integer if and only if $\text{Tr}(\alpha), \mathcal{N}(\alpha) \in \mathbb{Z}$ if and only if $2a, a^2 - db^2 \in \mathbb{Z}$. Now, let $a' = 2a, b' = 2b$. Then, a' must be an integer, and since

$$(a')^2 - d(b')^2 \in 4\mathbb{Z},$$

it must be that b' is also an integer since d is square free. By considering quadratic residues, we can then find out that a' and b' are both even or a' and b' are both odd if $d \equiv 1 \pmod{4}$. This can be summarized (along with the exercise below) in the following:

$$\mathcal{O}_d := \begin{cases} \mathbb{Z}[\sqrt{d}], & \text{if } d \not\equiv 1 \pmod{4} \\ \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right], & \text{if } d \equiv 1 \pmod{4} \end{cases}$$

Exercise 2.3.11. Show that $\mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$ is equal to the set

$$\left\{ \frac{a + b\sqrt{d}}{2} \mid a, b \in \mathbb{Z}, a \equiv b \pmod{2} \right\},$$

if $d \equiv 1 \pmod{4}$.

§2.3.ii Another Characterization of Algebraic Integers

Our aim from here is to show that algebraic integers in $\mathbb{C}$ is a ring.

Definition 2.3.12. An abelian group M is **finitely generated** if there exists $m_1, \dots, m_n \in M$ such that for all $m \in M$, we can write

$$m = \lambda_1 m_1 + \dots + \lambda_n m_n, \quad \lambda_i \in \mathbb{Z}.$$

Proposition 2.3.13

Let $\alpha \in \mathbb{C}$. Then, the following are equivalent.

1. α is an algebraic integer.
2. $\mathbb{Z}[\alpha]$ is finitely generated as an abelian group (under addition).
3. There exists a nonzero finitely generated subgroup $M \subseteq \mathbb{C}$ such that $\alpha \cdot M \subseteq M$

Proof. (1 $\implies$ 2): Let $a_0, \dots, a_{n-1} \in \mathbb{Z}$ such that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0.$$

We want to show that $\{1, \dots, \alpha, \dots, \alpha^{n-1}\}$ generated $\mathbb{Z}[\alpha]$ as an abelian group. A set of generators for $\mathbb{Z}[\alpha]$ is given by $\{\alpha^k \mid k \in \mathbb{N}\}$. We need to show that α^k can be expressed as a $\mathbb{Z}$ -linear combination of $\{1, \dots, \alpha^{n-1}\}$ if $k \geq n$. We do this via induction on k . If $k = n$, then the equation above shows this. Now suppose we can write

$$\alpha^k = \sum_{i=0}^{n-1} c_i \alpha^i,$$

for $c_i \in \mathbb{Z}$. Then,

$$\alpha^{k+1} = \sum_{i=0}^{n-2} c_i \alpha^{i+1} + c_{n-1} \alpha^n,$$

but then using the equation above we can express $c_{n-1} \alpha^n$ using lower powered terms. Thus, we've proven true.

(2 $\implies$ 3): Take $\mathbb{Z}[\alpha] = M$, and $\alpha \mathbb{Z}[\alpha] \subseteq \mathbb{Z}[\alpha]$, and by assumption, this is finitely generated.

(3 $\implies$ 1): Let $m_1, \dots, m_n$ be a set of generators of M . $\alpha M \subseteq M$ implies that

$$\alpha \cdot m_i = \sum_{j=1}^n a_{ij} m_j, \quad a_{ij} \in \mathbb{Z}.$$

Let $A = (a_{ij}) \in M_n(\mathbb{Z})$. What we get is a surjective morphism:

$$p: \mathbb{Z}^n \rightarrow M$$

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \mapsto \sum_{i=1}^n a_i m_i$$

For every $\mathbf{a} \in \mathbb{Z}^n$, we have

$$\alpha p(\mathbf{a}) = \alpha \cdot \left(\sum_{i=1}^n a_i m_i \right) = p(A \cdot \mathbf{a}).$$

Claim 2.3.14. For any $f \in \mathbb{Z}[x]$, we have that $f(\alpha)p(\mathbf{b}) = p(f(A) \cdot \mathbf{b})$.

Proof. We have as follows:

$$\begin{aligned} \alpha p(\mathbf{a}) &= \sum_{i=1}^n \alpha m_i a_i \\ &= \sum_{i,j=1}^n a_{ij} m_i b_i \\ &= \sum_{j=1}^n \left(\sum_{i=1}^n a_{ij} b_i \right) m_j \\ &= p(A\mathbf{b}). \end{aligned}$$

For any $\lambda \in \mathbb{Z}$, note that p is a morphism of abelian groups, so $\lambda p(\mathbf{b}) = p(\lambda \mathbf{b})$. Next, for any $n \in \mathbb{Z}_{>0}$, we have

$$\alpha^n p(\mathbf{b}) = \alpha^{n-1}(\alpha p(\mathbf{b})) = \alpha^{n-1} p(A\mathbf{b}) = \alpha^{n-2}(\alpha p(A\mathbf{b})) = \dots = p(A^n \mathbf{b}).$$

□

Consider the characteristic polynomial of A :

$$\chi_A(x) = \det(x \cdot I - A) \in \mathbb{Z}[x]$$

Note that this is also monic. Then, by the Cayley-Hamilton theorem, we get that $\chi_A(A) = 0$. It follows that $\chi_A(\alpha) \cdot M = 0$. This follows because for all $m \in M$, $m = p(\mathbf{a})$, and so

$$\chi_A(\alpha) \cdot p(\mathbf{a}) = p(\chi_A(A) \cdot \mathbf{a}) = 0.$$

Since M is not zero, there exists $m \in M \setminus \{0\}$ and

$$\chi_A(\alpha) \cdot m = 0 \implies \chi_A(\alpha) = 0.$$

This implication follows from $M \subseteq \mathbb{C}$. This implication then implies that α is an algebraic integer. $\square$

Remark 2.3.15 — Note that $\mathbb{Z}[\alpha]$ is the smallest subring of $\mathbb{C}$ containing α and can be described as

$$\mathbb{Z}[\alpha] = \{f(\alpha) \mid f \in \mathbb{Z}[x]\} \subseteq \mathbb{C}.$$

Exercise 2.3.16. Show that $\mathbb{Z}[\frac{1}{2}]$ is not finitely generated as an abelian group.

§2.3.iii Number Fields

Theorem 2.3.17

We have:

1. If α, β are algebraic integers, then so are $\alpha + \beta, \alpha - \beta, \alpha \cdot \beta$.
2. If

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0,$$

where $a_i \in \mathbb{C}$ are algebraic integers and $\alpha \in \mathbb{C}$ with $f(\alpha) = 0$ then α is an algebraic integer.

Proof. (1): We consider the ring

$$\mathbb{Z}[\alpha, \beta] = \{\sum a_{ij}\alpha^i\beta^j \mid a_{ij} \in \mathbb{Z}\}.$$

This is generated as an abelian group by $\{\alpha^i\beta^j \mid i, j \geq 0\}$. If α satisfies a monic polynomial with integer coefficients of degree n and β satisfies a monic polynomial with integer coefficients of degree m , then, in particular,

$$\{\alpha^i\beta^j \mid 0 \leq i \leq n, 0 \leq j \leq m\}$$

generates $\mathbb{Z}[\alpha, \beta]$. Now, for any $\gamma \in \mathbb{Z}[\alpha, \beta]$, it follows that $\gamma \cdot \mathbb{Z}[\alpha, \beta] \subseteq \mathbb{Z}[\alpha, \beta]$, so we can apply proposition 2.3.13. This tells us that γ is an algebraic integer, so in particular, $\alpha + \beta, \alpha - \beta, \alpha\beta \in \mathbb{Z}[\alpha, \beta]$.

(2): Consider $\alpha \in \mathbb{C}$ such that $a_{n-1}, \dots, a_0 \in \mathbb{C}$ are algebraic integers that satisfy the relation

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0.$$

Consider the ring $\mathbb{Z}[a_0, \dots, a_{n-1}, \alpha]$. If a_i satisfies a monic polynomial of degree m_i with integer coefficients, then the set

$$\{a_0^{i_0}, a_1^{i_1} \cdots a_{n-1}^{i_{n-1}} \alpha^k \mid 0 \leq i_j \leq m_j, 0 \leq k \leq n\}$$

generates the ring $\mathbb{Z}[a_0, \dots, a_{n-1}, \alpha]$. However, $\alpha \in \mathbb{Z}[a_0, \dots, a_{n-1}, \alpha]$, proving the result (by proposition 2.3.13). $\square$

Remark 2.3.18 — If $\overline{\mathbb{Z}}$ is the set of algebraic integers of $\mathbb{C}$, then the first part of the theorem says that $\mathbb{Z}$ is a ring.

Remark 2.3.19 — We don't get anything now if we consider the roots of monic polynomials whose coefficients are algebraic integers.

Definition 2.3.20. A **number field** is a subfield $K \subseteq \mathbb{C}$ that is finitely dimensional as a $\mathbb{Q}$ vector space. The dimension of K as a $\mathbb{Q}$ vector space is called the **degree** of K and is denoted $[K : \mathbb{Q}]$.

Example 2.3.21

Given an algebraic integer $\alpha \in \mathbb{C}$, $\mathbb{Q}(\alpha)$ (the smallest subfield of $\mathbb{C}$ containing $\mathbb{Q}$ and α) is a number field. Let m_α be the minimal polynomial of α with degree n . Then, $\{1, \alpha, \dots, \alpha^{n-1}\}$ is a basis of K as a $\mathbb{Q}$ vector space. In particular, we have

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = \deg(m_\alpha).$$

Notation 2.3.22. Given a number field K we denote the **set of algebraic integers** in K by $\mathcal{O}_K$.

Remark 2.3.23 — By the first part of theorem 2.3.17, it is a ring. $\mathcal{O}_K$ is called the **ring of integers** of K .

Remark 2.3.24 — Note that $\mathcal{O}_K = \overline{\mathbb{Z}} \cap K$, where $\overline{\mathbb{Z}}$ is defined as the set of algebraic integers of $\mathbb{C}$.

Remark 2.3.25 — $\text{Frac}(\mathcal{O}_K) = K$. It is enough to show that for all $\alpha \in K$, there exists $\lambda \in \mathbb{Z}$ such that $\lambda \cdot \alpha \in \mathcal{O}_K$. We can do this by the procedure below:

For $\alpha \in K$, $\exists a_0, \dots, a_{n-1} \in \mathbb{Q}$ such that

$$m_\alpha(\alpha) = \alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0.$$

Now, define λ as the least common multiple of the denominators of $a_{n-1}, \dots, a_0$. Define

$$p(x) = x^n + \lambda \cdot a_{n-1}x^{n-1} + \dots + \lambda^n \cdot a_0 \in \mathbb{Z}[x].$$

Then, we see that $p(\lambda\alpha) = \lambda^n m_\alpha(\alpha) = 0$.

Example 2.3.26

We've seen that $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$, or that $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$.

Recall the following definition:

Definition 2.3.27. An integral domain R is said to be **integrally closed** if for any monic polynomial $f \in R[X]$ and any $\alpha \in \text{Frac}(R)$ with $f(\alpha) = 0$, we have $\alpha \in R$.

Proposition 2.3.28

Let K be a number field. Then, $\mathcal{O}_K$ is integrally closed.

Proof. If $f(x) \in \mathcal{O}_K[x]$ is monic and $\alpha \in K$ with $f(\alpha) = 0$, then theorem 2.3.17 tells us that α is an algebraic integer:

$$\alpha \in \overline{\mathbb{Z}} \cap K = \mathcal{O}_K.$$

□

Corollary 2.3.29

Let K be a number field and $\mathcal{O} \subsetneq \mathcal{O}_K$ be any subring with $\text{Frac}(\mathcal{O}) = K$. Then, $\mathcal{O}$ is not integrally closed (and hence is not a UFD).

Proof. Take $\alpha \in \mathcal{O}_K \setminus \mathcal{O}$. Then, α satisfies a monic polynomial with coefficients in $\mathbb{Z}[x] \subseteq \mathcal{O}[x]$ (since α is algebraic). Therefore, $\mathcal{O}$ is not integrally closed. □

Example 2.3.30

In particular, if $d \equiv 1 \pmod{4}$, the ring $\mathbb{Z}[\sqrt{d}] \subsetneq \mathcal{O}_d$ is never a UFD.

§2.3.iv Trace and Norm

Let K be a number field with

$$\text{Tr}_K: K \rightarrow \mathbb{Q}, \quad \mathcal{N}_K: K \rightarrow \mathbb{Q}.$$

For $\alpha \in K$, we can consider the map

$$\text{mult}_\alpha: K \rightarrow K.$$

We can see mult_α as an endomorphism of $\mathbb{Q}$ -vector spaces, then

$$\mathcal{N}_K(\alpha) = \det(\text{mult}_\alpha)$$

and

$$\text{Tr}_K(\alpha) = \text{Tr}(\text{mult}_\alpha).$$

Remark 2.3.31 — These do not depend on a choice of basis of K as a $\mathbb{Q}$ vector space.

Properties 2.3.32 (Properties of Trace and Norm)

We have:

- $\forall \lambda \in \mathbb{Q}, \operatorname{Tr}_K(\lambda\alpha) = \lambda \cdot \operatorname{Tr}_K(\alpha).$
- $\forall \alpha, \beta \in K, \operatorname{Tr}_K(\alpha + \beta) = \operatorname{Tr}_K(\alpha) + \operatorname{Tr}_K(\beta).$
- $\forall \alpha, \beta \in K, \mathcal{N}_K(\alpha \cdot \beta) = \mathcal{N}_K(\alpha) \cdot \mathcal{N}_K(\beta).$

Exercise 2.3.33. Check that for $\mathbb{Q}(\sqrt{d})$, we recover the norm and trace introduced previously.

Example 2.3.34

Let $K = \mathbb{Q}(\sqrt[3]{2})$. We compute $\operatorname{Tr}_K(\sqrt[3]{2})$ and $\mathcal{N}_K(\sqrt[3]{2})$.

A basis of K as a $\mathbb{Q}$ vector space is given by $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$. Then, $\operatorname{mult}_{\sqrt[3]{2}}: K \rightarrow K$ sends

$$1 \mapsto \sqrt[3]{2}, \quad \sqrt[3]{2} \mapsto \sqrt[3]{4}, \quad \sqrt[3]{4} \mapsto 1.$$

The matrix of multiplication by $\sqrt[3]{2}$ with respect to this basis is then

$$\begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

We can then compute the trace and determinant of this matrix to conclude that

$$\operatorname{Tr}_K(\sqrt[3]{2}) = 0, \quad \mathcal{N}_K(\sqrt[3]{2}) = 2.$$

Proposition 2.3.35

Let $\alpha \in \mathbb{C}$ be an algebraic number with monic polynomial

$$m_\alpha = x^n + a_{n-1}x^{n-1} + \cdots + a_0.$$

Then,

$$\begin{aligned} \operatorname{Tr}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha) &= -a_{n-1} \\ \mathcal{N}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha) &= (-1)^n a_0. \end{aligned}$$

Proof. Take the set $\{1, \alpha, \dots, \alpha^{n-1}\}$ as a basis for $\mathbb{Q}(\alpha)$. Then, $\operatorname{mult}_\alpha: \mathbb{Q}(\alpha) \rightarrow \mathbb{Q}(\alpha)$ sends

$$\alpha^i \mapsto \alpha^{i+1}, \quad i < n-1$$

and

$$\alpha^{n-1} \mapsto -a_{n-1}\alpha^{n-1} - \cdots - a_0.$$

This then provides the companion matrix of m_α , $C(m_\alpha)$, so computing the trace and norm of this provides the result. $\square$

Remark 2.3.36 — In this case, if α is an algebra integer, then $m_\alpha(x) \in \mathbb{Z}[x]$. Therefore,

$$\mathrm{Tr}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha) \in \mathbb{Z}, \quad \mathcal{N}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha) \in \mathbb{Z}.$$

Remark 2.3.37 — We may write $\mathrm{Tr}_{K/L}: K \rightarrow L, \mathcal{N}_{K/L}: K \rightarrow L$, so it is explicit what field our trace and norm are working over.

Lemma 2.3.38 (Tower Law)

Let $K \subseteq L \subseteq M$ be an inclusion of fields and suppose $[M : L], [L : K]$ are finite. Then,

$$[M : K] = [M : L][L : K].$$

More precisely, if $\alpha_1, \dots, \alpha_n$ is a basis for L as a K vector space and $\beta_1, \dots, \beta_m$ is a basis for M as a L vector space, then a basis for M as a K vector space is give by $\alpha_i \beta_j$, for $1 \leq i \leq n, 1 \leq j \leq m$.

Here is now a generalization of proposition 2.3.35.

Proposition 2.3.39

Let K be a number field for $\alpha \in K$. We have:

$$\begin{aligned} \mathrm{Tr}_K(\alpha) &= [K : \mathbb{Q}(\alpha)] \cdot \mathrm{Tr}_{\mathbb{Q}(\alpha)}(\alpha) \\ \mathcal{N}_K(\alpha) &= (\mathcal{N}_{\mathbb{Q}(\alpha)}(\alpha))^{[K:\mathbb{Q}(\alpha)]} \end{aligned}$$

Proof. Let $n = \deg(\alpha) = [\mathbb{Q}(\alpha) : \mathbb{Q}]$. Note that a basis of $\mathbb{Q}(\alpha)$ as a $\mathbb{Q}$ vector space is given by $\{1, \alpha, \dots, \alpha^{n-1}\}$. Then, let $m = [K : \mathbb{Q}(\alpha)]$ and pick a basis $\beta_1, \dots, \beta_m$ of K as a $\mathbb{Q}(\alpha)$ vector space. By the previous lemma, we have that

$$\beta_1, \alpha\beta_1, \dots, \alpha^{n-1}\beta_1, \beta_2, \dots, \alpha^{n-1}\beta_2, \dots, \beta_m, \dots, \alpha^{n-1}\beta_m$$

is a basis of K as a $\mathbb{Q}$ vector space. Note that for all $i \in 1, \dots, m$, the subspace

$$\{\beta_i, \alpha\beta_i, \dots, \alpha^{n-1}\beta_i\}$$

is invariant by mult_α . Moreover, the matrix mult_α by α on these vectors is the companion matrix $C(m_\alpha)$ (as in proposition 2.3.35). It follows that the matrix of mult_α is a block diagonal matrix with m blocks which are $C(m_\alpha)$. Finally, computing the trace and determinant of this matrix provides the result. $\square$

Corollary 2.3.40

Let K be a number field. If $\alpha \in \mathcal{O}_K$, then $\mathrm{Tr}_K(\alpha)$ and $\mathcal{N}_K(\alpha)$ are in $\mathbb{Z}$.

§2.3.v Modules

We now work towards the following theorem:

Theorem 2.3.41

Let K be a number field. Then $\mathcal{O}_K$ is finitely generated abelian group (free of rank $[K : \mathbb{Q}]$).

Remark 2.3.42 — Recall that $\mathbb{Z}$ -modules are the same as abelian groups.

Definition 2.3.43. Given a ring R , an R -**module** M is an abelian group together with a map $R \times M \rightarrow M$ such that for $r, s \in R, x, y \in M$, the following hold:

- $r \cdot (x + y) = r \cdot x + r \cdot y$
- $(r + s) \cdot x = r \cdot x + s \cdot x$
- $(r \cdot s) \cdot x = r \cdot (s \cdot x)$
- $1 \cdot x = x$

Remark 2.3.44 — In particular, if M is an abelian group, then $\mathbb{Z} \times M \rightarrow M$ has the map

$$\lambda \cdot m = \begin{cases} m + \cdots + m & \lambda > 0 \\ 0 & \lambda = 0 \\ -(m + \cdots + m) & \lambda < 0 \end{cases}$$

Definition 2.3.45. We say that a $\mathbb{Z}$ -module is **free of rank** n if it is isomorphic (as a $\mathbb{Z}$ -module) to $\mathbb{Z}^n$.

Exercise 2.3.46. Show that the $\mathbb{Z}$ -module $\mathbb{Q}$ is not free of finite rank.

Theorem 2.3.47 (Structure Theorem, Weak Form)

Let M be a free $\mathbb{Z}$ -module of rank n , and let $M' \subseteq M$ be a submodule. Then, M' is free of rank $q \leq n$.

Proof. We induct on n . By assumption, $M \cong \mathbb{Z}^n$. WLOG, we assume $M = \mathbb{Z}^n$. For every i , we define $p_i: \mathbb{Z}^n \rightarrow \mathbb{Z}$ to be the projection into the i -th coordinate. If $M' \neq 0$, there exists an i such that $p_i(M') \neq 0$. Then, $\exists \beta \in \mathbb{Z}$ such that $p_i(M') = (\beta)$ (note that such a β exists since $\mathbb{Z}$ is a PID). This implies that there exists $b \in M'$ such that $(\beta) = (p_i(b))$. Now letting $N = \ker(p_i)$, we claim that

$$M' = b\mathbb{Z} \oplus M' \cap N. \quad (*)$$

We now prove this equality $(*)$. $m \in b\mathbb{Z}$, If $m = bk$, for $k \in \mathbb{Z}$, and $m \in M' \cap N$. Then, $p_i(m) = 0$, implying that $p_i(bk) = kp_i(b) = 0$. This implies that $k = 0$, so we have $b\mathbb{Z} \cap (M' \cap N) = 0$. Next, given $m \in M'$, we have $p_i(m) = kp_i(b)$, for some $k \in \mathbb{Z}$. Thus, we have $p_i(m - kb) = 0$. This implies that $m - kb = m_0 \in \ker(p_i) \cap M'$. Finally, this implies

$$m = \underbrace{m_0}_{\in N \cap M'} + \underbrace{kb}_{\in p_i(M')},$$

so our claim is true.

From the equality in (*), $M' \cap N$ is a submodule of N , which is free of rank $n - 1$. By induction, $M' \cap N$ is free of rank at most $n - 1$, and $b \cdot \mathbb{Z}$ is free of rank 1, so it follows that M' is of rank at most n . $\square$

We now work to prove the strong form of the theorem above.

Lemma 2.3.48

Let M be a free $\mathbb{Z}$ -module of rank n , and let $x \in M$. Let $p_1: M \rightarrow \mathbb{Z}$ and $p_2: M \rightarrow \mathbb{Z}$ be two homomorphisms. Then there is a homomorphism $p: M \rightarrow \mathbb{Z}$ such that $p(x)$ divides both $p_1(x)$ and $p_2(x)$.

Proof. Let $b_1, b_2 \in \mathbb{Z}$ such that $d = b_1 a_1 + b_2 a_2$ is the greatest common divisor of $a_1 = p_1(x)$ and $a_2 = p_2(x)$. Let $p: M \rightarrow \mathbb{Z}$ be the homomorphism $b_1 p_1 + b_2 p_2$. Then

$$p(x) = b_1 p_1(x) + b_2 p_2(x) = b_1 a_1 + b_2 a_2 = d.$$

$\square$

Lemma 2.3.49

Let R be a PID and let S be a non-empty set of ideals of R . Then S has an element which is maximal with respect to inclusion.

Proof. Assume that the claim is false. We are going to construct a sequence of ideals

$$I_1 \subsetneq I_2 \subsetneq \cdots \subsetneq I_n \subsetneq \cdots$$

such that $I_n \in S$ for each n as follows. Assume that $I_1, I_2, \dots, I_{n-1}$ is already constructed. Since I_{n-1} is not maximal, there is a $I_n \in S$ such that $I_{n-1} \subsetneq I_n$. However this is a contradiction, since R is a PID, and hence Noetherian. $\square$

Lemma 2.3.50

Let M be a free $\mathbb{Z}$ -module of rank n , and let $x \in M$. Then there is a homomorphism $p: M \rightarrow \mathbb{Z}$ such that $p(x)$ divides $q(x)$ for every homomorphism $q: M \rightarrow \mathbb{Z}$.

Proof. Let $p: M \rightarrow \mathbb{Z}$ be a homomorphism such that the ideal $(p(x))$ is maximal with respect to inclusion among all ideals of the form $(q(x))$, where $q: M \rightarrow \mathbb{Z}$ is a homomorphism. There is such a homomorphism by lemma 2.3.49. Now let $q: M \rightarrow \mathbb{Z}$ be any homomorphism. By lemma 2.3.48 there is a homomorphism $r: M \rightarrow \mathbb{Z}$ such that $r(x)$ divides both $p(x)$ and $q(x)$. Since $(p(x))$ is maximal this is only possible if $(r(x)) = (p(x))$. Therefore $(p(x))$ contains $(q(x))$, that is, we get that $p(x)$ divides $q(x)$. $\square$

Theorem 2.3.51 (Structure Theorem, Strong Form)

Let M be a free $\mathbb{Z}$ -module of rank n and let $M' \subseteq M$ be a submodule. If M' is not zero, then there is a basis of M $\{e_1, \dots, e_n\}$ and $a_1, \dots, a_q \in \mathbb{Z} \setminus \{0\}$ with $q \leq n$ such that

$$\{a_1 e_1, \dots, a_q e_q\}$$

is a basis for M' as a $\mathbb{Z}$ -module and $a_1 \mid a_2 \mid \dots \mid a_q$.

Proof. Again we will prove the claim by induction on n . Let $p: M \rightarrow \mathbb{Z}$ be a homomorphism such that $p(M')$ is maximal; there is such a homomorphism by the previous lemma. Since M' is non-zero, there is a non-zero $a_1 \in \mathbb{Z}$ such that $p(M') = a_1 \mathbb{Z}$. Let $x \in M'$ be such that $p(x) = a_1$. Let $p': M \rightarrow \mathbb{Z}$ be a homomorphism such that $p'(x)$ divides $q(x)$ for every homomorphism $q: M \rightarrow \mathbb{Z}$. Then $p(M') = (p(x))$ is contained by $(p'(x))$, and obviously $(p'(x)) \subseteq p'(M')$, so $p(M') \subseteq p'(M')$. By the maximality of $p(M')$ we get that

$$p(M') = p'(M') = (p(x)) = (p'(x)).$$

Therefore, without the loss of generality we may assume that $p = p'$, that is, the homomorphism $p: M \rightarrow \mathbb{Z}$ is such that $p(M')$ is maximal, and there is an $x \in M'$ such that $p(M') = (p(x))$ and $p(x)$ divides $q(x)$ for every homomorphism $q: M \rightarrow \mathbb{Z}$. We may even assume that $p(M) = \mathbb{Z}$; if not, there is an integer $a > 1$ such that $p(M) = a\mathbb{Z}$, but then $p' = a^{-1}p$ would be a homomorphism such that $p'(M')$ is strictly larger than $p(M')$, a contradiction.

Let N be the kernel of p in M . By Theorem 2.3.47 the $\mathbb{Z}$ -module N is free of rank $n - 1$. Set $N' = M' \cap N$. By the induction hypothesis there is $\mathbb{Z}$ -basis $(e_2, \dots, e_n)$ of N and $a_2, \dots, a_q \in \mathbb{Z} \setminus \{0\}$ such that $(a_2 e_2, \dots, a_q e_q)$ is a basis of N' and $a_2 \mid a_3 \mid \dots \mid a_q$. (Note that the list of the a_i might be empty.) Let $e'_1 \in M$ be such that $p(e'_1) = 1$. Then $e'_1, e_2, \dots, e_n$ is a $\mathbb{Z}$ -basis of M . Let $p_i: M \rightarrow \mathbb{Z}$ be the projection onto the i -th coordinate with respect to this basis, where $i = 1, \dots, n$. As $a_1 = p(x) \mid p_i(x)$ for each i , we get that there is an $e_1 \in M$ such that $x = a_1 e_1$. Now $e_1, e_2, \dots, e_n$ is also a $\mathbb{Z}$ -basis of M , and $(a_1 e_1, \dots, a_q e_q)$ is a basis of M' .

We claim that $a_1 \mid a_2$, and this would complete the proof. Let $b_1, b_2 \in \mathbb{Z}$ be such that $d = b_1 a_1 + b_2 a_2$ is the greatest common divisor of a_1 and a_2 . Let p'_2 be the projection onto the second coordinate in the basis $e_1, e_2, \dots, e_n$; then $q = b_1 p + b'_2 p$ is a homomorphism such that $q(M')$ contains $d = q(a_1 e_1 + a_2 e_2)$. (Here we use that $p(e_2) = 0 = p'_2(e_1)$.) Therefore $q(M')$ contains $p(M') = (a_1)$, so by the maximality of $p(M')$ we get that $q(M') = p(M')$, and hence $d \in (a_1)$. Therefore $a_1 \mid d$, and as $d \mid a_2$, we get that $a_1 \mid a_2$. $\square$

Corollary 2.3.52

Let E be a finitely generated abelian group. Then,

$$E \cong (\mathbb{Z}/a_1) \times \dots \times (\mathbb{Z}/a_n),$$

for some $a_i \in \mathbb{Z}$ such that $a_1 \mid \dots \mid a_n$.

Proof. By assumption, there exists $x_1, \dots, x_n \in E$ that generate E as a $\mathbb{Z}$ -module. Thus, if we consider the map $\phi: \mathbb{Z}^n \rightarrow E$ which sends

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \mapsto a_1x_1 + \dots + a_nx_n,$$

then ϕ is surjective. By the strong form of the structure theorem, we know then that there exists a $q \leq n$ and $\lambda_1, \dots, \lambda_q \in \mathbb{Z}$ such that $\{a_1e_1, \dots, a_qe_q\}$ is a basis of $\ker \phi$ and $\lambda_1 \mid \dots \mid \lambda_q$. Define $a_k = 0$, for $k > q$. Then,

$$E \cong \mathbb{Z}^n / \ker \phi \cong \prod_{i=1}^n e_i\mathbb{Z} / \lambda_i e_i\mathbb{Z} \cong \prod_{i=1}^n \mathbb{Z} / \lambda_i\mathbb{Z}$$

□

Corollary 2.3.53 (Smith Normal Form)

Let A be a nonzero $m \times n$ matrix with coefficients in $\mathbb{Z}$. Then, $\exists S \in GL_m(\mathbb{Z})$ and $T \in GL_n(\mathbb{Z})$, and an integer $r > 0$ and positive integers $d_1 \mid d_2 \mid \dots \mid d_r$ such that

$$SAT = \begin{pmatrix} \text{diag}(d_1 d_2 \dots d_r) & \dots & 0 \\ \vdots & 0 & \vdots \\ 0 & \dots & 0 \end{pmatrix}$$

Lemma 2.3.54

Let K be a number field, and let $\alpha_1, \dots, \alpha_n \in K$ be any basis of K as a $\mathbb{Q}$ vector space. Form the matrix $A = (\text{Tr}_K(\alpha_i \alpha_j))_{ij}$. Then, $\det(A) \neq 0$.

Proof. Consider the linear transformations associated to A , $A: \mathbb{Q}^n \rightarrow \mathbb{Q}^n$. We want to show that A is injective. Let

$$\mathbf{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} \in \ker(A).$$

Then, $A \cdot \mathbf{c} = 0$.

$$\begin{aligned} (A \cdot \mathbf{c})_i &= \sum_{j=1}^n a_{ij} c_j \\ &= \sum_{j=1}^n \text{Tr}_K(\alpha_i \alpha_j) c_j \\ &= \sum_{j=1}^n \text{Tr}_K(\alpha_i \alpha_j c_j) \\ &= \text{Tr}_K \left(\sum_{j=1}^n c_j \alpha_i \alpha_j \right) \end{aligned}$$

Note that we've used the $\mathbb{Q}$ -linearity of the trace and linearity of the trace in the above computations. Writing $\alpha = \sum_{j=1}^n c_j \alpha_j$, we see that

$$0 = (A \cdot \mathbf{c})_i = \text{Tr}_K(\alpha \cdot \alpha_i).$$

Then, $\forall i, \text{Tr}_K(\alpha \alpha_i) = 0$. Let $\beta \in K$. Then, $\beta = d_1 \alpha_1 + \cdots + d_n \alpha_n$ for some $d_1, \dots, d_n \in \mathbb{Q}$, and

$$\text{Tr}_K(\alpha \beta) = \sum_{i=1}^n d_i \text{Tr}(\alpha \alpha_i) = 0.$$

Take $\beta = \alpha^{-1}$, $\text{Tr}(\alpha \beta) = \text{Tr}(1) = n$. This implies that $\alpha = 0$, and so $c_j = 0$ for all j , and so $\mathbf{c} = 0$. $\square$

Theorem 2.3.55

Let K be a number field. Then, as a group under addition, $\mathcal{O}_K$ is free of rank $[K : \mathbb{Q}]$.

Proof. Let $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$ such that $\{\alpha_1, \dots, \alpha_n\}$ is a basis of K as a $\mathbb{Q}$ vector space (in fact, we've shown that $\forall \alpha \in K, \exists \lambda \in \mathbb{Z}$ such that $\lambda \alpha \in \mathcal{O}_K$). Every element $\alpha \in \mathcal{O}_K$ can be written as

$$\alpha = c_1 \alpha_1 + \cdots + c_n \alpha_n$$

for some $c_i \in \mathbb{Q}$. We want to now bound the denominators of α_i . Let $A = (\text{Tr}(\alpha_i \alpha_j))_{ij}$, then we've seen in the proof of lemma 2.3.54 that

$$A \cdot \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}, \quad m_i = \sum_j c_j \text{Tr}(\alpha_i \alpha_j).$$

This implies that

$$\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = A^{-1} \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix},$$

as the determinant is nonzero. Pick $d \in \mathbb{Z}$ such that $d \cdot A^{-1} \in M_n(\mathbb{Z})$. Then,

$$\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = d^{-1} \cdot d A^{-1} \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} \in d^{-1} \mathbb{Z}^n.$$

In particular, we get that for all $\alpha \in \mathcal{O}_K$, $\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} \in d^{-1} \mathbb{Z}^n$, so

$$\mathcal{O}_K \subseteq \mathbb{Z} \cdot \frac{\alpha_1}{d} \oplus \cdots \oplus \mathbb{Z} \frac{\alpha_n}{d} \cong \mathbb{Z}^n$$

This implies that the rank of $\mathcal{O}_K$ is at most n . By the weak form of the structure theorem, the above implies that $\mathcal{O}_K$ is free of rank at most n . Moreover,

$$\mathbb{Z}^n \cong \mathbb{Z} \alpha_1 \oplus \cdots \oplus \mathbb{Z} \alpha_n \subseteq \mathcal{O}_K,$$

so the rank of $\mathcal{O}_K$ is at least n . Thus, it follows that $\mathcal{O}_K$ is free of rank equal n . $\square$

§2.3.vi Embeddings in Number Fields

Definition 2.3.56. Let K be a number field. An **embedding** of K is a ring homomorphism $\tau: K \rightarrow \mathbb{C}$. We write Σ_K for the set of embeddings of K .

Remark 2.3.57 — Ring homomorphisms always send $1 \mapsto 1$, so embeddings are always injective.

Remark 2.3.58 — Recall that we defined number fields as subfields $K \subseteq \mathbb{C}$ that are finitely generated over $\mathbb{Q}$. Then, by definition, we have that $\text{ind}: K \hookrightarrow \mathbb{C}$, $\alpha \mapsto \alpha$ is an embedding.

Recall the following facts about field extensions.

- $\mathbb{Q}(\alpha) \cong \mathbb{Q}[x]/(m_\alpha(x))$
- If $f(x) \in \mathbb{Q}[x]$ is irreducible of degree n , then it has n distinct roots.

Proposition 2.3.59

Let α be an algebraic number with minimal polynomial $m_\alpha(x)$ of degree n . Then

$$|\Sigma_{\mathbb{Q}(\alpha)}| = n = [\mathbb{Q}(\alpha) : \mathbb{Q}].$$

Moreover, for each of the distinct roots $\alpha_1, \dots, \alpha_n$ of $m_\alpha(x)$, there is an embedding $\mathbb{Q}(\alpha) \rightarrow \mathbb{C}$ given by $\alpha \mapsto \alpha_i$.

Proof. To give a ring homomorphism from $\mathbb{Q}(\alpha)$ to $\mathbb{C}$ is the same as giving a ring homomorphism $\mathbb{Q}[x] \xrightarrow{\phi} \mathbb{C}$ such that $\ker(\phi) = m_\alpha$. A ring homomorphism from $\mathbb{Q}[x]$ to $\mathbb{C}$ is uniquely determined by the image of x : $x \xrightarrow{\phi} \beta \in \mathbb{C}$. Then, $\ker(\phi) = m_\alpha$ if and only if $\phi(m_\alpha(x)) = 0$. This implies that

$$m_\alpha(\beta) = 0,$$

and so β is a root of m_α . Finally, we see that the correspondence morphism

$$\begin{aligned} \mathbb{Q}[x]/(m_\alpha(x)) &\cong \mathbb{Q}(\alpha) \rightarrow \mathbb{C} \\ \alpha &\mapsto \beta \end{aligned}$$

□

Remark 2.3.60 — All the roots need not to be in $\mathbb{Q}(\alpha)$.

Example 2.3.61

Let $K = \mathbb{Q}(\sqrt{d})$. Then, $m_{\sqrt{d}} = T^2 - d$. This gives

$$\Sigma_{\mathbb{Q}(\sqrt{d})} = \{\tau_1, \tau_2\},$$

where

$$\begin{aligned}\tau_1: \mathbb{Q}(\sqrt{d}) &\rightarrow \mathbb{C} \\ \sqrt{d} &\mapsto \sqrt{d} \\ \tau_2: \mathbb{Q}(\sqrt{d}) &\rightarrow \mathbb{C} \\ \sqrt{d} &\mapsto -\sqrt{d}.\end{aligned}$$

Remark 2.3.62 — We have seen, previously, that if

$$m_\alpha = x^n + a_{n-1}x^{n-1} + \cdots + a_0,$$

then

$$\begin{aligned}\mathrm{Tr}_{\mathbb{Q}(\alpha)}(\alpha) &= -a_{n-1} \\ \mathcal{N}_{\mathbb{Q}(\alpha)}(\alpha) &= (-1)^n \cdot a_0.\end{aligned}$$

However, we can see (after factoring) that

$$\mathrm{Tr}_{\mathbb{Q}(\alpha)}(\alpha) = -a_{n-1} = \sum \alpha_i = \sum_{\tau: \mathbb{Q}(\alpha) \rightarrow \mathbb{C}} \tau(\alpha).$$

We can also see that

$$\mathcal{N}_{\mathbb{Q}(\alpha)}(\alpha) = (-1)^n \cdot a_0 = \prod \alpha_i = \prod_{\tau: \mathbb{Q}(\alpha) \rightarrow \mathbb{C}} \tau(\alpha).$$

Lemma 2.3.63

Let K be a number field and $\tau_0: K \hookrightarrow \mathbb{C}$ be an embedding. Let α be an algebraic number. Then, there are $[K(\alpha) : K]$ embeddings $\tau: K(\alpha) \rightarrow \mathbb{C}$ that restrict to $\tau_0: K \rightarrow \mathbb{C}$.

Proof. The proof is similar to the proof of lemma 2.3.59. □

Corollary 2.3.64

Let K and L be number fields and $\tau_0: K \rightarrow \mathbb{C}$ be an embedding. Then, there are $[L : K]$ embeddings $\tau: L \rightarrow \mathbb{C}$ that restrict to τ_0 on K .

Proof. We proceed by induction. If $[L : K] = 1$, then $L = K$ and we are done. Next, pick $\alpha \in L \setminus K$. Then, $K \subsetneq K(\alpha) \subseteq L$. Then,

$$[L : K(\alpha)] < [L : K].$$

By induction, the conclusion of the corollary is true for $K(\alpha) \subseteq L$. By lemma 2.3.63, we have τ_0 extends to $[K(\alpha) : K]$ embeddings of $K(\alpha)$, and by the induction hypothesis, they each extend to $[L : K(\alpha)]$ embeddings of L , giving

$$[L : K(\alpha)][K(\alpha) : K] = [L : K].$$

□

Theorem 2.3.65

Let K be a number field. Then, for $\alpha \in K$ we have:

$$\begin{aligned} \text{Tr}_K(\alpha) &= \sum_{\tau \in \Sigma_K} \tau(\alpha) \\ \mathcal{N}_K(\alpha) &= \prod_{\tau \in \Sigma_K} \tau(\alpha) \end{aligned}$$

Proof. We've already shown that

$$\text{Tr}_K(\alpha) = [K : \mathbb{Q}(\alpha)] \cdot \text{Tr}_{\mathbb{Q}(\alpha)}(\alpha).$$

Then,

$$\begin{aligned} \text{Tr}_K(\alpha) &= [K : \mathbb{Q}(\alpha)] \cdot \sum_{\tau_0 \in \Sigma_{\mathbb{Q}(\alpha)}} \tau_0(\alpha) \\ &= \sum_{\tau \in \Sigma_K} \tau(\alpha) \end{aligned} \quad (*)$$

The equality $(*)$ follows from the fact that if τ extends τ_0 , then $\tau(\alpha) = \tau_0(\alpha)$ and $\tau|_{\mathbb{Q}(\alpha)} = \tau_0$. □

Proposition 2.3.66

Let K be a number field and $\alpha \in \mathcal{O}_K$. Then, $\alpha \in \mathcal{O}_K^\times$ if and only if $\mathcal{N}(\alpha) \in \mathbb{Z}^\times = \{\pm 1\}$.

Proof. If $\alpha \in \mathcal{O}_K^\times$, then there exists $\beta \in \mathcal{O}_K$ such that $\alpha \cdot \beta = 1$. Then,

$$\mathcal{N}(\alpha) \cdot \mathcal{N}(\beta) = 1,$$

implying that $\mathcal{N}(\alpha) \in \mathbb{Z}^\times$.

Conversely, assume that $\mathcal{N}_K(\alpha) \in \mathbb{Z}^\times$. Then,

$$\mathcal{N}_K(\alpha) = \prod_{\tau \in \Sigma_K} \tau(\alpha) = \alpha \cdot \prod_{\substack{\tau \in \Sigma_K \\ \tau \neq \text{id}}} \tau(\alpha).$$

Then define

$$\beta := \mathcal{N}(\alpha) \cdot \prod_{\substack{\tau \in \Sigma_K \\ \tau \neq \text{id}}} \tau(\alpha).$$

This implies that $\alpha \cdot \beta = 1$, since $\alpha \cdot \beta$ is just $\mathcal{N}_K(\alpha)^2$. This implies that $\alpha^{-1} = \beta \in K$. Now note that $\tau(\alpha)$ is an algebraic integer for all $\tau \in \Sigma_K$ ($\tau(\alpha)$ is a root of $m_\alpha(x) \in \mathbb{Z}[x]$ and as α is an algebraic integer). Then β is a product of algebraic integers and $\beta \in K$, ultimately implying that $\beta \in \mathcal{O}_K$. Therefore, $\alpha \in \mathcal{O}_K^\times$. □

Remark 2.3.67 — If $(\alpha_1, \dots, \alpha_n)$ is a basis of K over $\mathbb{Q}$ such that $\alpha_i \in \mathcal{O}_K$, then

$$\mathbb{Z}\alpha_1 \oplus \dots \oplus \mathbb{Z}\alpha_n \subseteq \mathcal{O}_K.$$

§2.3.vii The Discriminant

Definition 2.3.68. Let K be a number field of degree $n = [K : \mathbb{Q}]$. The **discriminant** of $\alpha_1, \dots, \alpha_n \in K$ as

$$\text{disc}(\alpha_1, \dots, \alpha_n) = \det((\text{Tr}_K(\alpha_i \alpha_j))_{ij}) \in \mathbb{Q}.$$

Definition 2.3.69. We call a basis $\alpha_1, \dots, \alpha_n$ for $\mathcal{O}_K$ an **integral basis**.

Example 2.3.70

Let $K = \mathbb{Q}(\sqrt{d})$. If $d \not\equiv 1 \pmod{4}$, then we have $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ and an integral basis is given by $1, \sqrt{d}$. Then,

$$\text{disc}(1, \sqrt{d}) = \det \begin{pmatrix} 2 & 0 \\ 0 & 2d \end{pmatrix} = 4d.$$

If $d \equiv 1 \pmod{4}$, then we have $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$, and an integral basis is given by $1, \frac{1+\sqrt{d}}{2}$. Then,

$$\text{disc}(1, \frac{1+\sqrt{d}}{2}) = \det \begin{pmatrix} 2 & 1 \\ 1 & \frac{1+d}{2} \end{pmatrix} = d.$$

Remark 2.3.71 — We've seen in the proof of lemma 2.3.54, that if we take $\alpha_1, \dots, \alpha_n$ to be a basis of K as a $\mathbb{Q}$ vector space, then

$$\text{disc}(\alpha_1, \dots, \alpha_n) \neq 0.$$

Remark 2.3.72 — If $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$, then $\text{disc}(\alpha_1, \dots, \alpha_n) \in \mathbb{Z}$.

Lemma 2.3.73

Suppose $\alpha_1, \dots, \alpha_n \in K$ and $\beta_1, \dots, \beta_n \in K$ such that

$$\begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix} = S \cdot \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}, \quad S \in M_n(\mathbb{Q}).$$

Then, $\text{disc}(\beta_1, \dots, \beta_n) = \det(S)^2 \cdot \text{disc}(\alpha_1, \dots, \alpha_n)$.

Proof. This is just a computation. Show that if

$$\begin{aligned} A &= (\text{Tr}_K(\alpha_i \alpha_j))_{ij} \\ B &= (\text{Tr}_K(\beta_i \beta_j))_{ij} \\ S &= (c_{ij}), \end{aligned}$$

then

$$\mathrm{Tr}_K(\beta_i \beta_j) = \sum_{k,l} s_{ik} \mathrm{Tr}(\alpha_k \alpha_l) c_{jl}.$$

This implies that $B = SAS^T$, so then take the determinant. $\square$

Corollary 2.3.74

If $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$ and $\beta_1, \dots, \beta_n \in \mathcal{O}_K$ are two $\mathbb{Z}$ -bases of $\mathcal{O}_K$ as an abelian group, then

$$\mathrm{disc}(\alpha_1, \dots, \alpha_n) = \mathrm{disc}(\beta_1, \dots, \beta_n).$$

Proof. In this case, the change of base matrix S is in $GL_n(\mathbb{Z})$, so $\det(S) = \pm 1$. Thus, follows from lemma 2.3.73. $\square$

From this, we can now define the following.

Definition 2.3.75. If $M \subset K$ is a free abelian subgroup of rank n we define $\mathrm{disc}(M) = \mathrm{disc}(\alpha_1, \dots, \alpha_n)$ for any $\mathbb{Z}$ -basis of M .

Definition 2.3.76. We define the **discriminant of K** to be $D_K = \mathrm{disc}(\mathcal{O}_K) \in \mathbb{Z}$. In other words, $D_K = \mathrm{disc}(\alpha_1, \dots, \alpha_n)$ for any integral basis $\alpha_1, \dots, \alpha_n$.

Proposition 2.3.77

Suppose $M \subseteq N \subseteq K$ with M, N being free abelian subgroups of K of rank n . Then,

$$\mathrm{disc}(M) = [N : M]^2 \cdot \mathrm{disc}(N).$$

Proof. By the strong form of the structure theorem, there exists a basis $\alpha_1, \dots, \alpha_n$ of N and $d_1, \dots, d_n \in \mathbb{Z}$ such that $d_1 \alpha_1, \dots, d_n \alpha_n$ is a basis for M . Then.

$$\mathrm{disc}(M) = \det(\mathrm{Tr}_K(d_i \alpha_i d_j \alpha_j)_{ij}) = \det(d_i d_j \cdot \mathrm{Tr}_K(\alpha_i \alpha_j)) = (d_1 \dots d_n)^2 \det(\mathrm{Tr}_K(\alpha_i \alpha_j)).$$

Thus, we've shown that

$$\mathrm{disc}(M) = (d_1 \dots d_n)^2 \mathrm{disc}(N).$$

Finally, we have

$$N/M \cong (\mathbb{Z}\alpha_1 \oplus \dots \mathbb{Z}\alpha_n) / (\mathbb{Z}d_1\alpha_1 \oplus \dots \mathbb{Z}d_n\alpha_n) \cong \mathbb{Z}/d_1 \oplus \dots \oplus \mathbb{Z}/d_n.$$

Thus, $[N : M] = d_1 \dots d_n$. $\square$

Remark 2.3.78 — In particular, if $M \subseteq \mathcal{O}_K$ is free of rank n , then:

- $D_K \mid \mathrm{disc}(M)$.
- If $p \mid [\mathcal{O}_K : M]$ for some p , then $p^2 \mid \mathrm{disc}(M)$.
- If $\mathrm{disc}(M)$ is square free, then $\mathcal{O}_K = M$.

From the above, if $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$ such that

$$\mathrm{disc}(\alpha_1, \dots, \alpha_n) = \mathrm{disc}(\mathbb{Z}\alpha_1 \oplus \dots \oplus \mathbb{Z}\alpha_n)$$

is square free implies that $\alpha_1, \dots, \alpha_n$ is an integral basis of $\mathcal{O}_K$.

Remark 2.3.79 — We've seen in the case that $d \not\equiv 1 \pmod{4}$ that this discriminant can have a square in it.

Proposition 2.3.80

Let K be a number field and $\Sigma_K = \{\tau_1, \dots, \tau_n\}$. Letting $\alpha_1, \dots, \alpha_n \in K$, then

$$\text{disc}(\alpha_1, \dots, \alpha_n) = \det(\tau_i(\alpha_j)_{ij})^2.$$

Proof. Let $A = (\text{Tr}_K(\alpha_i \alpha_j)_{ij})$ and $T = (\tau_i(\alpha_j)_{ij})$. Then,

$$\text{Tr}_K(\alpha_i \alpha_j) = \sum_{l=1}^n \tau_l(\alpha_i \alpha_j) = \sum_l \tau_l(\alpha_i) \tau_l(\alpha_j),$$

but the final term is the (i, j) -th entry of $T^T \cdot T$, so, taking determinants, we're done. $\square$

Remark 2.3.81 — If $K = \mathbb{Q}(\alpha)$ with a homogeneous polynomial $m_\alpha(x) = (x - \alpha_1) \dots (x - \alpha_n)$, take $\{1, \alpha, \dots, \alpha^{n-1}\}$ as a basis of K as a $\mathbb{Q}$ vector space. Then, $\tau_1, \dots, \tau_n$ are such that $\tau_i(\alpha) = \alpha_i$. Then,

$$\text{disc}(1, \alpha, \dots, \alpha^{n-1}) = \det \begin{pmatrix} 1 & \alpha_1 & \dots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \dots & \alpha_2^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \alpha_n & \dots & \alpha_n^{n-1} \end{pmatrix} = \prod_{i < j} (\alpha_i - \alpha_j)^2 = \text{disc}(m_\alpha).$$

Proposition 2.3.82

We have that

$$\text{disc}(1, \dots, \alpha^{n-1}) = (-1)^{\frac{n(n-1)}{2}} \mathcal{N}_{\mathbb{Q}(\alpha)}(m'_\alpha(\alpha)).$$

Proof. We have that

$$\begin{aligned} \mathcal{N}_{\mathbb{Q}(\alpha)}(m'_\alpha(\alpha)) &= \prod_{i=1}^m \tau_i(m'_\alpha(\alpha)) \\ &= \prod_{i=1}^m (m'_\alpha(\tau_i \alpha)) \\ &= \prod_{i=1}^m (m'_\alpha(\alpha_i)). \end{aligned}$$

Note that we have

$$m'_\alpha(x) = (x - \alpha_1) \dots (x - \alpha_{i-1})(x - \alpha_{i+1}) \dots (x - \alpha_n) + (x - \alpha_i) \cdot \left(\frac{m_\alpha(x)}{(x - \alpha_i)} \right)',$$

so

$$m'_\alpha(\alpha_i) = \prod_{j \neq i} (\alpha_i - \alpha_j).$$

Therefore,

$$\begin{aligned}\mathcal{N}_{\mathbb{Q}(\alpha)}(m'_\alpha(\alpha)) &= \prod_{i=1}^n \prod_{j \neq i} (\alpha_i - \alpha_j) \\ &= (-1)^{\frac{n(n-1)}{2}} \prod_{i < j} (\alpha_i - \alpha_j)^2.\end{aligned}$$

□

§2.3.viii Prime Ideals in $\mathcal{O}_K$

The motivation now is to understand prime ideals in $\mathcal{O}_K$. Recall that:

Definition 2.3.83. An ideal $I \subsetneq R$ is **prime** if $\forall a, b \in R$ such that $a \cdot b \in I \implies a \in I$ or $b \in I$.

Notation 2.3.84. We denote elements of the quotient ring R/I as $[r] := r + I$.

Notation 2.3.85. Given a subset of R , say $S \subseteq R$, we denote by (S) the smallest ideal containing S :

$$(S) = \left\{ \sum_i r_i s_i \mid r_i \in R, s_i \in S \right\}.$$

Here are some operations with ideals. Given $I, J \subseteq R$ ideals, then the following are ideals:

1. $I \cap J$
2. $I + J = \{a + b \mid a \in I, b \in J\}$
3. $I \cdot J = \{\sum_i a_i b_i \mid a_i \in I, b_i \in J\}$

Recall the following:

Proposition 2.3.86

An ideal $I \subseteq R$ is prime if and only if R/I is an integral domain.

Notation 2.3.87. We denote by $(a_1, \dots, a_n)$ the ideal generated by the set $\{a_1, \dots, a_n\}$.

Exercise 2.3.88. Show that:

- $(a_1, \dots, a_n) + (b_1, \dots, b_m) = (a_1, \dots, a_n, b_1, \dots, b_m)$
- $(a_1, \dots, a_n)(b_1, \dots, b_m) = (\{a_i b_j\}_{i=1, \dots, n, j=1, \dots, m})$.

Example 2.3.89

Take $R = \mathbb{Z}[\sqrt{-5}]$, which is not a UFD. Define the following:

$$\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$$

$$\mathfrak{p}_3 = (3, 1 + \sqrt{-5})$$

$$\bar{\mathfrak{p}}_3 = (3, 1 - \sqrt{-5})$$

Then,

$$\mathfrak{p}_2^2 = (2), \mathfrak{p}_3 \bar{\mathfrak{p}}_3 = (3).$$

Then, $\mathfrak{p}_2$ is a prime ideal and $\mathfrak{p}_2$ is not principal.

Solution. Suppose $\mathfrak{p}_2$ is principal; particularly, that $\mathfrak{p}_2 = (\alpha)$, where $\alpha \in \mathbb{Z}[\sqrt{-5}]$. Then, we have the following.

$$(2) = \mathfrak{p}_2^2 = (\alpha)^2 \implies \alpha^2 = u \cdot z.$$

From this fact, we have that $N(\alpha^2) = N(\alpha)^2 = 4$, so $N(\alpha) = 2$. However, the equation

$$a^2 + 5b^2 = N(\alpha) = 2$$

has no solution. Similarly, we have that $\mathfrak{p}_3$ is not principal.

We now show that $\mathfrak{p}_3$ is a prime ideal by using the isomorphism:

$$\begin{aligned} \mathbb{Z}[\sqrt{-5}] &\xrightarrow{\sim} \mathbb{Z}[T]/(T^2 + 5) \\ \sqrt{-5} &\mapsto T \\ \mathfrak{p}_3 &\mapsto (3, 1 + T) \end{aligned}$$

Then, we have that

$$\begin{aligned} (\mathbb{Z}[T]/(T^2 + 5))/(3, 1 + T) &\cong \mathbb{Z}[T]/(T^2 + 5, 3, 1 + T) \cong (\mathbb{Z}[T]/3)/(T^2 + 5, 1 + T) \\ &\cong (\mathbb{Z}/3[T])/(T^2 - 1, T + 1) \cong (\mathbb{Z}/3[T])/(T + 1) \cong \mathbb{Z}/3. \end{aligned}$$

Thus, $\mathfrak{p}_3$ is a prime ideal.

We now show that $\mathfrak{p}_3 \cdot \bar{\mathfrak{p}}_3 = (3)$. We first have

$$\mathfrak{p}_3 \cdot \bar{\mathfrak{p}}_3 = (9, 3(1 - \sqrt{-5}), 3(1 + \sqrt{-5}), 6).$$

We claim that the above is equal to (3) . To see that $\mathfrak{p}_3 \cdot \bar{\mathfrak{p}}_3 \subseteq (3)$, we see that all generators are multiples of 3. To see the other inclusion, we note that

$$3 = 9 - 6.$$

□

3 Dedekind Domains

We will now move on to study a new class of rings, called Dedekind domains.

§3.1 Definitions and Introductory Properties

Definition 3.1.1. Let R be an integral domain. R is a **Dedekind domain** if it satisfies the following conditions:

- R is integrally closed.
- Every nonzero prime ideal is maximal.
- R is Noetherian.

Exercise 3.1.2. Provide examples of rings that satisfy only two of the three conditions above.

Solution: Let property (1), (2), (3) be the properties above listed in order.

- (1) + (3): $\mathbb{Z}[x]$
- (2) + (3): $\mathbb{Z}[\sqrt{-3}]$
- (1) + (2): $\overline{\mathbb{Z}}$ (ring of algebraic integers; hard to show this!)

□

Recall that a ring is Noetherian if it satisfies the following three equivalent conditions:

Proposition 3.1.3

A ring R is Noetherian if and only if one of the following hold:

1. R satisfies that ascending chain condition.
2. Any nonempty set of ideals in R has a maximal element.
3. Any ideal $I \subseteq R$ is finitely generated.

Remark 3.1.4 — These three conditions are equivalent.

§3.2 Number Fields and Dedekind Domains

We now work towards the following proposition.

Proposition 3.2.1

Let K be a number field. Then, $\mathcal{O}_K$ is a Dedekind domain.

We now introduce some lemmas.

Lemma 3.2.2

Let $I \subseteq \mathcal{O}_K$ be a nonzero ideal. Then, I has finite index in $\mathcal{O}_K$. That is,

$$[\mathcal{O}_K : I] < \infty.$$

Proof. Choose a nonzero $\alpha \in I$. Then,

$$(\mathcal{N}_K(\alpha)) \subseteq (\alpha) \subseteq I.$$

In fact,

$$\mathcal{N}_K(\alpha) = \prod_{\tau \in \Sigma_K} \tau(\alpha) = \alpha \cdot \underbrace{\prod_{\tau \neq \text{id}} \tau(\alpha)}_{=\beta}.$$

We have that $\beta = \frac{\mathcal{N}_K(\alpha)}{\alpha}$, but the numerator is in $\mathbb{Q}$ and the denominator is in K , implying that $\beta \in K$. Moreover, β is a product of algebraic integers, $\forall \tau$, $\tau(\alpha)$ is a root of

$$m_\alpha(T) \in \mathbb{Z}[T].$$

This implies that β lies in $\mathcal{O}_K$. We use this to bound the index.

The fact that $(\mathcal{N}_K(\alpha)) \subseteq (\alpha) \subseteq I$ implies that

$$[\mathcal{O}_K : I] \leq [\mathcal{O}_K : \mathcal{N}_K(\alpha)].$$

Therefore, it's enough to show that for all $N \in \mathbb{Z}$ and $N \neq 0$, $\mathcal{O}_K/N$ is finite. We know that $\mathcal{O}_K \cong \mathbb{Z}^n$ as abelian groups, where $n = [K : \mathbb{Q}]$. Therefore,

$$\mathcal{O}_K/N \cong \mathbb{Z}^n/N = (\mathbb{Z}/N)^n,$$

which is finite. □

Remark 3.2.3 — We note that if $I \subseteq J$, where I, J are ideals of R , then we get a surjective morphism

$$R/I \rightarrow R/J \cong (R/I)/(J(R/I)).$$

This implies that that

$$|(R/J)| \leq |(R/I)|.$$

Proof of Proposition 3.2.1. There are three criterion to check. We've already seen that $\mathcal{O}_K$ is integrally closed. Now, let $\mathfrak{p} \subseteq \mathcal{O}_K$ be a nonzero prime ideal. Then, the previous lemma tells us that $\mathcal{O}_K/\mathfrak{p}$ is finite. Then, $\mathcal{O}_K/\mathfrak{p}$ is an integral domain and finite, implying that $\mathcal{O}_K/\mathfrak{p}$ is a field. Therefore, $\mathfrak{p}$ is maximal. Finally, we show that $\mathcal{O}_K$ is Noetherian. Consider the following ascending chain of ideals:

$$I_1 \subseteq I_2 \subseteq \dots$$

Then,

$$\infty > [\mathcal{O}_K : I_1] \geq [\mathcal{O}_K : I_2] \geq \dots,$$

by the previous remark and lemma. Then, there exists an n such that

$$[\mathcal{O}_K : I_n] = [\mathcal{O}_K : I_{n+1}] = \dots,$$

implying that $I_n = I_{n+1} = \dots$, we had a surjective morphism $R/I_n \rightarrow R/I_{n+1}$ as well. Thus, $\mathcal{O}_K$ is Noetherian. □

Proposition 3.2.4

Let R be a PID. Then, R is a Dedekind domain.

Proof. We have that PIDs are UFDs, and UFDs are integrally closed. If $(a) \subseteq R$ is a prime ideal with $0 \neq a$, then a must be a prime element and hence irreducible. Assume that $(a) \subseteq (c) \subseteq R$, for some $c \in R$. Then, $a = b \cdot c$, for some b . However, a is irreducible, so b is a unit or c is a unit. If b is a unit, that $(a) = (c)$, and if c is a unit, then $(c) = R$, so (a) is maximal ideal. Finally, PIDs are Noetherian, since every ideal is principal (so finitely generated). $\square$

Remark 3.2.5 — The coordinate ring of smooth affine curves are Dedekind domains (non-examinable).

Here are some examples that illustrate the remark above.

Example 3.2.6

For instance, take $\mathbb{C}[x, y]/(x^n + y^n - 1)$.

Example 3.2.7

$\mathbb{C}[x, y]/(y^2 - x^3)$ is an example of a regular curve and the ring $(y^2 - x^3)$ is not integrally closed.

§3.3 Factorization into Prime Ideals

This is the next big result.

Theorem 3.3.1

Let R be a Dedekind domain. Then, any nonzero ideal $I \subsetneq R$ can be written in the form

$$I = \mathfrak{p}_1 \cdot \mathfrak{p}_2 \cdot \dots \cdot \mathfrak{p}_r,$$

where $\mathfrak{p}_i$ are prime ideals. Moreover, this expression is unique up to permuting the factors.

Remark 3.3.2 — For PIDs, this theorem is the same as proving that PIDs are UFDs.

To prove this, we are going to need a series of important results for a Dedekind domain R . The main results that will be crucial are lemma 3.3.5, corollary 3.3.9 and corollary 3.3.10.

Lemma 3.3.3

Let $I \subseteq R$ be an ideal and let $\alpha \in \text{Frac}(R)$ such that $\alpha \cdot I \subseteq I$. Then, $\alpha \in R$.

Proof. As R is Noetherian, we have that $I = (a_1, \dots, a_n)$, where $a_i \in R$. Then,

$$\alpha a_j = \sum_{i=1}^n c_{ij} a_i, \quad c_{ij} \in R.$$

We build the matrix $A = (c_{ij}) \in M_n(R)$, and consider the characteristic polynomial of A , which we'll denote as $\chi_A(T) \in R[T]$. Then,

$$\chi_A(\alpha) \cdot I = \chi_A(A) \cdot I,$$

which follows by a similar calculation to that in proposition 2.3.13. The Cayley-Hamilton theorem then tells us that $\chi_A(A) = 0$, implying that $\chi_A(\alpha) = 0$. Finally, $\chi_A(T) \in R[T]$ is monic and R is integrally closed, implying that $\alpha \in R$. $\square$

Lemma 3.3.4

Let $I \subseteq R$ be a nonzero ideal. Then, there are nonzero prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ of R such that

$$\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq I.$$

Proof. Let S be the set of nonzero ideals of R for which the conclusion of the lemma fails. If $S = \emptyset$, we're done, so assume that S is nonempty. In this case, pick a maximal element I of S . I can't be prime; otherwise, the conclusion of the lemma trivially holds for prime ideals. This implies that we can find $a, b \in R$ such that $a \cdot b \in I$, but neither $a \in I$ and $b \in I$. As $I \subsetneq (a) + I$ and $I \subsetneq (b) + I$. Since I was the maximal ideal for which the conclusion of the lemma fails, this implies that $(a) + I$ and $(b) + I$ contain a product of nonzero prime ideals. In particular,

$$\mathfrak{p}_1 \dots \mathfrak{p}_n \subseteq (a) + I, \quad \mathfrak{q}_1 \dots \mathfrak{q}_m \subseteq (b) + I.$$

Taking the product, we have that

$$\begin{aligned} \mathfrak{p}_1 \dots \mathfrak{p}_n \cdot \mathfrak{q}_1 \dots \mathfrak{q}_m &= ((a) + I)((b) + I) \\ &= (ab) + (a)I + (b)I + I^2 \subseteq I, \end{aligned}$$

which is a contradiction, as $a \cdot b \in I$. $\square$

Lemma 3.3.5

If $\mathfrak{p}$ is a prime ideal and $I, J \subseteq R$ are ideals, then $I \cdot J \subseteq \mathfrak{p}$ implies that $I \subseteq \mathfrak{p}$ or $J \subseteq \mathfrak{p}$.

Proof. Suppose that $I \not\subseteq \mathfrak{p}$. Then, there exists an element $a \in I$ and $a \notin \mathfrak{p}$. Then, for all $b \in J$, we have that

$$a \cdot b \in I \cdot J \subseteq \mathfrak{p}.$$

This implies that $b \in \mathfrak{p}$, implying that $J \subseteq \mathfrak{p}$. $\square$

Remark 3.3.6 — We didn't use anything regarding Dedekind domains here; this is a statement about rings.

Lemma 3.3.7

Let $I \subsetneq R$ be a proper nonzero ideal. Then, there exists an element $x \in \text{Frac}(R)$ and $x \notin R$ such that $x \cdot I \subseteq R$.

Proof. Pick a nonzero element $\alpha \in I$. By lemma 3.3.4, there exists $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ which are all nonzero prime ideals such that

$$\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq (\alpha).$$

We can choose an r that is small as possible. Let $\mathfrak{m}$ be a maximal ideal such that

$$\mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq (\alpha) \subseteq I \subseteq \mathfrak{m}.$$

By lemma 3.3.5, we know that there exists an i such that $\mathfrak{p}_i \subseteq \mathfrak{m}$, implying that $\mathfrak{p}_i = \mathfrak{m}$. WLOG, let $i = 1$. Since r is minimal,

$$\mathfrak{p}_2 \dots \mathfrak{p}_r \not\subseteq (\alpha).$$

Let $b \in \mathfrak{p}_2 \dots \mathfrak{p}_r \setminus (\alpha)$ now. Then,

$$b \cdot I \subseteq b \cdot \mathfrak{p}_1 \subseteq \mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq (\alpha),$$

implying that $\frac{b}{\alpha} \cdot I \subseteq R$. This implication follows from the fact that for all $i \in I$, $b \cdot i = \alpha r$, for some $r \in R$, and so $\frac{b}{\alpha} \cdot i \in R$. This implies that $x = \frac{b}{\alpha}$ is an element such that $x \cdot I \subseteq R$ and $x \notin R$, since $b \notin (\alpha)$. (If $x \in R$, $\frac{b}{\alpha} = r$ implies that $b = \alpha \cdot r \in (\alpha)$.) $\square$

Proposition 3.3.8

Let R be a Dedekind domain, $I \subseteq R$ be a proper nonzero ideal, $0 \neq a \in I$. Then, there exists an ideal $J \subseteq R$ such that $I \cdot J = (a)$.

Proof. Let $J = \{b \in R \mid b \cdot I \subseteq (a)\}$. This is an ideal (check). Clearly, $I \cdot J \subseteq (a)$. We now show the converse. Assume that $I \cdot J \subsetneq (a)$. Then, define

$$K := \frac{1}{a} \cdot I \cdot J \subsetneq R.$$

K is then a nonzero proper ideal. We can now apply lemma 3.3.7 and find an element $x \in \text{Frac}(R)$ and $x \notin R$, such that $x \cdot K \subseteq R$. Therefore, $x \cdot I \cdot J \subseteq (a)$. Moreover, $a \cdot J \subseteq I \cdot J$, because $a \in I$. Therefore,

$$J \subseteq \frac{1}{a} \cdot I \cdot J = K,$$

by definition of K . Thus, $x \cdot J \subseteq x \cdot K \subseteq R$. Now, if $b \in J$, then $x \cdot b \in R$. $x \cdot b \cdot I \subseteq x \cdot J \cdot I \subseteq (a)$. This means that $x \cdot b \in J$, implying that $x \cdot J \subseteq J$. Lemma 3.3.3 then tells us that $x \in R$. $\square$

Corollary 3.3.9

If $I, J, K \subseteq R$ are nonzero ideals such that $I \cdot K = J \cdot K$, then $I = J$.

Proof. By proposition 3.3.8, we can find K' such that $K \cdot K'$ is a principal ideal, i.e., $K \cdot K' = (a)$. This gives

$$I \cdot K \cdot K' = J \cdot K \cdot K' \implies I \cdot (a) = J \cdot (a),$$

and now we can cancel to get that $I = J$. $\square$

Corollary 3.3.10

If $I, J \subseteq R$ are nonzero ideals, then $I \subseteq J$ if and only if there exists an ideal $K \subseteq R$ such that $I = J \cdot K$.

Proof. ($\Leftarrow$): We have that $I = J \cdot K \subseteq J$, so true.

($\Rightarrow$): By the previous proposition, $\exists J'$ such that $J \cdot J' = (b)$, hence

$$I \cdot J' \subseteq J \cdot J' = (b).$$

Thus, have $K := \frac{1}{a} I J' \subseteq R$ is an ideal. Moreover,

$$JK = \frac{1}{a} I J J' = a^{-1} I (a) = I,$$

so we're done. $\square$

Proof of Theorem 3.3.1.

Claim 3.3.11. Every ideal is a product of prime ideals.

Proof. Let S be a set of ideals which are not a product of prime ideals. If $S = \emptyset$, then we are done. Thus, suppose that S is nonempty. As R is Noetherian, we can pick an ideal I that is maximal in S . We may pick $\mathfrak{p} \subset R$ such that $\mathfrak{p}$ is a maximal ideal such that $I \not\subseteq \mathfrak{p}$. We now apply corollary 3.3.10 to say that there exists an ideal $J \subseteq R$ such that $I = \mathfrak{p} \cdot J$. However, $I = \mathfrak{p} \cdot J \subseteq J$ if $I = J$, implying that $\mathfrak{p} = R$, which is a contradiction. Therefore, $I \not\subseteq J$ and hence J can be expressed as a product of prime ideals, and $I = \mathfrak{p} \cdot J$, which is a contradiction. $\square$

Claim 3.3.12 (Uniqueness of Factorization). Suppose we have

$$\mathfrak{p}_1 \dots \mathfrak{p}_r = \mathfrak{q}_1 \dots \mathfrak{q}_s.$$

Then, $r = s$ and $\mathfrak{p}_i$ and $\mathfrak{q}_i$ are the same up to reordering.

Proof. We do this by induction. We have that

$$\mathfrak{q}_1 \dots \mathfrak{q}_s = \mathfrak{p}_1 \dots \mathfrak{p}_r \subseteq \mathfrak{p}_1,$$

which implies (by lemma 3.3.5), that there exists an i such that $\mathfrak{q}_i \subseteq \mathfrak{p}_i$. After reordering, assume that $\mathfrak{q}_1 \subseteq \mathfrak{p}_1$. Since $\mathfrak{q}_1$ is maximal, we have that $\mathfrak{q}_1 = \mathfrak{p}_1$. By corollary 3.3.9 allows us to cancel, giving us

$$\mathfrak{q}_2 \dots \mathfrak{q}_s = \mathfrak{p}_2 \dots \mathfrak{p}_r,$$

and so induction finishes it off. $\square$

Thus, we've proven true. $\square$

§3.4 The Class Group

Here, we quantify how far a ring is from being a Dedekind domain.

Definition 3.4.1. Let R be a Dedekind domain. We say that two nonzero ideals $I, J \subseteq R$ are in the **same class** (and we make $I \sim J$) if there exists $\alpha \in \text{Frac}(R)^\times$ such that $\alpha I = J$. This defines an equivalence relation on the set of ideals of R .

Definition 3.4.2. We denote by $\text{Cl}(R)$ the set of equivalence classes. We call this the **class group** of R .

Notation 3.4.3. For an ideal I , we write $[I]$ for the class of I .

Remark 3.4.4 — We note that $[I] = [R] \iff [I] = \alpha[R]$ for some $\alpha \in R \iff I$ is principal.

Remark 3.4.5 — We define an operation on $\text{Cl}(R)$ given by

$$[I] \cdot [J] = [I \cdot J].$$

(check that this is well defined, associative, and commutative.)

Proposition 3.4.6

The set $\text{Cl}(R)$ together with this operation is an abelian group.

Proof. The identity will be the class of principal ideals, $[R]$:

$$[I] \cdot [R] = [I \cdot R] = [I].$$

Given $I \subseteq R$ that is nonzero, let $0 \neq \alpha \in I$. Then, by proposition 3.3.8, there exists J such that $I \cdot J = (\alpha)$. This gives that

$$[I] \cdot [J] = [I \cdot J] = [(\alpha)] = [R].$$

□

Example 3.4.7

Take $\mathbb{Z}[\sqrt{-5}]$, and the ideal $\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$. Then,

$$\begin{aligned} \mathfrak{p}_2 \mathfrak{p}_3 &= (3, 1 + \sqrt{-5})(2, 1 + \sqrt{-5}) \\ &= (6, 3(1 + \sqrt{-5}), 2(1 + \sqrt{-5}), 6) \\ &= (1 + \sqrt{-5}). \end{aligned}$$

Therefore, $[\mathfrak{p}_2] = [\mathfrak{p}_3]$ in $\text{Cl}(\mathbb{Z}[\sqrt{-5}])$. Moreover, we can check that $\mathfrak{p}_2^2 = (2)$. This implies that $[\mathfrak{p}_2] = [\mathfrak{p}_3]$ has order 2 in $\text{Cl}(\mathbb{Z}[\sqrt{-5}])$.

Proposition 3.4.8

Let R be a Dedekind domain. Then, the following are equivalent:

1. $\text{Cl}(R) = \{1\}$
2. R is a PID
3. R is a UFD.

Proof. (1 $\iff$ 2): This follows from $[I] = [R]$ if and only if I is principal.

(2 $\implies$ 3): This is because PIDs are UFDs.

(3 $\implies$ 2): Let $\mathfrak{p} \subseteq R$ be a nonzero prime ideal. Let $0 \neq \alpha \in \mathfrak{p}$. Since R is a UFD, by assumption, we have that

$$\alpha = u \Pi_1 \dots \Pi_r,$$

where $u \in R^\times$ and Π_i are irreducible. In UFDs, the Π_i s are prime. Then, $(\Pi_i) \subseteq \mathfrak{p}$, and (Π_i) is a prime ideal. This means that $(\Pi_i) = \mathfrak{p}$, so $\mathfrak{p}$ is principal. For a general ideal $I \subseteq R$, then lemma theorem 3.3.1 tells us that $I = \mathfrak{p}_1 \dots \mathfrak{p}_r$, with $\mathfrak{p}_i$ prime (and hence principal), so I is principal. $\square$

§3.5 Computations with Ideals**Proposition 3.5.1**

Let R be a Dedekind domain and let $I, J \subseteq R$ be nonzero ideals. Suppose

$$I = \mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r}, \quad J = \mathfrak{p}_1^{b_1} \dots \mathfrak{p}_r^{b_r}.$$

(Convention: $\mathfrak{p}^0 = R$) Here, the $\mathfrak{p}_i$ are distinct prime ideals, and a_i, b_i are nonnegative. Then:

1. $I \cdot J = \mathfrak{p}_1^{a_1+b_1} \dots \mathfrak{p}_r^{a_r+b_r}$
2. If $I \subseteq J$ if and only if $a_i \geq b_i$, for all i
3. $I + J = \mathfrak{p}_1^{\min(a_1, b_1)} \dots \mathfrak{p}_r^{\min(a_r, b_r)}$
4. $I \cap J = \mathfrak{p}_1^{\max(a_1, b_1)} \dots \mathfrak{p}_r^{\max(a_r, b_r)}$

Proof. (1): This follows by the commutativity of ideals.

(2): We know that $I \subseteq J$ if and only if there exists a $K \subseteq R$ ideal with $I = JK$. Let $K = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_r^{e_r} \cdot K'$. Using theorem 3.3.1, we get that $a_i = b_i + e_i$, for all i , and that $K' = R$. This implies that $a_i \geq b_i$, for all i .

(3): Assume that

$$I + J = \mathfrak{p}_1^{c_1} \dots \mathfrak{p}_r^{c_r}.$$

Since $I, J \subseteq I + J$, so we get that $c_i \leq a_i, b_i$. Therefore, $c_i \leq \min(a_i, b_i)$. On the other hand, $I, J \subseteq \mathfrak{p}_1^{\min(a_1, b_1)} \dots \mathfrak{p}_r^{\min(a_r, b_r)}$. Therefore, $I + J \subseteq \mathfrak{p}_1^{\min(a_1, b_1)} \dots \mathfrak{p}_r^{\min(a_r, b_r)}$. And so

the reverse inequality holds.

(4): Let $I \cap J = \mathfrak{p}_1^{c_1} \dots \mathfrak{p}_r^{c_r}$. Then, as $I \cap J \subseteq I, J$, we get that $a_i, b_i \leq c_i$, so $\max(a_i, b_i) \leq c_i$. On the other hand, $\mathfrak{p}_1^{\max(a_1, b_1)} \dots \mathfrak{p}_r^{\max(a_r, b_r)} \subseteq I, J$, so the reverse inequality holds. $\square$

Theorem 3.5.2 (Chinese Remainder Theorem)

Let R be a commutative ring. Let $I, J \subseteq R$ be ideals. If $I + J = R$, then we have that $I \cap J = I \cdot J$ and there is an isomorphism of rings

$$\begin{aligned} R/(I \cdot J) &\rightarrow R/I \times R/J \\ (a + IJ) &\mapsto (a + I, a + J). \end{aligned}$$

Proof. Covered before. $\square$

More in general, we have as follows.

Theorem 3.5.3 (General Chinese Remainder Theorem)

Let $I_1, \dots, I_r \subseteq R$ be ideals with $I_i + I_j = R$, for every $i \neq j$. Then,

$$I_1 \cap \dots \cap I_r = I_1 \dots I_r$$

and

$$R/(I_1 \dots I_r) = R/I_1 \times \dots \times R/I_r.$$

Consider the Chinese remainder theorem in the context of Dedekind domain. If $\mathfrak{p}_1, \mathfrak{p}_r$ are distinct prime ideals, then

$$\mathfrak{p}_1^{a_1} + \mathfrak{p}_1^{a_2} = R,$$

for all $a_1, a_2 \geq 0$. By the Chinese remainder theorem, we have that

$$R/(\mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r}) \xrightarrow{\sim} R/\mathfrak{p}_1^{a_1} \times \dots \times R/\mathfrak{p}_r^{a_r},$$

where $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are distinct prime ideals and $a_1, \dots, a_r \geq 0$. We will often use that we have a surjection map

$$R \rightarrow R/\mathfrak{p}_1^{a_1} \times \dots \times R/\mathfrak{p}_r^{a_r}.$$

Namely, the system of equations

$$\begin{aligned} x &= x_1 \pmod{\mathfrak{p}_1^{a_1}} \\ &\vdots \\ x &= x_r \pmod{\mathfrak{p}_r^{a_r}} \end{aligned}$$

is solvable for all $x_1, \dots, x_r \in R$.

Proposition 3.5.4

Let R be a Dedekind domain and let $I \subseteq R$ be a nonzero ideal and $0 \neq \alpha \in I$. Then, there exists $\beta \in I$ with $I = (\alpha, \beta)$. In particular, every ideal in R can be generated by two elements.

Proof. Let $I = \mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r}$ and $(\alpha) = \mathfrak{p}_1^{b_1} \dots \mathfrak{p}_r^{b_r} \mathfrak{q}_1^{c_1} \dots \mathfrak{q}_s^{c_s}$, where $\mathfrak{p}_i, \mathfrak{q}_j$ are distinct prime ideals. Since $(\alpha) \subseteq I$, we have that $b_i \geq a_i$. For $1 \leq i \leq r$, we have that $\mathfrak{p}_i^{a_i+1} \subsetneq \mathfrak{p}_i^{a_i}$. If there is equality here, then we get that $\mathfrak{p}_i = R$, which is a contradiction. Now pick $x_i \in \mathfrak{p}_i^{a_i}$ such that $x_i \notin \mathfrak{p}_i^{a_i+1}$. We also have that for all j , $\mathfrak{q}_j \subsetneq R$, so we pick $x_j \in R$ such that $x_j \notin \mathfrak{q}_j$. We now apply the Chinese remainder theorem to find $\beta \in R$ such that

$$\begin{aligned} \beta &= x_i \pmod{\mathfrak{p}_1^{a_i+1}} \\ \beta &= x_j \pmod{\mathfrak{q}_j} \end{aligned}$$

In particular, $\beta \notin \mathfrak{p}_i^{a_i+1}$ and $\beta \notin \mathfrak{q}_j$ ($\beta \in \mathfrak{p}_1^{a_i}$). This implies that

$$(\beta) = \mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r} \tau_1^{d_1} \dots \tau_l^{d_l},$$

with τ_i being different from $\mathfrak{p}_i$ and $\mathfrak{q}_j$. This implies that

$$(\alpha, \beta) = (\alpha) + (\beta) = \mathfrak{p}_1^{\min(a_1, b_1)} \dots \mathfrak{p}_r^{\min(a_r, b_r)} = I.$$

□

§3.6 Splitting of Primes

We now ask how we can describe all prime ideals in $\mathcal{O}_K$.

Lemma 3.6.1

Let $\phi: A \rightarrow B$ be a morphism of rings. Then, for all prime ideals $\mathfrak{p} \subset B$, then $\phi^{-1}(\mathfrak{p})$ is a prime ideal of A . Moreover, if ϕ is injective, we denote by $\mathfrak{p} \cap A = \phi^{-1}(\mathfrak{p})$.

Proof. We need to check that for all ideals $I \subseteq B$, $\phi^{-1}(I) \subseteq A$ is an ideal. If $\phi^{-1}(\mathfrak{p}) \subseteq A$ is prime, then we take $a, b \in A$ such that $a \cdot b \in \phi^{-1}(\mathfrak{p})$ implies that $\phi(ab) \in \mathfrak{p}$, implying that $\phi(a) \in \mathfrak{p}$ or $\phi(b) \in \mathfrak{p}$, since $\mathfrak{p}$ is prime. This implies that $a \in \phi^{-1}(\mathfrak{p})$ or $b \in \phi^{-1}(\mathfrak{p})$. □

Notation 3.6.2. When ϕ is injective, we write $\mathfrak{p} \cap A$ instead of $\phi^{-1}(\mathfrak{p})$. In particular, given a number field K , we always have a natural inclusion

$$\iota: \mathbb{Z} \hookrightarrow \mathcal{O}_K$$

and given $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal,

$$\mathfrak{p} \cap \mathbb{Z} = \iota^{-1}(\mathfrak{p}) \subseteq \mathbb{Z}$$

is a prime ideal of $\mathbb{Z}$.

Lemma 3.6.3

Let K be a number field and $\mathfrak{p} \subseteq \mathcal{O}_K$ be a nonzero prime ideal. Then, there exists a unique prime number $p \in \mathbb{Z}$ such that $p \in \mathfrak{p}$.

Proof. There are two ways to show this:

- (1): We have just seen that $\mathfrak{p} \cap \mathbb{Z} \subseteq \mathbb{Z}$ is a prime ideal, so it is nonzero, so we are done.

(2): Take $\alpha \in \mathfrak{p}$ such that $\alpha \neq 0$. Then,

$$\mathcal{N}_K(\alpha) = \alpha \prod_{\substack{\tau \in \Sigma_K \\ \tau \neq id}}^{\in \mathcal{O}_K} \tau(\alpha) \in \mathfrak{p} \cap \mathbb{Z}.$$

In the above, we used that $\mathcal{N}_K(\alpha) \in \mathbb{Z}$. We know that $\mathcal{O}_K/\mathfrak{p}$ is finite, so this is a field of positive characteristic. This implies that $p = 0$ in $\mathcal{O}_K/\mathfrak{p}$, implying that $p \in \mathfrak{p}$. $\square$

Definition 3.6.4. We say that a nonzero prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ **lies over** the prime $p \in \mathbb{Z}$ if $p \in \mathfrak{p}$. Equivalently, if $\mathfrak{p} \cap \mathbb{Z} = (p)$.

Proposition 3.6.5

Let $p \in \mathbb{Z}$ be a prime number and suppose

$$(p) = p \cdot \mathcal{O}_K = \mathfrak{p}_1^{a_1} \cdots \mathfrak{p}_r^{a_r},$$

with $\mathfrak{p}_1, \dots, \mathfrak{p}_r \subseteq \mathcal{O}_K$ being distinct prime ideals. Then, the primes $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are exactly the primes which lie above $p \in \mathbb{Z}$.

Proof. $\mathfrak{p}_1^{a_1} \cdots \mathfrak{p}_r^{a_r} \subseteq \mathfrak{p}_i$ for all i , so $p \in \mathfrak{p}_i$, for all i . Conversely, if $\mathfrak{p} \subseteq \mathcal{O}_K$ is such that $p \in \mathfrak{p}$, then $(p) \subseteq \mathfrak{p}$ implies that there exists $I \subseteq \mathcal{O}_K$ such that $(p) = \mathfrak{p} \cdot I$ (corollary 3.3.10). Therefore, by the uniqueness of factorization, there exists an i such that $\mathfrak{p} = \mathfrak{p}_i$. $\square$

Example 3.6.6

Let $K = \mathbb{Q}(i)$, with $\mathcal{O}_K = \mathbb{Z}[i]$. We know that $\mathbb{Z}[i]$ is a Euclidean domain. Therefore, every prime ideal $\mathfrak{p}$ is generated by one irreducible element.

If $p \equiv 3 \pmod{4}$, then p is irreducible, so (p) is prime. If $p \equiv 1 \pmod{4}$, then $p = \Pi_p \cdot \overline{\Pi}_p$, for some Π_p irreducible with $\Pi_p \neq \overline{\Pi}_p$ and

$$\mathcal{N}_{\mathbb{Q}(i)}(\Pi_p) = \mathcal{N}_{\mathbb{Q}(i)}(\overline{\Pi}_p) = p.$$

This implies that

$$(p) = (\Pi_p) \cdot (\overline{\Pi}_p)$$

with $(\Pi_p), (\overline{\Pi}_p)$ being prime ideals. Moreover, $(\Pi_p) \neq (\overline{\Pi}_p)$. If $(\Pi_p) = (\overline{\Pi}_p)$, then there exists a unit $v \in \mathbb{Z}[i]^\times$ such that $\Pi_p = v \cdot \overline{\Pi}_p$. Writing $\Pi_p = x + iy$, then since $N(\Pi_p) = x^2 + y^2 = p$, we must have that x and y have different parity.

- $u = 1$: $x + iy = x - iy$, contradiction.
- $u = -1$: $x + iy = -x + iy$, contradiction.
- $u = i$: $x + iy = ix + y$, contradiction.
- $u = -i$: $x + iy = -ix - y$, contradiction.

For $p = 2$, we have that $2 = (1 + i)(1 - i)$ and $1 + i, 1 - i$ are irreducible. Therefore,

$$(2) = (1 + i)(1 - i),$$

but $(1 + i) = (1 - i)$, since $1 + i = i(1 - i)$. Therefore, $(2) = (1 + i)^2$.

§3.6.i Ramification/ Inertial Degrees and Norm of Ideals

Definition 3.6.7. Let K be a number field and $\mathfrak{p} \subseteq \mathcal{O}_K$ be a nonzero prime ideal lying over a prime number $p \in \mathbb{Z}$. We define:

- The **ramification index**, denoted $e(\mathfrak{p})$, of p is the multiplicity with which ideal $\mathfrak{p}$ occurs in the factorization of (p) . If $e(\mathfrak{p}) > 1$, we say that p **ramifies** in $\mathcal{O}_K$.
- The quotient $\mathcal{O}_K/\mathfrak{p}$ is a finite field of characteristic $p > 0$. The **inertial degree** of $\mathfrak{p}$ as

$$f(\mathfrak{p}) = [\mathcal{O}_K/\mathfrak{p} : \mathbb{F}_p] = \dim_{\mathbb{F}_p}(\mathcal{O}_K/\mathfrak{p}).$$

If $f(\mathfrak{p}) > 1$, we say that p **inerts** in $\mathcal{O}_K$.

- If $e(\mathfrak{p}) = f(\mathfrak{p}) = 1$, we say that p **splits** in $\mathcal{O}_K$.

Remark 3.6.8 — The cardinality of $\mathcal{O}_K/\mathfrak{p}$, $[\mathcal{O}_K : \mathfrak{p}]$, is equal to $p^{f(\mathfrak{p})}$:

$$|\mathcal{O}_K/\mathfrak{p}| = p^{\dim_{\mathbb{F}_p}(\mathcal{O}_K/\mathfrak{p})} = p^{f(\mathfrak{p})}.$$

Remark 3.6.9 — Note that we use the same $[\cdot : \cdot]$ to denote the index of a subgroup

and the degree of a field extension.

Example 3.6.10

Let $K = \mathbb{Q}(\sqrt{-5})$, with $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$. We've seen that

$$(2) = \mathfrak{p}_2^2, \quad \mathfrak{p}_2 = (2, 1 + \sqrt{-5}) \implies e(\mathfrak{p}_2) = 2.$$

Further, we've seen that

$$(3) = \mathfrak{p}_2 \bar{\mathfrak{p}}_4, \quad \mathfrak{p}_3 = (2, 1 + \sqrt{-5}) \implies e(\mathfrak{p}_3) = e(\bar{\mathfrak{p}}_3) = 1.$$

Next, we have

$$(5) = (\sqrt{-5})^2, \quad \mathfrak{p}_5 = (\sqrt{-5}) \implies e(\mathfrak{p}_5) = 2.$$

And the list continues...

We now work to the following theorem.

Theorem 3.6.11

Let K be a number field, with $p \in \mathbb{Z}$ a prime number. Then,

$$\sum_{\substack{\mathfrak{p} \in \mathcal{O}_K \\ p \in \mathfrak{p}}} e(\mathfrak{p}) \cdot f(\mathfrak{p}) = \deg_{\mathbb{Q}}(K).$$

We start with an example about what this theorem is telling us.

Example 3.6.12 (Quadratic Extensions)

What is this theorem saying for $K = \mathbb{Q}(\sqrt{d})$?

In this case, we have three possibilities:

1. There is just one prime $\mathfrak{p} \subseteq \mathcal{O}_K$, lying above $p \in \mathbb{Z}$ and is such that $e(\mathfrak{p}) = 2$ and $f(\mathfrak{p}) = 1$.
2. There is just one prime $\mathfrak{p} \subseteq \mathcal{O}_K$ lying above $p \in \mathbb{Z}$ and is such that $e(\mathfrak{p}) = 1$ and $f(\mathfrak{p}) = 2$.
3. There are two distinct primes $\mathfrak{p}_1 \neq \mathfrak{p}_2$, and $e(\mathfrak{p}_1) = e(\mathfrak{p}_2)$ and $f(\mathfrak{p}_1) = f(\mathfrak{p}_2) = 1$.

To prove this, we introduce the notion of a norm of an ideal.

Definition 3.6.13. Let $I \subset \mathcal{O}_K$ be a nonzero ideal, then we define

$$N_K(I) = [\mathcal{O}_K : I] = |(\mathcal{O}_K/I)| \in \mathbb{Z}_{\geq 0}.$$

We call this the **norm** of I .

Remark 3.6.14 — Note that by lemma 3.2.2, this is always a finite integer.

Remark 3.6.15 — Some may also denote the norm of an ideal as $\|I\|$.

Remark 3.6.16 — If $\mathfrak{p} \subseteq \mathcal{O}_K$ is a prime ideal, then

$$N_K(\mathfrak{p}) = |(\mathcal{O}_K/\mathfrak{p})| = p^{f(\mathfrak{p})}.$$

Remark 3.6.17 — If $n \in \mathbb{Z}$, then

$$N_K((n)) = N_K(n\mathcal{O}_K) = |n|^{\deg_{\mathbb{Q}}(K)} = |\mathcal{N}_K(n)|.$$

If $n \in \mathbb{Z}$, then we know that (after fixing an integral basis of $\mathcal{O}_K$), then

$$\mathcal{O}_K \cong \mathbb{Z}^{\deg_{\mathbb{Q}}(K)},$$

implying that

$$\mathcal{O}_K/n\mathcal{O}_K \cong (\mathbb{Z}/n\mathbb{Z})^{\deg_{\mathbb{Q}}(K)}.$$

This implies that

$$|(\mathcal{O}_K/n\mathcal{O}_K)| = |n|^{\deg_{\mathbb{Q}}(K)}.$$

Proposition 3.6.18

If I, J are nonzero ideals of $\mathcal{O}_K$, we have that

$$N_K(I \cdot J) = N_K(I)N_K(J)$$

Proof. By unique factorization of ideals, it suffices to prove that

$$N_K(\mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r}) = N_K(\mathfrak{p}_1)^{a_1} \dots N_K(\mathfrak{p}_r)^{a_r},$$

for $\mathfrak{p}_i \subseteq \mathcal{O}_K$ prime ideals.

By the Chinese remainder theorem, we have that

$$\mathcal{O}_K/\mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r} \xrightarrow{\sim} \mathcal{O}_K/\mathfrak{p}_1^{a_1} \times \dots \times \mathcal{O}_K/\mathfrak{p}_r^{a_r}.$$

This implies that the cardinalities are the same:

$$|(\mathcal{O}_K/\mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r})| = |(\mathcal{O}_K/\mathfrak{p}_1^{a_1})| \cdot \dots \cdot |(\mathcal{O}_K/\mathfrak{p}_r^{a_r})|.$$

That is, the norm holds:

$$N_K(\mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r}) = N_K(\mathfrak{p}_1)^{a_1} \dots N_K(\mathfrak{p}_r)^{a_r}.$$

We now need to show that $N_K(\mathfrak{p}^a) = N_K(\mathfrak{p})^a$, for each i . We prove this by induction on a . If $a = 1$, this is true. If otherwise, observe that we have $\mathfrak{p}^{a+1} \subseteq \mathfrak{p}^a \subseteq \mathcal{O}_K$. Then, we have by Lagrange's theorem that

$$N_K(\mathfrak{p}^{a+1}) = [\mathcal{O}_K : \mathfrak{p}^{a+1}] = [\mathcal{O}_K : \mathfrak{p}^a][\mathfrak{p}^a : \mathfrak{p}^{a+1}].$$

Note that by induction, we know that

$$[\mathcal{O}_K : \mathfrak{p}^a] = N_K(\mathfrak{p})^a.$$

Thus, we must show that $[\mathfrak{p}^a : \mathfrak{p}^{a+1}] = N_K(\mathfrak{p})$. We want to build an isomorphism of abelian groups

$$\phi: \mathcal{O}_K/\mathfrak{p} \rightarrow \mathfrak{p}^a/\mathfrak{p}^{a+1}.$$

We know that $\mathfrak{p}^{a+1} \subsetneq \mathfrak{p}^a$ (if they are equal, then $\mathfrak{p} = \mathcal{O}_K$, a contradiction). We take $b \in \mathfrak{p}^a \setminus \mathfrak{p}^{a+1}$ and define our isomorphism $\phi: \mathcal{O}_K/\mathfrak{p} \rightarrow \mathfrak{p}^a/\mathfrak{p}^{a+1}$ by

$$y + \mathfrak{p} \mapsto yb + \mathfrak{p}^{a+1}.$$

Firstly, note that this map is *well defined*: if $y' = y + x$, with $x \in \mathfrak{p}$, then

$$y' + \mathfrak{p} \mapsto y'b + \mathfrak{p}^{a+1} = yb + xb + \mathfrak{p}^{a+1} = yb + \mathfrak{p}^{a+1}.$$

Now we show that this map is *injective*. $\mathfrak{p}$ is prime, so it is maximal. Then, for all $x \notin \mathfrak{p}$, we have that $\mathfrak{p} + (x) = \mathcal{O}_K$. This implies that $1 = y + xx'$, for some $y \in \mathfrak{p}$, $x' \in \mathcal{O}_K$. If

$$x + \mathfrak{p} \xrightarrow{\phi} 0,$$

then $xb \in \mathfrak{p}^{a+1}$. However, then

$$b = b(y + xx') = \underbrace{by}_{\in \mathfrak{p}^{a+1}} + \underbrace{bxx'}_{\in \mathfrak{p}^{a+1}} \in \mathfrak{p}^{a+1}.$$

However, this implies that $b \in \mathfrak{p}^{a+1}$, which is a contradiction. Next, we show *surjectivity*. Note that $(b) = \mathfrak{p}^a I$, with I being coprime to $\mathfrak{p}$, since $b \notin \mathfrak{p}^{a+1}$. Then, using the formulae in proposition 3.5.1,

$$(b) + \mathfrak{p}^{a+1} = \mathfrak{p}^a.$$

If $x \in \mathfrak{p}^a$, then we can use the above to write $x = yb + y'$, with $y \in \mathcal{O}_K$ and $y' \in \mathfrak{p}^{a+1}$. then,

$$\phi(y + \mathfrak{p}) = yb + \mathfrak{p}^{a+1} = x + \mathfrak{p}^{a+1}.$$

Thus, surjective. Therefore, we're done. $\square$

Remark 3.6.19 — The key from this proof is that we have an isomorphism

$$\mathcal{O}_K/\mathfrak{p} \xrightarrow{\sim} \mathfrak{p}^a/\mathfrak{p}^{a+1}.$$

Remark 3.6.20 — If $I \subseteq \mathcal{O}_K$ is an ideal and if $\mathcal{N}_K(I) = p \in \mathbb{Z}$ is prime number, then I is a prime ideal.

Proof. If $I = \mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r}$, then

$$N_K(I) = N_K(\mathfrak{p}_1)^{a_1} \dots N_K(\mathfrak{p}_r)^{a_r} = p.$$

Since each of the terms are positive integers, it follows that, up to reordering, $\mathcal{N}_K(\mathfrak{p}_1) = p$, $a_1 = 1$, and $a_2 = \dots = a_r = 0$. $\square$

Remark 3.6.21 — If $I \subseteq \mathcal{O}_K$ is an ideal and if $\mathcal{N}_K(I) = 1$, then $I = \mathcal{O}_K$.

Proof. The proof follows similarly to the above. Let $I = \mathfrak{p}_1^{a_1} \dots \mathfrak{p}_r^{a_r}$. Then,

$$N_K(I) = N_K(\mathfrak{p}_1)^{a_1} \dots N_K(\mathfrak{p}_r)^{a_r} = 1,$$

and it follows that we must have $a_1 = \dots = a_r = 0$, so $I = (1) = \mathcal{O}_K$. $\square$

Proof of Theorem 3.6.11. We have that $p \cdot \mathcal{O}_K = \mathfrak{p}_1^{e(\mathfrak{p}_1)} \dots \mathfrak{p}_r^{e(\mathfrak{p}_r)}$. Then,

$$p^{\deg_{\mathbb{Q}}(K)} = N_K(p \cdot \mathcal{O}_K) = N_K(\mathfrak{p}_1)^{e(\mathfrak{p}_1)} \dots N_K(\mathfrak{p}_r)^{e(\mathfrak{p}_r)},$$

by proposition 3.6.18. Then, recall that

$$N_K(\mathfrak{p}_i) = |(\mathcal{O}_K/\mathfrak{p}_i)| = p^{f(\mathfrak{p}_i)}.$$

Therefore,

$$p^{\deg_{\mathbb{Q}}(K)} = p^{f(\mathfrak{p}_1)e(\mathfrak{p}_1)} \dots p^{f(\mathfrak{p}_r)e(\mathfrak{p}_r)} = p^{f(\mathfrak{p}_1)e(\mathfrak{p}_1) + \dots + f(\mathfrak{p}_r)e(\mathfrak{p}_r)}.$$

This implies that

$$\dim_{\mathbb{Q}}(K) = f(\mathfrak{p}_1)e(\mathfrak{p}_1) + \dots + f(\mathfrak{p}_r)e(\mathfrak{p}_r).$$

$\square$

Proposition 3.6.22

Let $0 \neq \alpha \in \mathcal{O}_K$. Then,

$$N_K(\alpha \mathcal{O}_K) = |\mathcal{N}_K(\alpha)| = \prod_{\tau \in \Sigma_K} \tau(\alpha).$$

Proof. We have that $\mathcal{N}_K(\alpha) = \det(A)$, where A is the matrix associated to $\text{mult}_{\alpha}: \mathcal{O}_K \rightarrow \mathcal{O}_K$. In particular, if we fix a $\mathbb{Z}$ -basis of $\mathcal{O}_K$, then $\text{mult}_{\alpha}: \mathbb{Z}^n \rightarrow \mathbb{Z}^n$. It's left to show that

$$|(\mathcal{O}_K/\alpha \mathcal{O}_K)| = |(\mathbb{Z}^n/A\mathbb{Z}^n)| = |\det(A)|.$$

This last equality can be justified by the strong form of the structure theorem (or even easier, by the Smith normal form). $\square$

Remark 3.6.23 — If $\alpha \in \mathcal{O}_K$ and $\mathcal{N}_K(\alpha) = \pm p$, then $N_K(\alpha \mathcal{O}_K) = p$ implies that $\alpha \mathcal{O}_K$ is a prime ideal, so α is a prime element. Note that we've already seen that if $\mathcal{N}_K(\alpha) = \pm p$, then α is irreducible, but we now have something stronger. (prime elements are generally irreducible)

Remark 3.6.24 — Let $I \subset \mathcal{O}_K$ be a nonzero ideal, with $a \in I$. Then, $(a) \subseteq I$, and we then have:

$$N_K(I) = |\mathcal{O}_K/I| \subseteq |\mathcal{O}_K/(a)| = N_K(a) = \mathcal{N}_K(a).$$

In particular, $\mathcal{O}_K/I$ is a quotient of $\mathcal{O}_K/(a)$, so the $N_K(I)$ divides $\mathcal{N}_K(a)$, for all $a \in I$.

§3.6.ii Quadratic Integer Rings

Theorem 3.6.25

Let $d \neq 0, 1$ be a square free integer and let $p \in \mathbb{Z}$ be an odd prime. Then, $p \cdot \mathcal{O}_K$ splits in $\mathcal{O}_K$ (the ring of integers of $\mathbb{Q}(\sqrt{d})$) as follows:

1. If $p \mid d$, then p ramifies as $p\mathcal{O}_K = (p, \sqrt{d})^2$.
2. If $p \nmid d$ and $\exists a \in \mathbb{Z}$ such that $a^2 \equiv d \pmod{p}$, then

$$p \cdot \mathcal{O}_K = (p, a + \sqrt{d})(p, a - \sqrt{d}).$$

Here, p splits in $\mathcal{O}_K$

3. If $p \nmid d$ and $\nexists a \in \mathbb{Z}$ such that $a^2 \equiv d \pmod{p}$, then $p\mathcal{O}_K$ is a prime ideal. Here, p is inert in $\mathcal{O}_K$.

Proof. (1): If $p \mid d$, then

$$(p, \sqrt{d})^2 = (p^2, p\sqrt{d}, d) \subseteq (p).$$

The ideal $(p^2, p\sqrt{d}, d)$ contains p^2 and d , and therefore contains the $\gcd(p^2, d)$. This is actually equal to p , so we have equality. Therefore, $(p) = (p, \sqrt{d})^2$. Moreover,

$$N_k(\mathfrak{p})^2 = N_k(\mathfrak{p}^2) = N_K(p\mathcal{O}_K) = |p|^{\dim_{\mathbb{Q}}(\mathbb{Q}(\sqrt{d}))} = p^2.$$

Thus, $N_k(\mathfrak{p}) = p$, so $\mathfrak{p}$ is a prime ideal.

- (2): If $p \nmid d$ and $a \in \mathbb{Z}$ such that $a^2 \equiv d \pmod{p}$, then

$$(p, a + \sqrt{d})(p, a - \sqrt{d}) = \underbrace{(p^2, p(a + \sqrt{d}), p(a - \sqrt{d}), a^2 - d)}_{=: I} \subseteq p\mathcal{O}_K.$$

Note that $2pa \in I$ and $\gcd(2pa, a^2) = p$, so we have equality. □

Definition 3.6.26. Let p be an odd prime. We say that $m \in \mathbb{Z}$ is a **quadratic residue** modulo p if there exists $a \in \mathbb{Z}$ such that $a^2 \equiv m \pmod{p}$.

Definition 3.6.27. The **Legendre Symbol** is defined as

$$\left(\frac{m}{p}\right) = \begin{cases} 1 & \text{if } m \text{ is a quadratic residue } \pmod{p} \\ -1 & \text{if } m \text{ is a quadratic non-residue } \pmod{p} \\ 0 & \text{if } p \mid m \end{cases}$$

We now restate theorem 3.6.25 using this notation and including $p = 2$:

Theorem 3.6.28

Let $d \neq 0, 1$ be a square free integer and let p be an odd prime number. Then (p) factors in $\mathcal{O}_d$ in the following way:

1. Then $\left(\frac{d}{p}\right) = 1$, p splits as $p \cdot (\mathcal{O}_K) = (p, a + \sqrt{d})(p, a - \sqrt{d})$, where $a \in \mathbb{Z}$ such that $a^2 \equiv d \pmod{p}$.
2. If $\left(\frac{d}{p}\right) = -1$, then p is inert.
3. If $\left(\frac{d}{p}\right) = 0$, then p ramifies in $\mathcal{O}_d$ as $p\mathcal{O}_d = (p, \sqrt{d})^2$

Moreover, if $p = 2$, then

1. If $d \not\equiv 1 \pmod{4}$, then 2 ramifies in $\mathcal{O}$. Then, precisely,
 - a) If $2 \mid d$, then $(2) = (2, \sqrt{d})^2$
 - b) If $2 \nmid d$, then $d \equiv 3 \pmod{4}$ and $(2) = (2, \sqrt{d} + 1)^2$.
2. If $d \equiv 1 \pmod{4}$, then we further have:
 - a) If $d \equiv 1 \pmod{8}$, then 2 splits as $2\mathcal{O}_d = (2, \frac{1+\sqrt{d}}{2}) \cdot (2, \frac{1-\sqrt{d}}{2})$.
 - b) If $d \equiv 5 \pmod{8}$, then 2 is inert in $\mathcal{O}_d$.

Proof. We just check the last three. We can first check that all the claimed factorizations are correct. By considering norms we see that all the ideals occurring in the factorization have norm 2 and thus are prime. We should also check that the two ideals in the case $d \equiv 1 \pmod{8}$ are actually different: If they were equal they would contain $\frac{1+\sqrt{d}}{2} + \frac{1-\sqrt{d}}{2} = 1$. Finally we should check that when $d \equiv 5 \pmod{8}$ then (2) is prime. For this, note that $\frac{1+\sqrt{d}}{2}$ satisfies the polynomial $x^2 - x + \frac{1-d}{4}$, whose reduction modulo 2 is $x^2 + x + 1 \in \mathbb{F}_2[x]$. No element of $\mathbb{F}_2$ satisfies this polynomial. On the other hand, if (2) were not prime, there would be a prime ideal $(2) \subset \mathfrak{p} \subset \mathcal{O}_d$ with $N_K(\mathfrak{p})$ and hence $\mathcal{O}_d/\mathfrak{p} \cong \mathbb{F}_2$. But then $\frac{1+\sqrt{d}}{2} + \mathfrak{p}$ would give an element of $\mathbb{F}_2$ satisfying the polynomial $x^2 + x + 1$, which is impossible. $\square$

We now recall the discriminant. Recall that the discriminant of $\mathbb{Q}(\sqrt{d})$ is given by

$$D_{\mathbb{Q}(\sqrt{d})} = \begin{cases} 4d & d \not\equiv 1 \pmod{4} \\ d & d \equiv 1 \pmod{4} \end{cases}.$$

This gives the following corollary.

Corollary 3.6.29

A prime p ramifies in $\mathcal{O}_d$ if and only if $p \mid D_{\mathbb{Q}(\sqrt{d})}$

Proof. Theorem 3.6.25 says that an odd prime ramifies if and only if it divides d , while theorem 3.6.28 ramifies if and only if $d \not\equiv 1 \pmod{4}$. $\square$

Definition 3.6.30. More generally, we say that a prime number $p \in \mathbb{Z}$ **ramifies** in a number field K if, in the factorization of $p \cdot \mathcal{O}_K$, there is at least one repeated factor, i.e.,

$$p\mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

with $\mathfrak{p}_i$ distinct, there exists i such that $e_i > 1$.

§3.6.iii Ramification of Primes

The aim of this work is to prove the following theorem for $K = \mathbb{Q}(\alpha)$, where K is not necessarily a quadratic extension of $\mathbb{Q}$.

Theorem 3.6.31

Let K be a number field. Then, a prime number $p \in \mathbb{Z}$ ramifies in $\mathcal{O}_K$ if and only if $p \mid D_K$.

Lemma 3.6.32

Let $\alpha \in \mathbb{C}$ be a nonzero algebraic integer and let $m_\alpha(x) \in \mathbb{Z}[x]$ be the minimal polynomial of α . Then, there is an isomorphism

$$\mathbb{Z}[\alpha] \xrightarrow{\sim} \mathbb{Z}[x]/(m_\alpha(x)).$$

Proof. Consider the morphism

$$\begin{aligned} \phi: \mathbb{Z}[x] &\rightarrow \mathbb{Z}[\alpha] \\ x &\mapsto \alpha. \end{aligned}$$

This is surjective by definition. We want to show that $\ker(\phi) = (m_\alpha(x))$. Clearly, $(m_\alpha) \subseteq \ker(\phi)$. If $f(x) \in \mathbb{Z}[x]$ is such that $\phi(f) = f(\alpha) = 0$, then we may write

$$f(x) = g(x) \cdot m_\alpha(x),$$

for some $g(x) \in \mathbb{Q}[x]$. If $\deg(f) < \deg(m_\alpha)$, then $f = 0$, so $f \in (m_\alpha)$. If $\deg(f) \geq \deg(m_\alpha)$, then writing

$$f(x) = a_n x^n + \dots + a_0 = a_n x^{n-\deg(m_\alpha)}(m_\alpha(x)) + h(x),$$

with $\deg(h) < \deg(f)$. Moreover, $f(\alpha) = 0$, so as $m_\alpha(\alpha) = 0$, then $h(\alpha) = 0$. $\square$

Remark 3.6.33 — Let $I \subseteq \mathbb{Z}[\alpha] = \{f(\alpha) \mid f \in \mathbb{Z}[x]\}$. Then, we may write $I = (f_1(\alpha), \dots, f_n(\alpha))$. Then, we have the isomorphism

$$\mathbb{Z}[\alpha]/I \xrightarrow{\sim} (\mathbb{Z}[x]/(m_\alpha))/[\phi^{-1}(I)] \cong \mathbb{Z}[x]/(m_\alpha, f_1, \dots, f_n),$$

where $\phi: \mathbb{Z}[x]/(m_\alpha) \rightarrow \mathbb{Z}[\alpha]$ is defined by $[(f_1(x), \dots, f_n(x))] \mapsto I$. The last isomorphism is by the third isomorphism theorem.

Dedekind's Factorization Criteria

Theorem 3.6.34 (Dedekind's Factorization Criteria)

Let $K = \mathbb{Q}(\alpha)$, for $\alpha \in \mathbb{C}$ a nonzero algebraic integer with minimal polynomial $m_\alpha(x)$. Let p be a prime number such that $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then, if $\overline{m_\alpha}(x) = \overline{g_1}^{e_1} \cdots \overline{g_r}^{e_r}$ is the factorization of m_α modulo p , with g_i distinct monic irreducible polynomials, then

$$p \cdot \mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

where $\mathfrak{p}_i = (p, g_i(\alpha))$ with $g_i(x)$ any lift to $\mathbb{Z}[x]$ of the polynomial $\overline{g_i}(x) \in \mathbb{F}_p[x]$. Moreover, the $\mathfrak{p}_i$ are distinct prime ideals and $f(\mathfrak{p}_i) = \deg(\overline{g_i})$.

Here is a preliminary lemma:

Lemma 3.6.35

Suppose $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then, the map

$$\mathbb{Z}[\alpha]/p\mathbb{Z}[\alpha] \rightarrow \mathcal{O}_K/p\mathcal{O}_K$$

induced by the inclusion $\mathbb{Z}[\alpha] \subseteq \mathcal{O}_K$ is an isomorphism.

Proof. We first show surjectivity. By Lagrange's theorem,

$$[\mathcal{O}_K : \mathbb{Z}[\alpha]] \cdot \mathcal{O}_K \subseteq \mathbb{Z}[\alpha].$$

Moreover, $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Thus, we can use Bezout's theorem to say that there exists $a, b \in \mathbb{Z}$ such that

$$p \cdot b + [\mathcal{O}_K : \mathbb{Z}[\alpha]] \cdot a = 1.$$

Now take $x \in \mathcal{O}_K$. Then,

$$\begin{aligned} x + p\mathcal{O}_K &= (p \cdot b + [\mathcal{O}_K : \mathbb{Z}[\alpha]] \cdot a)x + p\mathcal{O}_K \\ &= \underbrace{a \cdot [\mathcal{O}_K : \mathbb{Z}[\alpha]] \cdot x}_{\in \mathbb{Z}[\alpha]} + p\mathcal{O}_K \\ &= \underbrace{a'}_{\in \mathbb{Z}[\alpha]} + p\mathcal{O}_K. \end{aligned}$$

Thus, the map is surjective, as $a' + p\mathbb{Z}[\alpha] \mapsto a' + p\mathcal{O}_K$. We now show injectivity. Let $y' \in \mathcal{O}_K$ such that $py' \in \mathbb{Z}[\alpha]$. As $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$, we can find a, b such that

$$1 = a \cdot p + b \cdot [\mathcal{O}_K : \mathbb{Z}[\alpha]].$$

Then,

$$y' = \underbrace{p \cdot a \cdot y'}_{\in \mathbb{Z}[\alpha]} + \underbrace{[\mathcal{O}_K : \mathbb{Z}[\alpha]] \cdot b \cdot y'}_{\in \mathbb{Z}[\alpha]},$$

and so $y' \in \mathbb{Z}[\alpha]$. Thus, we've shown injectivity. $\square$

Example 3.6.36

Let $d \equiv 1 \pmod{4}$. Then, $\mathbb{Z}[\sqrt{d}] \subsetneq \mathcal{O}_d = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$. We have that

$$[\mathcal{O}_d : \mathbb{Z}[\sqrt{d}]] = 2,$$

so for $p \neq 2$, we have

$$\mathbb{Z}[\sqrt{d}]/p\mathbb{Z}[\sqrt{d}] \cong \mathcal{O}_d/p\mathcal{O}_d.$$

Note that if $p \neq 2$, then 2 is invertible in $\mathbb{Z}[\sqrt{d}]/p\mathbb{Z}[\sqrt{d}]$. This implies that $\sqrt{d} + p\mathbb{Z}[\sqrt{d}] \in \mathbb{Z}[\sqrt{d}]/p\mathbb{Z}[\sqrt{d}]$. Therefore,

$$\frac{1 + \sqrt{d}}{2} \in \mathbb{Z}[\sqrt{d}]/p\mathbb{Z}[\sqrt{d}],$$

so

$$\mathbb{Z}[\frac{1 + \sqrt{d}}{2}]/p\mathbb{Z}[\frac{1 + \sqrt{d}}{2}] \subseteq \mathbb{Z}[\sqrt{d}]/p\mathbb{Z}[\sqrt{d}].$$

Proof of Dedekind's Factorization Criteria (3.6.34). We know that $\mathbb{Z}[\alpha] \cong \mathbb{Z}[x]/(m_\alpha(x))$. Combining this with the previous lemma, we get

$$\mathcal{O}_K/p\mathcal{O}_k \cong \mathbb{Z}[\alpha]/p\mathbb{Z}[\alpha] \cong \mathbb{Z}[x]/(p, m_\alpha(x)) \cong \mathbb{F}_p[x]/(\overline{m_\alpha}(x)).$$

Claim 3.6.37. Let $\overline{m_\alpha}(x) = \overline{g_1}^{e_1}(x) \dots \overline{g_r}^{e_r}(x)$, with $\overline{g_i}$ being monic, irreducible, and distinct. Then, $\mathfrak{p}_i = (p, g_i(\alpha))$ is a prime ideal.

Proof. We have that

$$\mathcal{O}_K/\mathfrak{p}_i = \mathcal{O}_k/(p, g_i(\alpha)) \cong \mathbb{Z}[x]/(m_\alpha, p, g_i(\alpha)) \cong \mathbb{F}_p[x]/(\overline{m_\alpha}(x), \overline{g_i}(x)) \cong \mathbb{F}_p[x]/(\overline{g_i}(x)).$$

The last is a field, since $\overline{g_i}(x)$ is irreducible. Moreover, note that the

$$|p^{f(\mathfrak{p})} = \mathcal{O}_K/\mathfrak{p}_i| = |\mathbb{F}_p[x]/\overline{g_i}(x)| = p^{\deg(g_i)}.$$

□

Claim 3.6.38. $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are pairwise distinct.

Proof. For $i \neq j$, we have that

$$\begin{aligned} \mathcal{O}_K/(\mathfrak{p}_i + \mathfrak{p}_j) &= \mathcal{O}_K/(p, \overline{g_i}(\alpha), g_j(\alpha)) \\ &\cong \mathbb{Z}[x]/(p, m_\alpha(x), g_i(x), g_j(x)) \\ &\cong \mathbb{F}_p[x]/(\overline{g_i}(x), \overline{g_j}(x)) \\ &= \{0\}, \end{aligned}$$

since $(\overline{g_i}, \overline{g_j}) = \mathbb{F}_p[x]$ as g_i, g_j are coprime. Hence $\mathfrak{p}_i + \mathfrak{p}_j = \mathcal{O}_K$, and hence $\mathfrak{p}_i \neq \mathfrak{p}_j$. □

Claim 3.6.39. We have $p\mathcal{O}_K = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_r^{e_r}$.

Proof. We observe that

$$\mathfrak{p}_1^{e_1} \dots \mathfrak{p}_r^{e_r} \subseteq (p, g_1^{e_1}(\alpha), \dots, g_r^{e_r}(\alpha)) \subseteq p\mathcal{O}_K.$$

Now, $g_1^{e_1} \dots g_r^{e_r} \equiv m_\alpha(x) \pmod{p}$. Hence,

$$g_1^{e_1} \dots g_r^{e_r} = m_\alpha(\alpha) + \underbrace{(g_1^{e_1} \dots g_r^{e_r} - m_\alpha)}_{p\mathcal{O}_K} \in p\mathcal{O}_K,$$

since $m_\alpha(\alpha) = 0$. Taking norms, we have

$$N_K(\mathfrak{p}_1^{e_1} \dots \mathfrak{p}_r^{e_r}) = p^{\deg(m_\alpha)} = p^{[K:\mathbb{Q}(\alpha)]} = N_K(p\mathcal{O}_K).$$

Therefore, the inclusion $\mathfrak{p}_1^{e_1} \dots \mathfrak{p}_r^{e_r} \subseteq p\mathcal{O}_K$ is an equality. $\square$

This concludes the proof. $\square$

Remark 3.6.40 — $\mathfrak{p}_i = (p, g_i(\alpha))$, where g_i is any lift. We show here that the choice doesn't matter. Note that if $g_i(x), \tilde{g}_i(x)$ are two different lifts of $\overline{g_i}(x) \in \mathbb{F}_p[x]$, then

$$g_i(x) - \tilde{g}_i(x) \in (p)$$

Therefore, $(p, g_i(\alpha)) = (p, \tilde{g}_i(\alpha))$.

Example 3.6.41

Let $d \not\equiv 1 \pmod{4}$, and $\mathcal{O}_d = \mathbb{Z}[\sqrt{d}]$. $\alpha = \sqrt{d}$ in this case, so $m_\alpha = x^2 - d$. In particular,

$$\mathcal{O}_d/p\mathcal{O}_d \cong \mathbb{F}_p[x]/(x^2 - \bar{d}).$$

We have $\overline{m_\alpha} = x^2 - \bar{d} \in \mathbb{F}_p[x]$. We note a few cases:

1. We have $\overline{m_\alpha} = (x - a)(x - b)$, $a, b \in \mathbb{F}_p[x]$ and $a \neq b$.

In this case,

$$\mathcal{O}_d/p\mathcal{O}_d \cong \mathbb{F}_p[x]/(x^2 - \bar{d}) \cong \mathbb{F}_p[x]/(x - a)(x - b) \cong \mathbb{F}_p[x]/(x - a) \times \mathbb{F}_p[x]/(x - b).$$

We note $\overline{m_\alpha} = (x - a)(x - b)$ if and only if $p \nmid d$ and there exists $\tilde{a} \in \mathbb{Z}$ such that $\tilde{a} \equiv a \pmod{p}$. Then, $\tilde{a}^2 \equiv a^2 \equiv d \pmod{p}$, so $\left(\frac{d}{p}\right) = 1$. In this case, we know that $p\mathcal{O}_d = \mathfrak{p}_1 \cdot \mathfrak{p}_2$ (with $\mathfrak{p}_1 \neq \mathfrak{p}_2$) and

$$\mathfrak{p}_1 = (p, a - \sqrt{d}) = (p, g_1(\sqrt{d}))$$

$$\mathfrak{p}_2 = (p, a + \sqrt{d}) = (p, g_2(\sqrt{d}))$$

2. We can also have $\overline{m_\alpha} = (x - a)^2$ for some $a \in \mathbb{F}_p$. If $p \neq 2$, then $x^2 - \bar{d} = (x - a)^2$, implying that $2a = 0$, so $a = 0$. If $p = 2$, we get this factorization is possible if $a = 0$ or $a = 1$. Note that if $p \neq 2$, then we are in the first case, which occurs if and only if $a = 0$, which occurs if and only if $d \equiv 0 \pmod{p}$, implying that $p \mid d$. We know that $p\mathcal{O}_K = \mathfrak{p}^2$, where $\mathfrak{p} = (p, \sqrt{d})$, but $\overline{m_\alpha} = x^2 - \bar{d} = x^2 = (g_1(x))^2$, so $\mathfrak{p} = (p, g_1(\sqrt{d}))$.

Example 3.6.42

Let $K = \mathbb{Q}(\alpha)$, where $\alpha = \sqrt[3]{2}$ with minimal polynomial $m_\alpha = x^3 - 2$. We computed that

$$\text{disc}(\mathbb{Z}[\alpha]) = -27 \cdot 4.$$

We can apply Dedekind's factorization criterion for $p \neq 2, 3$. $\text{disc}(\mathbb{Z}[\alpha]) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \cdot D_K$, so only 2 or 3 may divide $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Try $p = 5$: We have

$$x^3 - 2 = (x - 3)(x^2 + 3x - 1),$$

implying that $5 \cdot \mathcal{O}_K = (5, \sqrt[3]{2} - 3) \cdot (5, \sqrt[3]{4} + 3\sqrt[3]{2} - 1)$. For $p = 7$, we have that $x^3 - 2$ is irreducible.

Back to the Main Theorem

Definition 3.6.43. We say that the prime p **ramifies** in K if in the unique factorization of (p) into prime ideals of $\mathcal{O}_K$ there is a repeated factor. Equivalently there exists some prime $\mathfrak{p}$ lying above p for which the ramification index $e(\mathfrak{p}) > 1$.

We now go back to the study of the following theorem:

Theorem 3.6.44

Let K be a number field. Then, a prime number $p \in \mathbb{Z}$ ramifies in $\mathcal{O}_K$ if and only if $p \mid D_K$.

Remark 3.6.45 — An immediate consequence of the theorem is that only finitely many primes ramify.

We are going to show today that:

1. If $K = \mathbb{Q}(\alpha)$, then just finitely many primes ramify.
2. If a prime p ramifies, then $p \mid D_K$.

Proposition 3.6.46

Let $f \in \mathbb{Z}[x]$ be an irreducible monic polynomial. Then, for all but finitely many primes p , $\bar{f} \in \mathbb{F}_p[x]$ has no repeated irreducible factors.

Proof. Let F be any field and $h(x) \in F[x]$. Then, if there exists some polynomial $g \in F[x]$ such that $g^2 \mid h$, then $g \mid h'$. This follows by the Leibnitz rule. As a consequence, we see that if $(h) + (h') = F[x]$, then h cannot have any repeated factors. Now since $f \in \mathbb{Z}[x]$ is irreducible, we know that $(f) + (f') = \mathbb{Q}[x]$. Therefore, there exists $g_1, g_2 \in \mathbb{Q}[x]$ such that $g_1 f + g_2 f' = 1$. Now, there exists $d \in \mathbb{Z}$ such that

$$d \cdot g_1 f + d \cdot g_2 f' = d.$$

For all $p \nmid d$, we have that

$$\overline{d g_1} \cdot \bar{f} + \overline{d g_2} \cdot \bar{f}' = \bar{d} \pmod{p},$$

and so $(\bar{f}) + (\bar{f}') = \mathbb{F}_p[x]$. □

Remark 3.6.47 — We can take d to be the discriminant of the polynomial f .

Remark 3.6.48 — If $\mathcal{O}_K = \mathbb{Z}[\alpha]$, if $p \mid D_K = \text{disc}(m_\alpha)$, then

$$(\overline{D_K}) = (\overline{m_\alpha}) + (\overline{m'_\alpha}) = 0 \pmod{p} \implies 0 = \overline{m_\alpha} \pmod{p}$$

will have a repeated factor.

Lemma 3.6.49

If $K = \mathbb{Q}(\alpha)$, then only finitely many primes ramify in $\mathcal{O}_K$.

Proof. We know that $\mathbb{Z}[\alpha] \subseteq \mathcal{O}_K$ has finite index ($\mathbb{Z}[\alpha], \mathcal{O}_K$ have the same rank). Now by Dedekind's factorization criteria (theorem 3.6.34), we know that for primes $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ ramify if and only if $\overline{m_\alpha} \pmod{p}$ has a repeated factor. This then follows by proposition 3.6.46. $\square$

Lemma 3.6.50

Let K be a number field. Then, if a prime number $p \in \mathbb{Z}$ ramifies in $\mathcal{O}_K$, then $p \mid D_K$.

Proof Sketch. p ramifies, so

$$p \cdot \mathcal{O}_K = \mathfrak{p}^2 \cdot \mathfrak{p}_1 \dots \mathfrak{p}_r,$$

where each $\mathfrak{p}_i$ are not necessarily distinct (and $\mathfrak{p}$ may appear again). Now take $x \in \mathfrak{p}\mathfrak{p}_1 \dots \mathfrak{p}_r$ but $x \notin p\mathcal{O}_K$. Then, $[x]$ in $\mathcal{O}_K/p\mathcal{O}_K$ is nilpotent. In fact,

$$[x]^2 = [x^2] \in \mathfrak{p}^2 \mathfrak{p}_1^2 \dots \mathfrak{p}_r^2 \subseteq p \cdot \mathcal{O}_K.$$

Take an integral basis $\{\alpha_1, \dots, \alpha_n\}$ of $\mathcal{O}_K$ with α_n such that $[\alpha_n] \in \mathcal{O}_K/p$ is nilpotent. We note that this always exists (check!). Then, $\alpha_1 \cdot \alpha_i$ is nilpotent, and so $p \mid \text{Tr}(\alpha_i \cdot \alpha_j)$, as the trace of a nilpotent matrix is zero and $(\overline{\alpha_j \alpha_j}) \pmod{p}$ will correspond to a nilpotent matrix. Then, $p \mid D_K = \det(\text{Tr}(\alpha_i \alpha_j))$. $\square$

§3.7 Cyclotomic Fields

Let p be an odd prime. We consider $\mathbb{Q}(\xi_p)$, where ξ_p is a primitive p th root of unity. We can take $\xi_p = e^{\frac{2\pi i}{p}}$. The minimal polynomial of ξ_p is

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = 1 + x + \dots + x^{p-1}.$$

Definition 3.7.1. Given a positive integer n , the n th cyclotomic polynomial is defined as the irreducible polynomial with integer coefficients that is a divisor of $x^n - 1$ and is not a divisor of $x^k - 1$ for any $k < n$.

Remark 3.7.2 — This may also be defined as the minimal polynomial of any primitive n root of unity over $\mathbb{Q}$.

Definition 3.7.3. A monic polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in \mathbb{Z}[x]$ is called **Eisenstein** at p if $p \mid a_i$ for all $i = 0, \dots, n-1$ and $p^2 \nmid a_0$.

Proposition 3.7.4 (Eisenstein's Criterion)

Eisenstein polynomials are irreducible.

Example 3.7.5

Take $\Phi_p(x+1)$. Then,

$$\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + px^{p-2} + \dots + p.$$

By Eisenstein's criterion, $\Phi_p(X+1)$ is irreducible, implying that $\Phi_p(x)$ is irreducible.

Proposition 3.7.6

Let $K = \mathbb{Q}(\alpha)$, where α is an algebraic integer and suppose that the minimal polynomial m_α of α is Eisenstein at p . Then, $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ and $p\mathcal{O}_K = (p, \alpha)^{[K:\mathbb{Q}]}$.

Proof. Write $m_\alpha(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$. If $p \mid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ then there exists $y \in \mathcal{O}_K$ such that $y \notin \mathbb{Z}[\alpha]$ but $p \cdot y \in \mathbb{Z}[\alpha]$ (just take an element of order p in $\mathcal{O}_K/\mathbb{Z}[\alpha]$). Then,

$$\begin{aligned} py &= c_0 + c_1\alpha + \dots + c_{n-1}\alpha^{n-1} \\ y &= \frac{c_0}{p} + \frac{c_1}{p}\alpha + \dots + \frac{c_{n-1}}{p}\alpha^{n-1}. \end{aligned}$$

Let $i \geq 0$ be the smallest number such that $c_i \notin p \cdot \mathbb{Z}$. Then,

$$\frac{c_i}{p}\alpha^i + \dots + \frac{c_{n-1}}{p}\alpha^{n-1} \in \mathcal{O}_K.$$

Now multiply by α^{n-1-i} and we get

$$\frac{c_i}{p}\alpha^{n-1} + \frac{\alpha^n}{p}(c_{i+1} + \dots + c_{n-1}\alpha^{n-i-2}) \in \mathcal{O}_K.$$

But, $\alpha^n = -(a_{n-1}\alpha^{n-1} + \dots + a_0) \in p\mathcal{O}_K$, since m_α is Eisenstein at p (all the coefficients are divisible in p). This implies that $\frac{\alpha^n}{p} \in \mathcal{O}_K$, implying that $\frac{c_i}{p}\alpha^{n-1} \in \mathcal{O}_K$. Now if we take the norm,

$$\begin{aligned} \mathcal{N}_K\left(\frac{c_i}{p}\alpha^{n-1}\right) &= \left(\frac{c_i}{p}\right)^n \mathcal{N}_K(\alpha)^{n-1} \\ &= \left(\frac{c_i}{p}\right)^n ((-1)^n a_0)^{n-1} \\ &= \pm \left(\frac{c_i}{p}\right)^n \cdot a_0^{n-1}. \end{aligned}$$

Since $p^2 \nmid a_0$, this implies that $p^n \nmid a_0^{n-1}$. It follows that $p \mid c_i$, since the norm quantity lies in $\mathbb{Z}$. This is a contradiction, since we chose i to be the smallest such that $c_i \notin p\mathbb{Z}$.

For the last part of the statement, we apply Dedekind's factorization. Suppose that

$$p\mathcal{O}_K = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_r^{e_r} \pmod{p},$$

with $\mathfrak{p}_i = (p, g_i(\alpha))$ and g_i such that

$$\overbrace{m_\alpha}^{x^n} = \overline{g_1}^{e_1} \dots \overline{g_r}^{e_r} \pmod{p}.$$

This implies that $r = 1$, $g_1 = x$, and $e_1 = n$. This implies that $p \cdot \mathcal{O}_K = \mathfrak{p}^n$, and $\mathfrak{p} = (p, g_1(\alpha)) = (p, \alpha)$. $\square$

Example 3.7.7

Let $K = \mathbb{Q}(\sqrt[3]{2})$. Then, $\mathcal{O}_K = \mathbb{Z}[\alpha]$. Recall that $\text{disc}(\mathbb{Z}[\sqrt[3]{2}]) = -3^2 \cdot 2^2$. It is enough to prove that $2, 3 \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$, as $\text{disc}(\mathbb{Z}[\alpha]) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \cdot D_K$. We note that $m_{\sqrt[3]{2}}(x) = x^3 - 2$ is Eisenstein at 2, implying that $2 \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Next we consider 3. We consider the minimal polynomial of $\alpha \neq \lambda$, where $\lambda \in \mathbb{Z}$. Note that $m_{\alpha+\lambda}(x) = m_\alpha(x - \lambda)$. Next, $\mathbb{Z}[\alpha] = \mathbb{Z}[\alpha + \lambda]$, so if $m_{\alpha+\lambda}(x)$ is Eisenstein at p , then $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha + \lambda]] = [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. For $p = 3$, note that

$$m_{\alpha-2}(x) = m_\alpha(x+2) = (x+2)^3 - 2 = x^3 + 6x^2 + 12x + 6,$$

which is Eisenstein at 3.

Exercise 3.7.8. Let $d \neq \pm 1$ be square free, and consider $K = \mathbb{Q}(\sqrt[3]{d})$. Then, show that $\text{disc}(\mathbb{Z}[\sqrt[3]{d}]) = -27 \cdot d^2$.

Example 3.7.9

Continuing from the previous exercise, if $p \mid \text{disc}(\mathbb{Z}[\sqrt[3]{d}])$ and $p \neq 3$, then $p \mid d$ but $m_{\sqrt[3]{d}}(x) = x^3 - d$ which is Eisenstein at p , implying that $p \nmid [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{d}]]$.

For $p = 3$, if $d \not\equiv \pm 1 \pmod{9}$, then $m_\alpha(x+d)$ has $\sqrt[3]{d} - d$ as a root and is Eisenstein at 3. Note that the constant term of $m_\alpha(x+d)$ is $d^3 - d = d(d-1)(d+1)$. This implies (show this!) that $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{d}]$.

For $p = 3$ and if $d \equiv \pm 1 \pmod{9}$, then one can check (exercise) that $\frac{(\sqrt[3]{d}-d)^2}{3}$ is an algebraic integer. Note that

$$\frac{(\sqrt[3]{d}-d)^2}{3} \notin \mathbb{Z}[\sqrt[3]{d}]$$

Now let p be an odd prime and consider $K = \mathbb{Q}(\xi_p)$, where $\xi_p := e^{\frac{2\pi i}{p}}$ (the primitive p -th root of unity). The minimal polynomial of ξ_p is

$$\Phi_p(x) := \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + 1$$

and the minimal polynomial of $\xi_p - 1$ is

$$\Phi(x+1) = x^{p-1} + px^{p-2} + \cdots + p,$$

which is Eisenstein at p .

Theorem 3.7.10

We have $\mathcal{O}_K = \mathbb{Z}[\xi_p]$, where $K = \mathbb{Q}(\xi_p)$ and $D_{\mathbb{Q}(\xi_p)} = (-1)^{\frac{p-1}{2}} p^{p-2}$ and

$$p \cdot \mathcal{O}_K = (\xi_p - 1)^{p-1}$$

is totally ramified.

Proof. Step 1: We use the derivative $\Phi_p(\xi_p)$. Note that $(x-1)\Phi_p(x) = x^p - 1$. Then, we use Leibnitz to say that

$$\Phi_p(x) + (x-1)\Phi_p'(x) = p \cdot x^{p-1}.$$

This implies that

$$\Phi_p(\xi_p) + (\xi_p - 1)\Phi_p'(\xi_p) = p \cdot \xi_p^{p-1} \implies \Phi_p'(\xi_p) = \frac{p \cdot \xi_p^{p-1}}{\xi_p - 1} = \frac{p \cdot \xi_p^{-1}}{\xi_p - 1}.$$

Noting that $\text{disc}(\mathbb{Z}[\xi_p]) = (-1)^{\frac{(p-1)(p-2)}{2}} \mathcal{N}_K(\Phi_p'(\xi_p))$, we now find the norm of the quantity above. Particularly, as the norm is multiplicative, we only need to solve for each component. We first have that $\mathcal{N}_K(p) = p^{[K:\mathbb{Q}]} = p^{p-1}$. Next, note that

$$\mathcal{N}_K(\xi_p^{-1}) = \mathcal{N}_K(\xi_p)^{-1}.$$

Note that from the minimal polynomial of ξ_p and $\xi_p - 1$, we get

$$\mathcal{N}_K(\xi_p) = 1, \quad \mathcal{N}_K(\xi_p - 1) = p.$$

Thus, we have that

$$\text{disc}(\mathbb{Z}[\xi_p]) = (-1)^{\frac{(p-1)(p-2)}{2}} \cdot \frac{\mathcal{N}_K(p)\mathcal{N}_K(\xi_p)^{-1}}{\mathcal{N}_K(\xi_p - 1)} = (-1)^{\frac{p-1}{2}} p^{p-2}.$$

If $q \mid [\mathcal{O}_K : \mathbb{Z}[\xi_p]]$, then $q = p$. However, $\Phi_p(x+1)$ is Eisenstein at p , so by proposition 3.7.6, $p \nmid [\mathcal{O}_K : \mathbb{Z}[\xi_p - 1]]$. As $\mathbb{Z}[\xi_p - 1] = \mathbb{Z}[\xi_p]$, we ultimately get that

$$\mathcal{O}_K = \mathbb{Z}[\xi_p]$$

and that

$$D_K = (-1)^{\frac{p-1}{2}} p^{p-2}.$$

Finally, by Dedekind's criteria, we know that

$$p \cdot \mathcal{O}_K = p \cdot \mathcal{O}_{\mathbb{Q}(\xi_p-1)} = (p, \xi_p - 1)^{p-1}.$$

However, note that $\xi_p \mid \mathcal{N}_K(\xi_p - 1) = p$, so $p \in (\xi_p - 1)$. Therefore, $(p, \xi_p - 1) = (\xi_p - 1)$. $\square$

Theorem 3.7.11

Let $q \in \mathbb{Z}$ be a prime number such that $q \neq p$. Let f be the order of $[q] \in (\mathbb{Z}/p\mathbb{Z})^\times$ and let $r = \frac{p-1}{f}$. Then,

$$q \cdot \mathcal{O}_{\mathbb{Q}(\xi_p)} = \mathfrak{q}_1 \cdots \mathfrak{q}_r,$$

where $\mathfrak{q}_i$ are distinct prime ideals with $f(\mathfrak{q}_i) = f$ for every i .

Proof. Let $h(x) = x^p - 1$. Then, $h'(x) = px^{p-1}$ and note that $-p \cdot h(x) + x \cdot h'(x) = p$. This implies that $h(x)$ and $h'(x)$ cannot have common roots modulo q . Otherwise, take α which is a common root modulo q . Then, it must be that

$$p \cdot h(\alpha) + \alpha \cdot h'(\alpha) = p \equiv 0 \pmod{q},$$

but this cannot occur as $p \neq q$. Therefore, $h(x)$ cannot have multiple roots modulo q implying that $\Phi_p(x) = \frac{h(x)}{x-1}$ cannot have multiple roots modulo q . Therefore,

$$\overline{\Phi_p(x)} = g_1 \cdots g_r$$

in $\mathbb{F}_q[x]$, with $g_1, \dots, g_r$ being distinct irreducible polynomials. Now we want to show that $f = \deg(g_i) = f(\mathfrak{q}_i)$. Note that $\mathbb{F}_q[x]/(g_i(x))$ is a field, which we'll call F . We have that $|F| = q^{\deg(g_i)}$. For any nonzero element $a \in F^\times$, we have

$$a^{q^{\deg(g_i)} - 1} = 1,$$

since $F^\times$ is cyclic of order $q^{\deg(g_i)} - 1$. Moreover, $a = [x] \in F^\times = (\mathbb{F}_q[x]/(g_i(x)))^\times$. Then, since x is a root of $\Phi_p(x)$, this implies that

$$a^p = [x]^p = 1,$$

implying that $p \mid q^{\deg(g_i)} - 1$. This is equivalent to saying that

$$q^{\deg(g_i)} \equiv 1 \pmod{p}.$$

As f is the order of q in $(\mathbb{Z}/p\mathbb{Z})^\times$, this implies that f divides $\deg(g_i)$. In particular, $f \leq \deg(g_i)$

On the other hand, let us define

$$\tilde{F} := \{b \in F \mid b^{q^f} - b\} \subseteq F.$$

Note that this is a subfield of F (we can check this easily). Next, we also have that $\tilde{F}$ contains $a = [x] \in F$. To show this, recall that f is the order of q in $(\mathbb{Z}/p\mathbb{Z})^\times$. Thus, $a^{q^f} - a = 0$, so $a(a^{q^f-1} - 1) = 0$, implying that

$$a^{q^f-1} - 1 = 0.$$

Noting that $q^f \equiv 1 \pmod{p}$, we have $q^f - 1 = p \cdot \lambda$. Therefore, $a^{p \cdot \lambda} - 1 = 0$, so $1 - 1 = 0$. Thus, it turns out that $\tilde{F} = F$. Therefore, we have that $q^{\deg(g_i)} = |F| = |\tilde{F}| \leq q^f$, so $\deg(g_i) \leq f$, and so $\deg(g_i) = f$.

Finally, we have shown that each irreducible factor g_i has degree f , and so there must be $r = \frac{p-1}{f}$ factors. The theorem now follows from Dedekind's factorization criteria. $\square$

Remark 3.7.12 — The key point of the proof is that if F is a finite field of characteristic q , then $F^\times$ is cyclic of order $|F| - 1$.

Remark 3.7.13 — The splitting of q in $\mathbb{Z}[\xi_p]$ depends only on the reduction of q modulo p . All the information encoded in $\gcd(\bar{q}, p-1) = r$.

§3.7.i Quadratic Reciprocity

We will use theorem 3.7.11 now to provide another proof of quadratic reciprocity. The idea goes as follows: if a quadratic field $K \subseteq \mathbb{Q}(\xi_p)$, we try to read information on the splitting on K from $\mathbb{Q}(\xi_p)$.

Lemma 3.7.14

We have that

$$\mathbb{Q}\left(\sqrt{(-1)^{\frac{p-1}{2}}p}\right) \subseteq \mathbb{Q}(\xi_p).$$

Proof. We know that the roots of $\Phi_p(x)$ are $\xi_p, \xi_p^2, \dots, \xi_p^{p-1}$. Therefore, the embeddings $\tau_1, \dots, \tau_{p-1}: \mathbb{Q}(\xi_p) \rightarrow \mathbb{C}$ is given by $\tau_i(\xi_p) = \xi_p^i$. We also have that $\{\xi_p, \dots, \xi_p^{p-1}\}$ is an integral basis of $\mathcal{O}_{\mathbb{Q}(\xi_p)} = \mathbb{Z}[\xi_p]$. We've seen that

$$D_{\mathbb{Q}(\xi_p)} = (-1)^{\frac{p-1}{2}} p^{p-2},$$

and so we get that

$$(-1)^{\frac{p-1}{2}} p^{p-2} = \det(\tau_i(\xi_p^j))^2 = \det((\xi_p^{ij}))^2.$$

Therefore,

$$\left(\frac{\det((\xi_p^{ij}))}{p^{\frac{p-3}{2}}}\right)^2 = \frac{\det((\xi_p^{ij}))^2}{p^{p-3}} = (-1)^{\frac{p-1}{2}} p.$$

Note, note that the quantity on the left hand side is in $\mathbb{Q}(\xi_p)$, and so it follows that $\sqrt{(-1)^{\frac{p-1}{2}}p} \in \mathbb{Q}(\xi_p)$. Therefore,

$$\mathbb{Q}\left(\sqrt{(-1)^{\frac{p-1}{2}}p}\right) \subseteq \mathbb{Q}(\xi_p).$$

□

Exercise 3.7.15. Show that if $K \subseteq \mathbb{Q}(\xi_p)$ is a quadratic field, then the only prime that can ramify in K is p . Deduce that

$$K = \mathbb{Q}\left(\sqrt{(-1)^{\frac{p-1}{2}}p}\right)$$

Notation 3.7.16. We will denote from now that $p^* = (-1)^{\frac{p-1}{2}}p$.

Note that $p^* \equiv 1 \pmod{4}$ then we can have:

- if $p \equiv 1 \pmod{4}$, then $\frac{p-1}{2}$ is even, so $p = p^*$.
- if $p \equiv 3 \pmod{4}$, then $\frac{p-1}{2}$ is odd, so $p = -p^*$ (so $p^* \equiv 1 \pmod{4}$).

Lemma 3.7.17

Let f be the order of q in $(\mathbb{Z}/p\mathbb{Z})^\times$ and let $r = \frac{p-1}{f}$ be the number of prime factors of $q\mathbb{Z}[\xi_p]$.

1. If q splits in $\mathbb{Q}(\sqrt{p^*})$, then r is even.
2. If $p \equiv 3 \pmod{4}$, then also the converse holds, i.e., q splits in $\mathbb{Q}(\sqrt{p^*})$ if and only if r is even.

Proof. (1): q splits, so we have

$$q \cdot \mathcal{O}_{\mathbb{Q}(\sqrt{p^*})} = \mathfrak{q} \cdot \bar{\mathfrak{q}}$$

and $q\mathbb{Z}[\xi_p] = (\mathfrak{q}\mathbb{Z}[\xi_p])(\bar{\mathfrak{q}}\mathbb{Z}[\xi_p])$. We want to prove that $\mathfrak{q}\mathbb{Z}[\xi_p]$ and $\bar{\mathfrak{q}}\mathbb{Z}[\xi_p]$ have the same number of factors. Take $\tau_0: \mathbb{Q}(\sqrt{p^*}) \rightarrow \mathbb{C}$ given by $x \mapsto \bar{x}$ (the conjugation map). This is the only nontrivial embedding. Now, we now that there is an embedding $\tau: \mathbb{Q}(\xi_p) \rightarrow \mathbb{C}$ such that $\tau|_{\mathbb{Q}(\sqrt{p^*})} = \tau_0$. In particular, there exists an i such that

$$\begin{aligned} \tau: \mathbb{Q}(\xi_p) &\rightarrow \mathbb{C} \\ \xi_p &\mapsto \xi_p^i. \end{aligned}$$

We note that $\tau(\mathbb{Q}(\xi_p)) \subseteq \mathbb{Q}(\xi_p)$. Moreover, the inverse of τ is τ^{-1} that send $\xi_p \mapsto \xi_p^j$, where $ij \equiv 1 \pmod{p}$. Therefore,

$$\tau(\mathbb{Q}(\xi_p)) = \mathbb{Q}(\xi_p).$$

Note that $\tau(\mathbb{Z}[\xi_p]) \subseteq \mathbb{Z}[\xi_p]$ and $\tau^{-1}(\mathbb{Z}[\xi_p]) \subseteq \mathbb{Z}[\xi_p]$. Then, τ gives us an automorphism

$$\tau: \mathbb{Z}[\xi_p] \rightarrow \mathbb{Z}[\xi_p].$$

Then,

$$\begin{aligned} \tau(\mathfrak{q} \cdot \mathbb{Z}[\xi_p]) &= \tau(\mathfrak{q})\mathbb{Z}[\xi_p] \\ &= \tau_0(\mathfrak{q}) \\ &= \bar{\mathfrak{q}}\mathbb{Z}[\xi_p]. \end{aligned}$$

This tells us that $\mathfrak{q}\mathbb{Z}[\xi_p]$ and $\bar{\mathfrak{q}}\mathbb{Z}[\xi_p]$ has the same number of factors.

(2): If q is inert in $\mathbb{Q}(\sqrt{p^*})$, then fix some $\mathfrak{q} \subseteq \mathbb{Z}[\xi_p]$ lying above q , then

$$\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z} \subseteq \mathcal{O}_{p^*}/\mathfrak{q} \cdot \mathcal{O}_{p^*} \subseteq \mathbb{Z}[\xi_p]/\mathfrak{q} \cdot \mathbb{Z}[\xi_p].$$

Note that $[\mathcal{O}_{p^*}/\mathfrak{q} \cdot \mathcal{O}_{p^*} : \mathbb{Z}/q\mathbb{Z}] = 2$ since q is inert. Then, we may apply the tower law:

$$2 = [\mathcal{O}_{p^*} : \mathbb{F}_q] \mid [\mathbb{Z}[\xi_p]/\mathfrak{q}\mathbb{Z}[\xi_p] : \mathbb{F}_p] = f(\mathfrak{q}) = f,$$

which is the order of q in $(\mathbb{Z}/p\mathbb{Z})^\times$. If q is inert, then f is even. Now, we look at the condition $p \equiv 3 \pmod{4}$. We can write $p-1 = 2k$ with k odd, since $r \cdot f = p-1$. Now, f and r have to be of different parity then. Therefore, if r is even, f is odd and so q cannot be inert, so q splits. $\square$

Remark 3.7.18 — Since $p^* \equiv 1 \pmod{4}$, we have that $D_{\mathbb{Q}(\sqrt{p^*})} = p^*$. If $q \neq p$, then $q \nmid D_{\mathbb{Q}(\sqrt{p^*})}$, implying that q doesn't ramify in $D_{\mathbb{Q}(\sqrt{p^*})}$.

Lemma 3.7.19

Let p be an odd prime. Then, for $a, b \in \mathbb{Z}$, we have:

1. $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$.
2. $\left(\frac{a}{p}\right) = 1$ if and only if $\frac{p-1}{f}$ is even, where f is the order of a in $(\mathbb{Z}/p\mathbb{Z})^\times$.
3. $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$.

Proof. (1): The key point is to see that $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$, so $\left((\mathbb{Z}/p\mathbb{Z})^\times\right)^2$ is a subgroup of $(\mathbb{Z}/p\mathbb{Z})^\times$ with index 2. Then, we can see there is a homomorphism $(\mathbb{Z}/p\mathbb{Z})^\times \rightarrow \{\pm 1\}$ sending $\left((\mathbb{Z}/p\mathbb{Z})^\times\right)^2$ to 1 and the non trivial coset of $\left((\mathbb{Z}/p\mathbb{Z})^\times\right)^2$ to -1 , and this homomorphism is exactly the Legendre symbol, by definition. This proves the first statement.

(2): We recall from group theory that the order of $r \in \mathbb{Z}/(p-1)\mathbb{Z}$ is $f = \frac{p-1}{\gcd(r, p-1)}$. In particular $\frac{p-1}{f} = \gcd(r, p-1)$ is even if and only if r is.

(3): This is an application of the second statement. The order of -1 is 2, so -1 is a quadratic residue mod p if and only if $\frac{p-1}{2}$ is even, i.e. if and only if $p \equiv 1 \pmod{4}$. $\square$

Theorem 3.7.20 (Gauss's Law of Quadratic Reciprocity)

Let p, q be distinct odd primes. then,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right).$$

Equivalently,

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Proof. Recall that if q splits in $\mathbb{Q}(\sqrt{p^*})$, then

$$\left(\frac{D_{\mathbb{Q}(\sqrt{p^*})}}{q}\right) = \left(\frac{p^*}{q}\right) = 1.$$

Moreover, lemma 3.7.17 is telling us that $\left(\frac{p^*}{q}\right) = 1$, implying that r is even. now, by lemma 3.7.19, we have that r is even, implying that $\left(\frac{q}{p}\right) = 1$. On the other hand, $\left(\frac{q}{p}\right) = 1$ if and only if r is even.

Now we move on to the proof. Assume that $p \equiv 1 \pmod{4}$ or $q \equiv 1 \pmod{4}$. Then, r is even if and only if q splits in $\mathbb{Q}(\sqrt{p^*})$. Therefore,

$$\left(\frac{p^*}{q}\right) = 1$$

and r being even further implies that $\left(\frac{q}{p}\right) = 1$. Therefore,

$$\left(\frac{p^*}{q}\right) = \left(\frac{q}{p}\right).$$

Now assume both $p, q \equiv 3 \pmod{4}$. Then, $\left(\frac{q}{p}\right) = 1$ implies that p splits in $\mathbb{Q}(\sqrt{q})$, so r is even. Lemma 3.7.19 then says that $\left(\frac{p^*}{q}\right) = 1$. Next, if $\left(\frac{q}{p}\right) = 1$, then p is inert in $\mathbb{Q}(\sqrt{q})$ and so r is odd (as f is even), implying that $\left(\frac{p^*}{q}\right) = 1$. $\square$

§3.8 Finiteness of the Class Group

The goal of this section will be to show the following theorem.

Theorem 3.8.1

Let K be a number field. Then $\text{Cl}(\mathcal{O}_K)$ is finite.

Recall that $\text{Cl}(\mathcal{O}_K) = \{\text{ideals } I \subseteq \mathcal{O}_K\} / \sim$, where $I \sim J$ if $\exists \alpha \in K$ such that $\alpha I = J$. The group operation is $[I][J] = [IJ]$.

Lemma 3.8.2

For any $C \in \mathbb{R}_{\geq 0}$, the number of ideals $I \subseteq \mathcal{O}_K$ with $N_K(I) \leq C$ is finite.

Proof. Consider $I \subseteq \mathcal{O}_K$, and consider the factorization

$$I = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}.$$

Step 1: Bound r with respect to C :

We now that $N_k(\mathfrak{p}_i) = |\mathcal{O}_K/\mathfrak{p}_i| \geq 2$. But then, we see that

$$2^r \leq N_K(I) \leq C \implies r \leq \log_2(C).$$

Therefore, we may have at most $\lfloor \log_2(C) \rfloor$ distinct factors.

Step 2: Bound the primes $p \in \mathbb{Z} \cap I$.

Note that if p lies in $\mathbb{Z} \cap I$, then there exists i such that $\mathfrak{p}_i$ lies above p . This tells us that

$$C > N_k(I) \geq N_k(\mathfrak{p}_i) = p^{f(\mathfrak{p}_i)} \geq p.$$

This tells us that we have a finite number of prime $p \in I \cap \mathbb{Z}$.

Thus, we're done, since there are finitely many prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ lying above a given primes p . $\square$

Lemma 3.8.3

Let K be a number field. Suppose that there is a constant $C \in \mathbb{R}_{>0}$ such that for any nonzero ideal $I \subseteq \mathcal{O}_K$, there exists $0 \neq \alpha \in I$ such that

$$|\mathcal{N}_K(\alpha)| \leq C \cdot N_K(I).$$

then, for all $[I] \in \text{Cl}(\mathcal{O}_K)$ there exists I' such that $[I] = [I']$ and $N_K(I') \leq C$.

Proof. Pick $J \subseteq \mathcal{O}_K$ such that $[J] = [I]^{-1}$ in $\text{Cl}(\mathcal{O}_K)$. Then, by assumption, there exists nonzero $\alpha \in J$ such that $|\mathcal{N}_K(\alpha)| \leq C \cdot N_K(J)$. Take I' now such that $J \cdot I' = (\alpha)$. Then, $[I'] = [J]^{-1} = [I]$ and

$$N_K(J \cdot I') = N_K(\alpha \cdot \mathcal{O}_K) = |\mathcal{N}_K(\alpha)|.$$

This implies that

$$N_K(I') = \frac{|\mathcal{N}_K(\alpha)|}{N_K(J)} \leq C.$$

□

We now want to determine this constant C , as we can then limit the ideals we want to consider.

Definition 3.8.4. Given a number field K , we call an embedding $\tau: K \hookrightarrow \mathbb{C}$ **real** if $\tau(K) \subseteq \mathbb{R}$. Otherwise, $\tau: K \hookrightarrow \mathbb{C}$ it is a **complex embedding**.

Remark 3.8.5 — From the definition, we write

$$n = [K : \mathbb{Q}] = |\Sigma_K| = r + 2s,$$

where r is the number of real embeddings and s is the number of complex embeddings.

Theorem 3.8.6

Let K be a number field with $n = [K : \mathbb{Q}] = r + 2s$ and define the **Minkowski constant**

$$C_K = \frac{n!}{n^n} \left(\frac{4}{\pi}\right)^s \cdot \sqrt{|D_K|}.$$

For any nonzero ideal $I \subseteq \mathcal{O}_K$, there exists $0 \neq \alpha \in I$ such that

$$|\mathcal{N}_K(\alpha)| \leq C_K \cdot N_K(I).$$

We explore some examples before proving this theorem.

§3.8.i Some Examples in the Quadratic Case

Let $K = \mathbb{Q}(\sqrt{d})$.

- If $d < 0$, then $r = 0$ and $s = 1$, so

$$C_K = \frac{2!}{2^2} \left(\frac{4}{\pi}\right)^2 \sqrt{|D_K|} = \frac{2}{\pi} \sqrt{|D_K|}.$$

- If $d > 0$, then $r = 2$ and $s = 0$, so

$$C_K = \frac{2!}{2^2} \left(\frac{4}{\pi}\right) \sqrt{|D_K|} = \frac{1}{2} \sqrt{|D_K|}.$$

Example 3.8.7

Let $K = \mathbb{Q}(\sqrt{-5})$ and so $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$. We know that $D_K = -4 \cdot 5 = -20$, as $-5 \equiv 3 \pmod{4}$. Hence,

$$C_K = \frac{2}{\pi} \sqrt{20} < 3.$$

Therefore, every element $[I] \in \text{Cl}(\mathcal{O}_K)$ can be represented by an ideal of $N_K(I) \leq 2$.

The only ideal of norm 1 is $\mathcal{O}_K$. The only ideal of norm 2 is $\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$. Therefore, $\text{Cl}(\mathcal{O}_{-5}) = \mathbb{Z}/2\mathbb{Z}$.

Theorem 3.8.8

The rings $\mathcal{O}_{-15}, \mathcal{O}_{-43}, \mathcal{O}_{-67}, \mathcal{O}_{-163}$ are PIDs.

Proof. Recall that $\mathcal{O}_K$ is a PID if and only if $\text{Cl}(\mathcal{O}_K) = \{1\}$. For $\mathcal{O}_{-163}$, the Minkowski constant $C_K = \frac{2}{\pi} \sqrt{163} < 9$. We want to see if there are non-principal ideals of norm less than or equal to 8. We need to see if there are no non-principal ideals above 2, 3, 5, 7.

- $p = 2$: We claim that 2 is inert.

$$\left(\frac{2}{163} \right) = -1,$$

as $163 \equiv -3 \pmod{8}$, and so 2 is inert. Thus, $2\mathcal{O}_K$ is prime.

- $p = 5$: We claim that 5 is inert.

$$\left(\frac{5}{163} \right) \cdot \left(\frac{163}{5} \right) = (-1)^{2 \cdot 81} = 1.$$

Now, $\left(\frac{3}{5} \right) = -1$, and so $\left(\frac{5}{163} \right) = -1$, and so 5 is inert.

- $p = 3, 7$ (exercise)

For the other cases, we can continue in this manner as well. □

§3.8.ii Lattices and Proving Theorem 3.8.6

To prove theorem 3.8.6, we need to introduce lattices.

Definition 3.8.9. A discrete subgroup $\Lambda \subseteq \mathbb{R}^n$ is called a **lattice** if it has a basis $v_1, \dots, v_n$ as a $\mathbb{Z}$ -module (abelian group) which is also a basis of $\mathbb{R}^n$ as an $\mathbb{R}$ -vector space.

Example 3.8.10

$\mathbb{Z}^n \subseteq \mathbb{R}^n$ is a lattice (take $e_1, \dots, e_n$).

Example 3.8.11

We consider $\mathcal{O}_d \subseteq \mathbb{C} \cong \mathbb{R}^2$. We note that

$$\mathcal{O}_d = \begin{cases} \text{span}_{\mathbb{Z}}\{1, \sqrt{d}\} & d \not\equiv 1 \pmod{4} \\ \text{span}_{\mathbb{Z}}\{1, \frac{1+\sqrt{d}}{2}\} & d \equiv 1 \pmod{4} \end{cases}.$$

Then, $\mathcal{O}_d$ is a lattice if and only if $d < 0$.

Definition 3.8.12. Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice with basis $v_1, \dots, v_n$. Then, the **fundamental parallelotope** is

$$P(v_1, \dots, v_n) = \{c_1 v_1 + \dots + c_n v_n \mid 0 \leq c_i < 1\}.$$

It is a **fundamental domain** for the lattice in the sense that

$$\mathbb{R}^n = \bigcup_{v \in \Lambda} (v + P(v_1, \dots, v_n))$$

and the sets in the union are pairwise disjoint. We also write

$$\mathbb{R}^n = \bigsqcup_{v \in \Lambda} (v + P(v_1, \dots, v_n))$$

to indicate that the union is disjoint. In other words we can subdivide $\mathbb{R}^n$ into translates of the fundamental parallelotope.

Recall from linear algebra, that we know

$$\text{vol}(P(v_1, \dots, v_n)) = |\det(A)|,$$

where $A = (v_1 \mid \dots \mid v_n)$.

Lemma 3.8.13

If $v_1, \dots, v_n$ and $w_1, \dots, w_n$ are two bases for a given lattice $\Lambda \subseteq \mathbb{R}^n$ then

$$\text{vol}(P(v_1, \dots, v_n)) = \text{vol}(P(w_1, \dots, w_n)).$$

Proof Sketch. Take the transition matrix from one basis of the other. □

Definition 3.8.14. For a lattice $\Lambda \subseteq \mathbb{R}^n$, we define the **covolume** of Λ as

$$\text{covol}(\Lambda) := \text{vol}(P(v_1, \dots, v_n))$$

for any basis $v_1, \dots, v_n$.

Remark 3.8.15 — $\text{covol}(\Lambda) = \text{vol}(\Lambda/\mathbb{R}^n)$.

Example 3.8.16

Take $\mathbb{Z}^n = \Lambda \subseteq \mathbb{R}^n$, then

$$\text{covol}(\mathbb{Z}^n) = 1.$$

Lemma 3.8.17 (Blichfeldt's Lemma)

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and let $S \subseteq \mathbb{R}^n$ be a measurable subset. Then if $\text{vol}(S) > \text{covol}(\Lambda)$, there exists $x, y \in S$ with $0 \neq x - y \in \Lambda$.

Proof. Fix a basis $v_1, \dots, v_n$ of Λ and $P = P(v_1, \dots, v_n)$. Then, we have that $\mathbb{R}^n = \bigsqcup_{v \in \Lambda} (v + P)$. This gives us that

$$S = \bigsqcup_{v \in \Lambda} (v + P) \cap S,$$

and so

$$\text{vol}(S) \stackrel{(1)}{=} \sum_{v \in \Lambda} \text{vol}((v + P) \cap S) \stackrel{(2)}{=} \sum_{v \in \Lambda} \text{vol}(P \cap (S - v)).$$

Note that the equality (1) is justified by countable additivity of volume, and (2) is justified by translational invariance. If the sets $P \cap (S - v)$ are pairwise disjoint, then

$$\text{vol}(S) = \text{vol}\left(\bigcup_{v \in \Lambda} (P \cap (S - v))\right) \leq \text{vol}(P) = \text{covol}(\Lambda).$$

However, this is a contradiction to our assumption. Therefore, $P \cap (S - v)$ are not pairwise disjoint. This means that there exists $v \neq w \in \Lambda$ such that

$$(P \cap (S - v)) \cap (P \cap (S - w)) \neq \emptyset.$$

However, the LHS is a subset of $(S - v) \cap (S - w)$, and so there exists an element $a \in (S - v) \cap (S - w)$. Taking $x := a + v$ and $y := a + w$, we have that $x \in S$ and $y \in S$, and $x - y \in \Lambda$. $\square$

Theorem 3.8.18 (Minkowski's convex body theorem)

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and $S \subseteq \mathbb{R}^n$ be a measurable subset which is:

- convex: $\forall x, y \in S, tx + (1 - t)y \in S$, for $t \in [0, 1]$
- symmetric with respect to 0: $x \in S$ implies $-x \in S$

Then if $\text{vol}(S) > 2^n \text{covol}(\Lambda)$, there exists $0 \neq x \in S \cap \Lambda$.

Proof. We apply Blichfeldt's lemma (3.8.17) to

$$\frac{1}{2}S = \left\{ \frac{s}{2} \mid s \in S \right\}.$$

We have that

$$\text{vol}\left(\frac{1}{2}S\right) = \frac{1}{2^n} \text{vol}(S) > \text{covol}(\Lambda).$$

Then, lemma 3.8.17 says that there exists $x, y \in \frac{1}{2}S$ with $0 \neq x - y \in \Lambda$. However, $x, y \in \frac{1}{2}S$ implies that $2x, 2y \in S$, and hence $-2y \in S$ by symmetry with respect to 0. Therefore,

$$\frac{1}{2}(2x) + \frac{1}{2}(-2y) \in S,$$

and so $x - y \in S \cap \Lambda$. $\square$

Lemma 3.8.19

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and $\Lambda' \subseteq \Lambda$ be a subgroup of finite index. Then, Λ' is also a lattice and

$$\text{covol}(\Lambda') = [\Lambda : \Lambda'] \text{covol}(\Lambda).$$

Proof. We have

$$[\Lambda : \Lambda']\Lambda \subseteq \Lambda' \subseteq \Lambda.$$

As Λ' is sandwiched between free abelian groups of rank n , it must itself be a free abelian group of rank n . Now let $v_1, \dots, v_n$ be a basis for Λ , and let $w_1, \dots, w_n$ be a basis for Λ' . If A is the matrix with $w_1, \dots, w_n$ as columns, and B is the matrix with $v_1, \dots, v_n$ as columns, then we have

$$A = SB,$$

for $S \in M_n(\mathbb{Z})$. Taking absolute values of determinants we obtain

$$\text{covol}(\Lambda') = |\det(S)| \cdot \text{covol}(\Lambda).$$

However, we have $|\det(S)| = [\Lambda : \Lambda']$. □

Remark 3.8.20 — This is very similar to what happens for discriminants. Instead of considering

$$[\mathbb{R}^n : \Lambda] = |(\mathbb{R}^n / \Lambda)|,$$

we take $\text{vol}(\mathbb{R}^n / \Lambda)$. (as the cardinality above is infinite)

Now assume that

$$\mathcal{O}_K \xrightarrow[\tau]{} \Lambda \subseteq \mathbb{R}^n,$$

where $n = [K : \mathbb{Q}]$. Here, τ is an isomorphism of abelian groups. Then, $I \subseteq \mathcal{O}_K$, we have

$$N_K(I) = |(\mathcal{O}_K / I)| = |(\Lambda / \tau(I))|,$$

where the second equality comes from τ being an isomorphism. $\tau(I) \subseteq \Lambda$ is a lattice, and

$$\text{covol}(\tau(I)) = N_K(I) \text{covol}(\mathcal{O}_K).$$

§3.8.iii Minkowski for Imaginary Quadratic Fields

Fix $d < 0$ and let $K = \mathbb{Q}(\sqrt{d})$. Then, $\mathcal{O}_d \subseteq \mathbb{C} \cong \mathbb{R}^2$ is a lattice.

Lemma 3.8.21

$\mathcal{O}_d \subseteq \mathbb{C} \cong \mathbb{R}^2$ is a lattice with covolume $\frac{\sqrt{|D_K|}}{2}$.

Proof. Recall that $\text{covol}(\mathcal{O}_d) = \sqrt{d}P(v_1, v_2)$ with v_1, v_2 a $\mathbb{Z}$ -basis of $\mathcal{O}_d$. Then, $\mathbb{Z}[\sqrt{d}] \subset \mathbb{R}^2$ is a lattice with basis $(1, 0), (0, \sqrt{-d})$. Therefore,

$$\text{covol}(\mathbb{Z}[\sqrt{d}]) = \sqrt{|d|}.$$

Now, if $d \not\equiv 1 \pmod{4}$, then $\mathcal{O}_d = \mathbb{Z}[\sqrt{d}]$ and we know that $D_K = 4d$, and so

$$\text{covol}(\mathcal{O}_d) = \sqrt{|d|} = \frac{\sqrt{|D_K|}}{2},$$

so we're done. If $d \equiv 1 \pmod{4}$, then $[\mathcal{O}_d : \mathbb{Z}[\sqrt{d}]] = 2$, as $\mathcal{O}_d = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$. This implies that

$$\text{covol}(\mathcal{O}_d) = \frac{\sqrt{|d|}}{[\mathcal{O}_d : \mathbb{Z}[\sqrt{d}]]} = \frac{\sqrt{|D_K|}}{2}.$$

□

Theorem 3.8.22 (Minkowski's bound for Imaginary Quadratic Fields)

For any nonzero ideal $I \subseteq \mathcal{O}_d$ with $d < 0$ square free, there exists $0 \neq \alpha \in I$ such that

$$|N_K(\alpha)| \leq \left(\frac{2}{\pi} \sqrt{|D_K|} \right) \cdot N_K(I)$$

Proof. Take $S_R = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq R\} \subseteq \mathbb{R}^2$. S_R is a circle with volume $\text{vol}(S_R) = \pi \cdot R$. S_R is convex and symmetric with respect to 0. Take $I \subseteq \mathcal{O}_d$. Noting that I has finite index in $\mathcal{O}_d$, we get that I is a lattice in $\mathcal{O}_d$ and

$$\text{covol}(I) = N_K(I) \cdot \text{covol}(\mathcal{O}_d).$$

We have

$$\pi R = \text{vol}(S_R) > 2^2 \cdot \text{covol}(I) = \frac{4\sqrt{|D_K|}}{2} N_K(I).$$

Equivalently, $R > \frac{2}{\pi} \sqrt{|D_K|} \cdot N_K(I)$. This implies, by the Minkowski body theorem, that there exists a nonzero element $\alpha \in I \cap S_R$. However, note that $\alpha \in S_R$ implies that

$$\mathcal{N}_K(\alpha) = \text{Re}(\alpha)^2 + \text{Im}(\alpha)^2 \leq R.$$

Therefore, if we pick $R = \lceil \frac{2}{\pi} \sqrt{|D_K|} \cdot N_K(I) \rceil$, then since $\mathcal{N}_K(\alpha) \in \mathbb{Z}$, $\mathcal{N}_K(\alpha) \leq R$ implies that

$$\mathcal{N}_K(\alpha) \leq \frac{2}{\pi} \sqrt{|D_K|} N_K(I).$$

□

Remark 3.8.23 — $\frac{2}{\pi} \sqrt{|D_K|}$ is the Minkowski constant for imaginary quadratic fields.

§3.8.iv Minkowski for Real Quadratic Fields

For $d > 0$, we consider

$$\begin{aligned} \iota: \mathcal{O}_d &\rightarrow \mathbb{R}^2 \\ \alpha &\mapsto (\tau_1(\alpha), \tau_2(\alpha)) \end{aligned}$$

with τ_1, τ_2 being embeddings of $K = \mathbb{Q}[\sqrt{d}] \rightarrow \mathbb{C}$. (τ_1, τ_2 are real embeddings, i.e. $\tau_i(K) \subseteq \mathbb{R}$).

Lemma 3.8.24

We have:

1. $\iota(\mathcal{O}_d) \subseteq \mathbb{R}^2$ is a lattice of $\text{covol} = \sqrt{|D_K|}$.
2. For all $I \subseteq \mathcal{O}_d$, $\iota(I) \subseteq \mathbb{R}^2$ is a lattice with $\text{covol} = \sqrt{|D_K|} \cdot N_K(I)$.

Proof Sketch. (1): Let α_1, α_2 be an integral basis of $\mathcal{O}_d$. Then,

$$\iota(\mathcal{O}_d) = \text{span} \left\{ \begin{pmatrix} \tau_1(\alpha_1) \\ \tau_2(\alpha_1) \end{pmatrix}, \begin{pmatrix} \tau_1(\alpha_2) \\ \tau_2(\alpha_2) \end{pmatrix} \right\}.$$

Then,

$$\text{covol}(\iota(\mathcal{O}_d)) = \left| \det \begin{pmatrix} \tau_1(\alpha_1) & \tau_1(\alpha_2) \\ \tau_2(\alpha_1) & \tau_2(\alpha_2) \end{pmatrix} \right| = \det \sqrt{|D_K|},$$

$\underbrace{\hspace{10em}}_{=A}$

as $D_K = \det(A)^2$. □

Since $N_K(\alpha) = \tau_1(\alpha)\tau_2(\alpha)$, so it's natural to consider

$$M_R = \{(x, y) \in \mathbb{R}^2 \mid |xy| \leq R\}.$$

This is the region bounded by two hyperbolas. The good news is that it has infinite volume, but the bad news is that this is not convex. Therefore, we cannot apply the Minkowski convex body theorem. The right region to consider will be

$$S_R = \{(x, y) \in \mathbb{R}^2 \mid |x| + |y| \leq 2\sqrt{R}\}.$$

This is convex and symmetric with respect to 0. We note that $S_R \subseteq M_R$, since the AM-GM inequality tells us that

$$|xy| \leq \left(\frac{|x| + |y|}{2} \right)^2 \leq R.$$

Note that S_R is a square with length $2\sqrt{2R}$. Therefore,

$$\text{vol}(S_R) = 8R.$$

If $\text{vol}(S_R) > 2^2 \text{covol}(I)$, then $8R > 4\sqrt{|D_K|}N_K(I)$, so

$$R > \frac{1}{2}\sqrt{|D_K|}N_K(I).$$

The Minkowski body theorem then tells us that there exists a nonzero $\alpha \in I \cap S_R$. In particular, $|\mathcal{N}_K(\alpha)| \leq R$. If we select $R > \frac{1}{2}\sqrt{|D_K|}N_K(I)$ such that

$$\lfloor R \rfloor = \lfloor \frac{1}{2}\sqrt{|D_K|}N_K(I) \rfloor,$$

then $|\mathcal{N}_K(\alpha)| \leq R$ and $\mathcal{N}_K(\alpha) \in \mathbb{Z}$, implying that

$$|\mathcal{N}_K(\alpha)| \leq \frac{1}{2}\sqrt{|D_K|}N_K(I).$$

Thus, we've proven:

Theorem 3.8.25 (Minkowski for Real Quadratic Fields)

For any nonzero ideal $I \subseteq \mathcal{O}_d$, with $d > 1$ square free, there exists $0 \neq \alpha \in I$ with

$$|\mathcal{N}_K(\alpha)| \leq \frac{1}{2}\sqrt{|D_K|}N_K(I).$$

§3.8.v Minkowski in General

Let K be a number field, with $[K : \mathbb{Q}] = n = r + 2s$. Write

$$\Sigma_K = \{\tau_1, \dots, \tau_r, \sigma_1, \overline{\sigma_1}, \dots, \sigma_s, \overline{\sigma_s}\}.$$

Then, we consider the following:

$$\begin{aligned} \iota: K &\hookrightarrow \mathbb{R}^n \\ \alpha &\mapsto (\tau_1(\alpha), \dots, \tau_r(\alpha), \operatorname{Re}(\sigma_1(\alpha)), \operatorname{Im}(\sigma_1(\alpha)), \dots, \operatorname{Re}(\sigma_s(\alpha)), \operatorname{Im}(\sigma_s(\alpha))) \end{aligned}$$

With this we can determine the covolume of an ideal:

Lemma 3.8.26

$\iota(\mathcal{O}_K) \subseteq \mathbb{R}^n$ is a lattice and $\operatorname{covol}(\iota(\mathcal{O}_K)) = \frac{\sqrt{|D_K|}}{2^s}$.

Proof. Pick $\{\alpha_1, \dots, \alpha_n\}$ an integral basis of $\mathcal{O}_K$. Then, we have the following:

$$\begin{aligned} &\begin{pmatrix} \tau_1(\alpha_1) & \dots & \tau_1(\alpha_n) \\ \vdots & & \vdots \\ \tau_r(\alpha_1) & \dots & \tau_r(\alpha_n) \\ \operatorname{Re}(\sigma_1(\alpha_1)) & \dots & \operatorname{Re}(\sigma_1(\alpha_n)) \\ \operatorname{Im}(\sigma_1(\alpha_1)) & \dots & \operatorname{Im}(\sigma_1(\alpha_n)) \\ \vdots & & \vdots \\ \operatorname{Re}(\sigma_s(\alpha_1)) & \dots & \operatorname{Re}(\sigma_s(\alpha_n)) \\ \operatorname{Im}(\sigma_s(\alpha_1)) & \dots & \operatorname{Im}(\sigma_s(\alpha_n)) \end{pmatrix} \\ &= \begin{pmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & \frac{1}{2} & \frac{1}{2} & & & \\ & & & \frac{1}{2i} & -\frac{1}{2i} & & & \\ & & & & & \ddots & & \\ & & & & & & \frac{1}{2} & \frac{1}{2} \\ & & & & & & \frac{1}{2i} & -\frac{1}{2i} \end{pmatrix} \cdot \begin{pmatrix} \tau_1(\alpha_1) & \dots & \tau_1(\alpha_n) \\ \vdots & & \vdots \\ \tau_r(\alpha_1) & \dots & \tau_r(\alpha_n) \\ \sigma_1(\alpha_1) & \dots & \sigma_1(\alpha_n) \\ \overline{\sigma_1}(\alpha_1) & \dots & \overline{\sigma_1}(\alpha_n) \\ \vdots & & \vdots \\ \sigma_s(\alpha_1) & \dots & \sigma_s(\alpha_n) \\ \overline{\sigma_s}(\alpha_1) & \dots & \overline{\sigma_s}(\alpha_n) \end{pmatrix} \end{aligned}$$

Taking absolute values of determinants and noting that

$$\det \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2i} & -\frac{1}{2i} \end{pmatrix} = \frac{i}{2},$$

we obtain $\operatorname{covol}(\iota(\mathcal{O}_K)) = \frac{\sqrt{|D_K|}}{2^s}$. □

Note that we have

$$|\mathcal{N}_K(\alpha)| = \prod_{i=1}^r |\tau_i(\alpha)| \cdot \prod_{j=1}^s (\operatorname{Re}(\sigma_j(\alpha))^2 + \operatorname{Im}(\sigma_j(\alpha))^2),$$

which motivates the following definition:

$$M_R = \{(x_1, \dots, x_r, y_1, z_1, \dots, y_s, z_s) \in \mathbb{R}^n \mid |x_1 \dots x_r| \cdot (y_1^2 + z_1^2) \dots (y_s^2 + z_s^2) \leq R\}$$

As in the totally real case this region is not convex (unless K is imaginary quadratic). So we should try to pick a convex centrally symmetric region $S_R \subseteq M_R$ with as large volume as possible. This is the following:

$$S_R = \{(x_1, \dots, x_r, y_1, z_1, \dots, y_s, z_s) \in \mathbb{R}^n \mid |x_1| + \dots + |x_r| + 2\sqrt{y_1^2 + z_1^2} + \dots + 2\sqrt{y_s^2 + z_s^2} \leq nR^{\frac{1}{n}}\} \subseteq M_R.$$

Exercise 3.8.27. Show the following:

1. S_R is convex.
2. Verify that $S_R \subseteq M_R$.
3. $\text{vol}(S_R) = \frac{n^n}{n!} 2^r \left(\frac{\pi}{2}\right)^s R$.

Solution: (1): Let $p = (x_1, \dots, x_r, y_1, z_1, \dots, y_s, z_s)$ and $q = (x'_1, \dots, x'_r, y'_1, z'_1, \dots, y'_s, z'_s)$. Then, consider the line $L = t \cdot p + (1-t) \cdot q$. We want to show all points in this line are in S_R . We then check the condition. First, we have

$$|t(x_i) + (1-t)x'_i| \leq |t(x_i)| + |(1-t)x'_i|, \quad 1 \leq i \leq r.$$

Next, we focus on the inside of $\sqrt{(t(y_i) + (1-t)(y'_i))^2 + (t(z_i) + (1-t)(z'_i))^2}$. First, note that

$$(t(y_i) + (1-t)(y'_i))^2 + (t(z_i) + (1-t)(z'_i))^2 = t^2(y_i^2 + z_i^2) + 2t(1-t)(y_i y'_i + z_i z'_i) + (1-t)^2(y_i'^2 + z_i'^2).$$

We then apply Cauchy-Schwarz to the middle term to get:

$$y_i y'_i + z_i z'_i \leq \sqrt{y_i^2 + z_i^2} \cdot \sqrt{y_i'^2 + z_i'^2}.$$

Thus, we now have:

$$\begin{aligned} & t^2(y_i^2 + z_i^2) + 2t(1-t)(y_i y'_i + z_i z'_i) + (1-t)^2(y_i'^2 + z_i'^2) \\ & \leq t^2(y_i^2 + z_i^2) + 2t(1-t)\sqrt{y_i^2 + z_i^2} \cdot \sqrt{y_i'^2 + z_i'^2} + (1-t)^2(y_i'^2 + z_i'^2), \end{aligned}$$

which we can factor the RHS as

$$\left(t\sqrt{y_i^2 + z_i^2} + (1-t)\sqrt{y_i'^2 + z_i'^2} \right)^2.$$

Thus, the square root unravels and so we get:

$$\sqrt{(t(y_i) + (1-t)(y'_i))^2 + (t(z_i) + (1-t)(z'_i))^2} \leq t\sqrt{y_i^2 + z_i^2} + (1-t)\sqrt{y_i'^2 + z_i'^2}.$$

Thus, if we consider the condition in S_R , then we have that the sum is less than or equal to:

$$\sum_{i=1}^r t|x_i| + (1-t)|x'_i| + 2 \sum_{i=1}^s t\sqrt{y_i^2 + z_i^2} + (1-t)\sqrt{y_i'^2 + z_i'^2}.$$

This is further less than or equal to

$$t \cdot nR^{\frac{1}{n}} + (1-t)nR^{\frac{1}{n}} = nR^{\frac{1}{n}},$$

so this set is convex.

(2): First note that by AM-GM, $nR^{\frac{1}{n}} \leq R$. We will now prove that the condition on the left hand side of the inequality in M_R is less than or equal to that of the one in S_R . We apply the AM-GM inequality once more:

$$r^{+2s} \sqrt{|x_1 \dots x_r| \cdot (y_1^2 + z_1^2) \dots (y_s^2 + z_s^2)} = \frac{1}{r+2s} \left(\sum_{i=1}^r |x_i| + 2 \sum_{i=1}^s \sqrt{x_i^2 + y_i^2} \right)$$

The RHS of the above is less than or equal to $nR^{\frac{1}{n}}$, which is less than or equal to R .

(3): This third part requires proving the formula

$$V_{r,s}(t) = \frac{1}{(r+2s)!} 2^r \left(\frac{\pi}{2} \right)^2 t^{r+2s},$$

where

$$V_{r,s} = \text{vol}(\{(x_1, \dots, x_r, y_1, z_1, \dots, y_s, z_s) \in \mathbb{R}^n \mid |x_1| + \dots + |x_r| + 2\sqrt{y_1^2 + z_1^2} + \dots + 2\sqrt{y_s^2 + z_s^2} \leq t\}).$$

To prove this, we use induction on r and s and reduce to a problem of single variable calculus; we do this by first scaling we have $V_{r,s}(t) = t^{r+2s} V_{r,s}(1)$. Then, we have the formulae:

$$\begin{aligned} V_{r,s}(1) &= \int_0^1 V_{r-1,s}(1-x) dx \\ V_{r,s}(1) &= \int_{2\sqrt{y^2+z^2} < 1} V_{r,s-1}(1-2\sqrt{y^2+z^2}) dy dz. \end{aligned}$$

□

Proof of Theorem 3.8.6. Now, if

$$\text{vol}(S_R) > 2^n \cdot \text{covol}(I) = 2^n \text{covol}(\iota(\mathcal{O}_K)) N_K(I) = \frac{2^{r+2s}}{2^s} \sqrt{|D_K|} N_K(I).$$

In other words, that $R > \frac{n!}{n^n} \left(\frac{4}{\pi} \right)^s \sqrt{|D_K|} N_K(I)$. Note that $C_K = \frac{n!}{n^n} \left(\frac{4}{\pi} \right)^s \sqrt{|D_K|}$. This implies that there exists a nonzero $\alpha \in I \cap S_R$ such that $|N_K(\alpha)| \leq R$, by the Minkowski body theorem. Therefore, taking $R > C_K N_K(I)$ such that

$$\lfloor R \rfloor = \lfloor C_K N_K(I) \rfloor,$$

we get the result. □

Example 3.8.28

Let us compute $\text{Cl}(\mathcal{O}_{-14})$.

Solution. Recall that $\mathcal{O}_{-14} = \mathbb{Z}[\sqrt{-14}]$. We first compute the Minkowski constant.

$$C_K = \frac{2}{\pi} \sqrt{|D_K|} = \frac{2}{\pi} \sqrt{4 \cdot 14} < 5.$$

Therefore, $\text{Cl}(\mathcal{O}_{-14})$ is generated by $[\mathfrak{p}]$ with $\mathfrak{p} \subseteq \mathcal{O}_{-14}$, a prime ideal with $N_K(\mathfrak{p}) < 5$.

Next, we study the factorization of prime numbers $p \in \mathbb{Z}$, where $p \leq C_K$. In our case, we study the factorization of $p = 2, 3$. In particular,

- $2 \cdot \mathcal{O}_{-14} = (2, \sqrt{-14})^2 = \mathfrak{p}_2^2 \implies N_K(\mathfrak{p}_2) = 2.$
- $3 \cdot \mathcal{O}_{-14} = (3, 1 + \sqrt{-14})(3, 1 - \sqrt{-14}) = \mathfrak{p}_3 \bar{\mathfrak{p}}_3 \implies N_K(\mathfrak{p}_3) = N_K(\bar{\mathfrak{p}}_3) = 3.$

Next, we check if we need them all. To do this, we first check whether the ideals we found above are not principal. In our case, assume that $\mathfrak{p}_2 = (\alpha)$, so assume that $\mathcal{N}_K(\alpha) = N_K(\alpha \cdot \mathcal{O}_{-14}) = 2$, which gives the equation $a^2 + 14b^2 = 2$, which has no solution. We can do a similar argument with 3 to check that $\mathfrak{p}_3$ and $\bar{\mathfrak{p}}_3$ are not similar. We now check whether they can be related by some constant. For this, assume that $[\mathfrak{p}_2] = [\mathfrak{p}_3]$. This then gives that

$$[\mathfrak{p}_2]^{-1} \cdot [\mathfrak{p}_3] = [1].$$

Noting that $\mathfrak{p}_2^2$ is $2 \cdot \mathcal{O}_K$, the above translates to say that $[\mathfrak{p}_2] \cdot [\mathfrak{p}_3] = [1]$. Therefore, $\mathfrak{p}_2 \cdot \mathfrak{p}_3 = (\alpha)$, but then

$$\mathcal{N}_K(\alpha) = N_K(\mathfrak{p}_2)N_K(\mathfrak{p}_3) = 6,$$

but this has no solution. Next, for the assumptions $[\mathfrak{p}_2] = [\bar{\mathfrak{p}}_3]$, a similar argument as before provides a contradiction. Finally, assume $[\mathfrak{p}_3] = [\bar{\mathfrak{p}}_3]$. Then,

$$[\mathfrak{p}_3] \cdot [\mathfrak{p}_3] = [\mathfrak{p}_3] \cdot [\bar{\mathfrak{p}}_3]^{-1} = [1].$$

Therefore, $\mathfrak{p}_3^2 = (\alpha)$, implying that $\mathcal{N}_K(\alpha) = 9$, which tells us that $\alpha = \pm 3$. This then means that $\mathfrak{p}_3 \bar{\mathfrak{p}}_3 = (3) = \mathfrak{p}_3^2$, which is not true.

With this, we know that $[\mathfrak{p}_2] \neq [\mathfrak{p}_3]$, $[\mathfrak{p}_2] \neq [\bar{\mathfrak{p}}_3]$, and $[\mathfrak{p}_3] \neq [\bar{\mathfrak{p}}_3]$. Thus, $\text{Cl}(\mathcal{O}_{-14})$ is of order 4, so there are two possibilities (cyclic group or Klein-four). However, $[\mathfrak{p}_3]^2 \neq [1]$, implying that we have an element of order 4, so

$$\text{Cl}(\mathcal{O}_{-14}) \cong \mathbb{Z}/4\mathbb{Z}.$$

□

Example 3.8.29

Let us compute the class group for $\mathcal{O}_{33}$.

Solution. $33 \equiv 1 \pmod{4}$, so

$$\mathcal{O}_{33} = \mathbb{Z}\left[\frac{1 + \sqrt{d}}{2}\right].$$

The Minkowski bound is

$$C_K = \frac{\sqrt{|D_K|}}{2} < 3.$$

The primes $p \leq C_K$ is just 2. We can factorize $2 \cdot \mathcal{O}_K$ as

$$2 \cdot \mathcal{O}_K = \left(2, \frac{1 + \sqrt{33}}{2}\right) \left(2, \frac{1 - \sqrt{33}}{2}\right),$$

so $N_K(\mathfrak{p}_2) = N_K(\bar{\mathfrak{p}}_2) = 2$. We now ask whether these are principal. If $\mathfrak{p}_2 = (\alpha)$, then (writing $\alpha = a + b(\frac{1 + \sqrt{33}}{2})$):

$$\mathcal{N}_K(a + b(\frac{1 + \sqrt{33}}{2})) = a^2 + ab - 8b^2 = 2.$$

Then, $a = 2, b = 1$ is a solution. This gives the element

$$2 + \frac{1 + \sqrt{33}}{2} = \frac{5 + \sqrt{33}}{2} \in \mathfrak{p}_2,$$

implying that $(\frac{5+\sqrt{33}}{2}) \subseteq \mathfrak{p}_2$. As the norm of this ideal is equal to 2, we actually find that

$$\mathfrak{p}_2 = (\frac{5 + \sqrt{33}}{2}) \cdot \mathcal{O}_{33},$$

so $[\mathfrak{p}_2] = [1]$. □

§3.9 Using the Class Group and Minkowski for Diophantine Equations

Here we use the class group and the Minkowski convex body theorem to solve Diophantine equations. First, recall that we have the following theorem by Fermat (proven in problems class):

Theorem 3.9.1 (Fermat's Theorem)

Let p be a prime. Then,

1. $p = x^2 + y^2$ for $x, y \in \mathbb{Z}$ if and only if $p = 2$ or $p \equiv 1 \pmod{4}$.
2. $p = x^2 + 2y^2$ for $x, y \in \mathbb{Z}$ if and only if $p = 2$ or $p \equiv 1, 3 \pmod{8}$.

Proof. 1. We have that $p = x^2 + y^2 = (x + iy)(x - iy)$ in $\mathbb{Z}[i]$, which is a PID. Therefore, this happens if and only if there is a prime ideal $\mathfrak{p} \in \mathbb{Z}[i]$ such that $N_k(\mathfrak{p}) = p$. This occurs if and only if p ramifies or splits in $\mathbb{Z}[i]$. Applying what we know about primes in quadratic fields, it follows that this occurs if and only if $p = 2$ or $\left(\frac{-1}{p}\right) = 1$, which is equivalent to stating $p = 2$ or $p \equiv 1 \pmod{4}$.

2. By an analogous argument, we can prove this with $\mathbb{Z}[\sqrt{-2}]$, which is also a PID. □

§3.9.i $p = x^2 + 5y^2$

Let us now shift our focus to when we don't work over PIDs. For instance, we may study the equation

$$p = x^2 + 5y^2 = (x + \sqrt{-5}y)(x - \sqrt{-5}y) \in \mathbb{Z}[\sqrt{-5}].$$

Note that $\mathbb{Z}[\sqrt{-5}] = \mathcal{O}_{-5}$ and we know that $\text{Cl}(\mathcal{O}_{-5}) = \mathbb{Z}/2\mathbb{Z}$ and it is generated by $[\mathfrak{p}_2]$, where $\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$.

Theorem 3.9.2

Let p be a prime.

1. If $p \equiv 1, 9 \pmod{20}$, then $p \cdot \mathcal{O}_{-5} = \mathfrak{p} \cdot \bar{\mathfrak{p}}$ splits and $\mathfrak{p}, \bar{\mathfrak{p}}$ are principal. In particular, $p = x^2 + 5y^2$ for some $x, y \in \mathbb{Z}$.
2. If $p \equiv 3, 7 \pmod{20}$, then $p \cdot \mathcal{O}_{-5} = \mathfrak{p} \cdot \bar{\mathfrak{p}}$ splits with $[\mathfrak{p}] = [\bar{\mathfrak{p}}] = [\mathfrak{p}_2]$. In particular, $2p = x^2 + 5y^2$ for some $x, y \in \mathbb{Z}$.
3. $2 \cdot \mathcal{O}_{-5} = \mathfrak{p}_2^2$ and $5 \cdot \mathcal{O}_{-5} = (\sqrt{-5})^2$.
4. In the other cases, p is inert.

Proof. For $p = 2, 5$, these ramify because 2 and 5 divide the discriminant. Now, let $p \neq 2, 5$. We know that p splits if and only if $\left(\frac{-5}{p}\right) = 1$. By quadratic reciprocity, we have that

$$\left(\frac{-5}{p}\right) = \left(\frac{-1}{p}\right) \cdot \left(\frac{5}{p}\right) = \left(\frac{-1}{p}\right) \cdot \left(\frac{p}{5}\right).$$

Thus to have $\left(\frac{-5}{p}\right) = 1$, there are two possibilities:

1. $\left(\frac{-1}{p}\right) = \left(\frac{p}{5}\right) = 1$. The first condition gives $p \equiv 1 \pmod{4}$ and the second gives $p \equiv \pm 1 \pmod{5}$, so $p \equiv 1, 9 \pmod{20}$ (by CRT).
2. We have $\left(\frac{-1}{p}\right) = \left(\frac{p}{5}\right) = -1$. The first condition gives $p \equiv 3 \pmod{4}$ and the second gives $p \equiv \pm 2 \pmod{5}$. Thus, $p \equiv 3, 7 \pmod{20}$ (by CRT).

What we get from this is that p splits if and only if $p \equiv 1, 3, 7, 9 \pmod{20}$: $p \cdot \mathcal{O}_{-5} = \mathfrak{p} \cdot \bar{\mathfrak{p}}$. There are two cases now:

1. $[\mathfrak{p}] = [1]$, implying that $\mathfrak{p}$ is principal, so there exists $x, y \in \mathbb{Z}$ with $\mathcal{N}(x + \sqrt{-5}y) = p$. If we consider this modulo 5, we see that $p \equiv \pm 1 \pmod{5}$.
2. If $[\mathfrak{p}] = [\mathfrak{p}_2]$ then $\mathfrak{p}\mathfrak{p}_2$ is principal (since $[\mathfrak{p}\mathfrak{p}_2] = [\mathfrak{p}_2]^2 = 1$). Thus there exists an element of $\mathcal{O}_{-5}$ with norm $2p$, and thus $2p = x^2 + 5y^2$. Considering this modulo 5, we see that $2p \equiv \pm 1 \pmod{5}$, or $p \equiv \pm 2 \pmod{5}$.

If $p \equiv 1, 9 \pmod{20}$, then the second possibility cannot occur so the first does, while if $p \equiv 3, 7 \pmod{20}$, then the first possibility does not occur so the second does. This proves the theorem. $\square$

Exercise 3.9.3. Show the following:

1. We have $p = x^2 + 5y^2$ if and only if $p = 5$ or $p \equiv 1, 9 \pmod{20}$.
2. We have $2p = x^2 + 5y^2$ if and only if $p = 2$ or $p \equiv 3, 7 \pmod{20}$.
3. If p, q are either 2 or 3, 7 $\pmod{20}$, then $pq = x^2 + ry^2$.

§3.9.ii Mordell's Equation

We want to see how to use what we did so far to study solutions to equations of the form

$$y^2 = x^3 + d.$$

The strategy to solve this is to rewrite the equation as

$$(y + \sqrt{d})(y - \sqrt{d}) = y^2 - d = x^3,$$

in $\mathbb{Z}[\sqrt{d}] \subseteq \mathcal{O}_d$.

Example 3.9.4

Let us consider $y^2 = x^3 - 1$.

Solution. We rewrite the equation as

$$(y - i)(y + i) = x^3.$$

Claim 3.9.5. The ideals $(y - i), (y + i) \subseteq \mathcal{O}_{-1} = \mathbb{Z}[i]$ are coprime.

Proof. If $\mathfrak{p} \mid (y - i), (y + i)$, then $\mathfrak{p} \mid (2i)$. Then, $2 \in \mathfrak{p}$ or $i \in \mathfrak{p}$. But, i is a unit so $2 \mid \mathfrak{p}$. This implies that $\mathfrak{p} = (1 + i)$. Thus, $(1 + i) \mid (y - i), (y + i)$, and so $(1 + i) \mid (y - i) \cdot (y + i) = x^3 \in \mathbb{Z}$. it then follows that

$$2 = \mathcal{N}(1 + i) \mid x^3 \implies 2 \mid x \implies 8 \mid x^3.$$

Therefore, $y^2 \equiv -1 \pmod{8}$, which is a contradiction, since -1 is not a quadratic residue of 8. $\square$

We now write $(y - i) = \mathfrak{p}_1 \dots \mathfrak{p}_n$ and $(y + i) = \mathfrak{q}_1 \dots \mathfrak{q}_m$, and $(x) = \mathfrak{b}_1 \dots \mathfrak{b}_t$. Thus,

$$(y - i)(y + i) = \mathfrak{b}_1^3 \dots \mathfrak{b}_t^3,$$

so $(y - i) = \mathfrak{q}^3$ and $(y + i) = (\bar{\mathfrak{q}})^3$, since we have that these two ideals are coprime. Here, $\mathfrak{q}, \bar{\mathfrak{q}} \subseteq \mathcal{O}_{-1} = \mathbb{Z}[i]$.

Now, we have that $\text{Cl}(\mathcal{O}_{-1}) = \{1\}$, and so $\mathfrak{q} = (a + ib)$, for some integers a, b . This implies that $(y - i) = (a + ib)^3$, so

$$y - i = u = (a + ib)^3,$$

where $u \in \mathbb{Z}[i]^\times$. Now, we expand the RHS to restrict what a and b are. This gives:

$$y - i = (a^3 - 2ab^2) - i(b^3 - 3a^2b).$$

This implies that

$$y = a^3 - 2ab^2, \quad 1 = b^3 - 3a^2b.$$

This tells us that $b = \pm 1$. If $b = 1$, then $a = 0$. If $b = -1$, there is no solution. Thus, we only have $y = 0$, and so the only solution of the original equation is $(x, y) = (1, 0)$. $\square$

Example 3.9.6

Consider the equation $y^2 = x^3 - 5$.

Solution. Write $x^3 = (y + \sqrt{-5})(y - \sqrt{-5})$. Similar to before, we claim that $(y + \sqrt{-5})(y - \sqrt{-5}) \subseteq \mathcal{O}_{-5}$. If $(y + \sqrt{-5}), (y - \sqrt{-5}) \subseteq \mathfrak{p}$, then, $2\sqrt{-5} \in \mathfrak{p}$, and so $N_K(\mathfrak{p}) \mid \mathcal{N}(2\sqrt{-5}) = 20$. Thus, $\mathfrak{p}$ lies over 2 or 5. Now,

$$x^3 = (y + \sqrt{-5})(y - \sqrt{-5}) \in \mathfrak{p} \implies x \in \mathfrak{p},$$

and $x \in \mathbb{Z}$ implies that $x \in \mathfrak{p} \cap \mathbb{Z}$. If $\mathfrak{p}$ lies over 2, then $x \in (2)$ (since $\mathfrak{p}$ lies over 2 if and only if $\mathfrak{p} \cap \mathbb{Z} = (2)$). Then, $2 \mid x$ and so $8 \mid x^3$, implying that $y^2 \equiv 5 \pmod{8}$, which is not a quadratic residue. Next, if $\mathfrak{p}$ lies over 5, then $x \in (5)$ and so $5 \mid x$ and $5 \mid x^3 + 5 = y^2$, so $5 \mid y^2$. Therefore, $25 \mid x^3 - y^2 = 5$, which is not true.

Next, we look at the factorization into prime ideals:

$$(y + \sqrt{-5})(y - \sqrt{-5}) = (x)^3.$$

Then, each ideal on the left is a cube, since they are coprime:

$$(y + \sqrt{-5}) = \mathfrak{q}^3, \quad (y - \sqrt{-5}) = \bar{\mathfrak{q}}^3,$$

where $\mathfrak{q}, \bar{\mathfrak{q}} \subseteq \mathcal{O}_{-5} = \mathbb{Z}[\sqrt{-5}]$. Now, we know that $\text{Cl}(\mathcal{O}_{-5}) = \mathbb{Z}/2\mathbb{Z}$, and $[\mathfrak{q}]^3 = [1]$, so $[\mathfrak{q}] = 1$. Thus, $\mathfrak{q}$ is principal, so write $\mathfrak{q} = (\alpha)$, where $\alpha = a + b\sqrt{-5}$. Therefore, we have that

$$(y + \sqrt{-5}) = (\alpha)^3 = (\alpha^3),$$

which implies that $y + \sqrt{-5} = \pm(a + \sqrt{-5}b)^3$. We have

$$y + \sqrt{-5} = \pm \left((a^3 - 15ab^2) + (3a^2b - 5b^3)\sqrt{-5} \right),$$

which tells us

$$\begin{aligned} y &= \pm(a^3 - 15ab^2) \\ 1 &= \pm b(3a^2 - 5b^2). \end{aligned}$$

In particular, $b = \pm 1$. If $b = 1$, then there's no solution for a . If $b = -1$, then there is no solution again for a . Therefore, there are no solutions to the Diophantine equations $y^2 = x^3 - 5$ over the integers. $\square$

Let's try to generalize our arguments in the previous two examples. Consider $d < 0$ which is square free and $d \not\equiv 1 \pmod{4}$. Then, $\mathcal{O}_d = \mathbb{Z}[\sqrt{d}]$. We consider

$$y^2 = x^3 + d.$$

We first write

$$(y + \sqrt{d})(y - \sqrt{d}) = x^3.$$

We then want to show that the two ideals $(y + \sqrt{d})$ and $(y - \sqrt{d})$ are coprime. Assume that there is a prime ideal $\mathfrak{p}$ such that $(y + \sqrt{d})(y - \sqrt{d}) \subseteq \mathfrak{p}$. Then, $2\sqrt{d} \in \mathfrak{p}$. Then, $\mathfrak{p}$ lies above 2 or a prime p such that $p \mid d$. If $\mathfrak{p}$ lies above 2, then $x^3 = (y + \sqrt{d})(y - \sqrt{d}) \in \mathfrak{p} \cap \mathbb{Z} = (2)$, but then $(x) \in (2)$. This then tells us that $2 \mid x$, so

$$y^2 \equiv d \pmod{8}.$$

This is not possible, since $d \not\equiv 1 \pmod{4}$ and d is a nonsquare. Now, suppose $\mathfrak{p}$ lies above $p \mid d$. Then,

$$x^3 = (y + \sqrt{d})(y - \sqrt{d}) \in \mathfrak{p} \cap \mathbb{Z} = (p).$$

Then, $p \mid x^3, d$, so $p \mid y^2$. In particular, $p \mid y$ then. This implies that

$$p^2 = x^3 - y^2 = -d,$$

and d is square free, so this is not possible. Therefore, the ideals are coprime. By coprimality, there exists an ideal $\mathfrak{a} \in \mathcal{O}_d$ such that

$$(y + \sqrt{d}) = \mathfrak{a}^3.$$

By Lagrange's theorem, if $3 \nmid |\text{Cl}(\mathcal{O}_d)|$, then we attain that $\mathfrak{a}$ is principal (note that if we don't make this additional assumption here, we are stuck). If $\mathfrak{a}$ is principal, i.e., if $\mathfrak{a} = (\alpha)$,

$$(y + \sqrt{d}) = (\alpha^3).$$

Note that

$$\mathbb{Z}[\sqrt{d}]^\times = \begin{cases} \{\pm 1\} & d \neq -1 \\ \{\pm 1, \pm i\} & d = -1 \end{cases}.$$

Then, $y + \sqrt{d} = (a + \sqrt{d}b)^3$, for some $a, b \in \mathbb{Z}$, and so

$$\begin{aligned} y &= a^3 + 3dab^2 \\ 1 &= (3a^2 + db^2). \end{aligned}$$

In particular, $b = \pm 1$. If $b = 1$, then $3a^2 + d = 1$ and if $b = -1$, then $3a^2 = -1$. Note that if $3a^2 + d = \pm 1$ has a solution, say $\pm a_0$, then

$$y = a_0^3 + 3da_0 = \pm a_0(a_0^2 + 3d).$$

We can find for x then by

$$x^3 = (a_0 + \sqrt{d})^3(a_0 - \sqrt{d})^3 = (a_0^2 - d)^3.$$

Thus, $x = a_0^2 - d$. Therefore,

$$(a_0^2 - d, \pm a_0(a_0^2 + 3d))$$

is a solution, if $y^2 = x^3 + d$ if $\pm a_0$ are solutions of

$$3a^2 + d = \pm 1.$$

Theorem 3.9.7

Suppose $d < 0$ is square free with $d \not\equiv 1 \pmod{4}$. If there exists $a \in \mathbb{Z}$ with

$$3a^2 + d = \pm 1,$$

then

$$(x, y) = (a^2 - d, \pm a(a^2 + 3d))$$

is a solution of the Diophantine equation

$$y^2 = x^3 + d.$$

Moreover, if $3 \nmid |\text{Cl}(\mathcal{O}_d)|$, then these are the only solution to this equation.

Example 3.9.8

Let $d = -13$. Then, $a = \pm 2$ works:

$$3(\pm 2)^2 - 13 = -1.$$

Thus, $(x, y) = (4 + 13, \pm 2((\pm 2)^2 - 3 \cdot 13)) = (17, \pm 70)$ is a solution of $y^2 = x^3 - 13$. Moreover,

$$C_K = \frac{2}{\pi} \sqrt{4 \cdot 13} < 5,$$

then $2 \cdot \mathcal{O}_{-13} = \mathfrak{p}_2^2$ and $N(\mathfrak{p}_2) = 2$, and $3 \cdot \mathcal{O}_{-13}$ is inert. Thus,

$$\text{Cl}(\mathcal{O}_{-13}) = \{[1], [\mathfrak{p}_2]\},$$

since $\mathfrak{p}_2$ cannot be principal. Thus, $\text{Cl}(\mathcal{O}_{-13}) \cong \mathbb{Z}/2\mathbb{Z}$. Therefore, the solutions above are the only solutions.

Example 3.9.9

Consider $d = -26$. Then, $a \pm 3$ works, so $(x, y) = (35, \pm 207)$ is a solution to $y^2 = x^3 - 26$. However, $(x, y) = (3, \pm 1)$ are also solution to the same equations, so we deduced that $3 \mid \text{Cl}(\mathcal{O}_{-26})$ (by the converse). If we do some more computation, this tells us that $\text{Cl}(\mathcal{O}_{-26})$ has order 6.

Here is a theorem that we won't prove, and we'll list for curiosity purposes.

Theorem 3.9.10

For $d \neq 0$, the Diophantine equation

$$y^2 = x^3 + d$$

has only finitely many solutions over the integers.

Remark 3.9.11 — This is quite different compared to the behavior of this equation over rationals, as there are infinitely many rational points for an elliptic curve.

§3.9.iii Units in Real Quadratic Fields

Note that the nice aspect of the above was that the units were ± 1 or $\pm i$.

Example 3.9.12

Let $d < 0$ be square free. Then,

$$\mathcal{O}_d^\times = \begin{cases} \{\pm 1\} & d \neq -1, -3 \\ \{\pm 1, \pm i\} & d = -1 \\ \{\zeta_6^k \mid 0 \leq k \leq 5\}, \text{ where } \zeta_6 \text{ is a 6-th root of unity} & d = -3 \end{cases}.$$

Example 3.9.13

Consider $\mathcal{O}_2^\times$. note that $1 + \sqrt{2} \in \mathcal{O}_2^\times = \mathbb{Z}[\sqrt{2}]^\times$. Then, $1 + \sqrt{2} \in \mathcal{O}_2^\times$, as $\mathcal{N}(1 + \sqrt{2}) = -1$. In particular, $(1 + \sqrt{2})^n \in \mathcal{O}_2^\times$, for all $n \in \mathbb{Z}$. This tells us that the set of units is infinite in $\mathcal{O}_2$.

Theorem 3.9.14

Let $d > 0$ be square free. Then, there exists a unique unit ϵ with $\epsilon > 1$ and such that

$$\mathcal{O}_d^\times = \{\pm \epsilon^n \mid n \in \mathbb{Z}\}.$$

The unit ϵ is called the **fundamental unit**.

Corollary 3.9.15

We have the isomorphism

$$\mathcal{O}_d^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z},$$

if we consider the map

$$\{\pm \epsilon^n \mid n \in \mathbb{Z}\} \mapsto (\text{sign}(\pm \epsilon^n), n).$$

Remark 3.9.16 — When $d \not\equiv 1 \pmod{4}$, the roots we get are the solution to Pell's equation:

$$x^2 - dy^2 = \pm 1.$$

Example 3.9.17 (Fundamental Units for Different d)

We have:

1. $1 + \sqrt{2} \in \mathcal{O}_2^\times$
2. $2 + \sqrt{3} \in \mathcal{O}_3^\times$
3. $\frac{1+\sqrt{6}}{2} \in \mathcal{O}_5^\times$
4. $170 + 39\sqrt{19} \in \mathcal{O}_{19}^\times$
5. $2143295 + 221069\sqrt{94} \in \mathcal{O}_{94}^\times$

The point of this example is that these fundamental units can grow really quickly.

Later, we will see that the fundamental unit comes from one of the following equations

$$\begin{aligned} \mathcal{N}(a + \sqrt{d}b) &= \pm 1 \\ \mathcal{N}\left(a + \left(\frac{1 + \sqrt{d}}{2}\right)b\right) &= \pm 1, \end{aligned}$$

with the smallest possible value of $b > 0$.

We now go on to prove the theorem in three steps.

Lemma 3.9.18

There exists $v \in \mathcal{O}_d^\times$ with $v \neq \pm 1$.

Proof. Let $K = \mathbb{Q}(\sqrt{d})$ and $\mathcal{O}_d$ to be the ring of integers of K .

Claim 3.9.19. There exists $c > 0$ such that there are infinitely many $\alpha \in \mathcal{O}_d$ with $|N_K(\alpha)| < c$.

Proof of Claim 3.9.19. Recall that $\iota: K \hookrightarrow \mathbb{R}^2$ given by $\tau \mapsto (\tau_1(\alpha), \tau_2(\alpha))$ and $N: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $(x, y) \mapsto |x \cdot y|$. Take $\{(x, y) \in \mathbb{R}^2 \mid N(x, y) = |x \cdot y| < c\}$. The following is a subset of this set:

$$S_r = \{(x, y) \in \mathbb{R}^2 \mid |x| < r, |y| < \frac{c}{r}\}.$$

Take $c > \sqrt{|D_K|} = \text{covol}(\iota(\mathcal{O}_d))$. Then,

$$\text{vol}(S_r) = (2r) \frac{2c}{r} = 4c > 2^2 \cdot \text{covol}(\iota(\mathcal{O}_d^\times)).$$

Therefore, for all $r > 0$, there exists $0 \neq \alpha \in \mathcal{O}_d$ such that $\iota(\alpha) \in S_r$ by the Minkowski body theorem. In other words, $|\tau_1(\alpha)| = |\alpha| < r$ and $|\tau_2(\alpha)| < \frac{c}{r}$. Therefore, $|\mathcal{N}_K(\alpha)| = |\tau_1(\alpha)\tau_2(\alpha)| < c$.

Now take $0 \neq \alpha_1 \in \mathcal{O}_d \cap S_1$. Next, take $0 \neq \alpha_2 \in \mathcal{O}_d \cap S_{|\alpha_1|}$. Next, take $0 \neq \alpha_3 \in \mathcal{O}_d \cap S_{|\alpha_2|}$. Continuing, we have a sequence of $\alpha_i \in \mathcal{O}_d$ with $|\alpha_1| > |\alpha_2| > |\alpha_3| > \dots$ and $|\mathcal{N}(\alpha_i)| < C$. $\square$

We note that the claim implies the lemma. In fact, we know that there are just finitely many ideals with norm less than C . Therefore, $\exists \alpha \neq \beta \neq \gamma$ such that $(\alpha) = (\beta) = (\gamma)$. Then, the first claim along with the pigeon hole principle tells us that at least one among β and γ will be also not equal to $-\alpha$. Then, $\beta \neq \pm \alpha$ and $\beta = v \cdot \alpha$, for some $v \in \mathcal{O}_d^\times$, so $v \neq \pm 1$. $\square$

Lemma 3.9.20

Let $u = a + b\sqrt{d} \in \mathcal{O}_d^\times$. Then, $u > 1 \iff a, b > 0$. Moreover, if $u' = a' + b'\sqrt{d} \in \mathcal{O}_d^\times$ is another unit with $u' > 1$. Then, $u \geq u'$ if and only if $a \geq a'$ and $b \geq b'$. In fact, $u \geq u'$ if and only if $b \geq b'$ with the only exception $d = 5$, $u = \frac{1+\sqrt{5}}{2}$, $u' = \frac{3+\sqrt{5}}{2}$.

Proof. We first want to show that $u = a + b\sqrt{d} > 1$ if and only if $a, b > 0$.

($\Leftarrow$): If $a, b > 0$, then $a, b \geq \frac{1}{2}$ ($a, b \in \mathbb{Z}$ or $2a, 2b \in \mathbb{Z}$). Then, $u = a + b\sqrt{d} \geq \frac{1+\sqrt{d}}{2} > 1$.

($\Rightarrow$): Let $u \in \mathcal{O}_d^\times$, $\mathcal{N}_K(u) = u \cdot \bar{u} = \pm 1$. Then, $u > 1$ implies that $|\bar{u}| < 1$. Now, $u + \bar{u} = 2a$ and $u - \bar{u} = 2b\sqrt{d}$. $u > 1$ and $|\bar{u}| < 1$, so $u + \bar{u}$, $u - \bar{u}$ are positive, so $a, b > 0$.

Let $u' = a' + b'\sqrt{d} \in \mathcal{O}_d$ and $u' > 1$. We now want to show the second equivalence.

($\Rightarrow$): We have $u - u' = 2\sqrt{d}(b - b') + \bar{u} - \bar{u}'$. If $b \neq b'$, then if $d \not\equiv 1 \pmod{4}$, we have $|b - b'| \geq 1$ and so $2\sqrt{d}|b - b'| > 2$. If $d \equiv 1 \pmod{4}$, then $|b - b'| \geq \frac{1}{2}$ and so $2\sqrt{d}|b - b'| \geq \sqrt{d} \geq 2$. In both cases, $b + b'$ implies that $2\sqrt{d}|b - b'| > 2$. Since $|\bar{u} - \bar{u}'| < 2$,

then $b > b'$ implies that $u > u'$ and $b < b'$ implies that $u < u'$. If $b = b'$, then $u \geq u'$ if and only if $a \geq a'$. Since $a^2 - db^2 = \pm 1$ and $(a')^2 - d(b')^2 = \pm 1$ and $a, a' > 0$ (by the first equivalence), we can assume that WLOG that $a^2 - db^2 = 1$ and $(a')^2 - d(b')^2 = -1$ and $a^2 - (a')^2 = 2$. If $d \not\equiv 1 \pmod{4}$, then $a, a' \in \mathbb{Z}$ and so $a^2 - (a')^2 = 2$ never happens. thus, $d \equiv 1 \pmod{4}$. Then, $b = b' \in \frac{1}{2} + \mathbb{Z}$, $a, a' \in \frac{1}{2}\mathbb{Z} \setminus \mathbb{Z}$. Then, $a' = \frac{2n+1}{2}$ and $a = \frac{2n+2k+1}{2}$. Then we get that the difference of the squares satisfies

$$(2n+1) \cdot k + k^2 = 2.$$

This equation has just one solution when $n > 0$ and $k \geq 1$. This solution is $n = 0$, $k = 1$, $a = \frac{1}{2}$, $b = \frac{3}{2}$. Therefore, the norm of u' gives

$$\frac{1}{4} - db^2 = 1,$$

so $d = 5$ $a' = \frac{5}{2}$ □

Proof of the Theorem. Take $\epsilon \in \mathcal{O}_d^\times$ with $\epsilon > 1$. This exists by lemma 3.9.18 and the fact that with u is a unit implies $-u, \frac{1}{u}, -\frac{1}{u}$ is a unit. Let $u \in \mathcal{O}_d^\times$ and $u \neq \pm 1$. Consider

$$u' = \begin{cases} u & u > 0 \\ -u & u < 0 \end{cases}.$$

Then, $u' > 0$. Then, $\epsilon^k < u'$ for some $k \ll 0$ and $\epsilon^k > u'$ for $k \gg 0$. Then, there exists a largest k with $\epsilon^k \leq u'$ and $u' < \epsilon^{k+1}$. This implies that $1 \leq \frac{u'}{\epsilon^k} < \epsilon$, so $\frac{u'}{\epsilon^k} = 1$ and so $u = \pm \epsilon^k$. □

Remark 3.9.21 — Note that lemma 3.9.18 and 3.9.20, we know that a unit bigger than 1 exists, and there there is a minimum.

Remark 3.9.22 — If $d = 5$, then $\epsilon = a + \sqrt{d}b$ will be the unit with $a, b > 0$ and b smallest possible such that $a^2 - db^2 = \pm 1$ has a solution.

Example 3.9.23

Consider $\mathcal{O}_{11}$.

Solution. We want to find solutions to $a^2 - 11b^2 = \pm 1$ with $a, b > 0$ and smallest b . For $b = 1$, we get $a^2 - 11 = \pm 1$, so there are no solution. Next, for $b = 2$, $a^2 - 11 \cdot 4 = \pm 1$, so there are no solution. Finally, with $b = 3$, we get $a^2 - 11 \cdot 9 = \pm 1$, and so $a = 10$ is a solution. Therefore, $\epsilon = 10 + 3\sqrt{11}$ is a fundamental unit. □

Exercise 3.9.24. Find the fundamental unit of $\mathcal{O}_{13}$.

Remark 3.9.25 — When we study the class group $\text{Cl}(\mathcal{O}_d)$, we need to be able to solve norm equations, i.e.,

$$\mathcal{N}_K(\alpha) = n, \quad n \in \mathbb{Z}.$$

If $\mathcal{N}_K(\alpha) = n$, then $\mathcal{N}(\pm \eta^k \cdot \alpha) = n$ if $\eta > 1$ is the smallest unit such that $\mathcal{N}_K(\eta) = 1$.

This tells us that we can replace α with $\alpha' = \pm\eta^k\alpha$. We can pick k to have

$$\sqrt{|n|} \cdot \eta^{-\frac{1}{2}} \leq \alpha' \leq \sqrt{|n|}\eta^{\frac{1}{2}}.$$

This implies that $|\overline{\alpha'}| = \frac{n}{\alpha'} \leq \sqrt{|n|} \cdot \eta^{\frac{1}{2}}$ and hence

$$|\alpha' - \overline{\alpha'}| \leq 2\sqrt{|n|}\eta^{\frac{1}{2}} \leq |\alpha'| + |\overline{\alpha'}| \leq 2\eta^{\frac{1}{2}}\sqrt{|n|}.$$

However, $\alpha' - \overline{\alpha'} = 2b\sqrt{d}$. Therefore, $b \leq \sqrt{\frac{|n|\eta}{d}}$, and we know every number appearing in the bound.

What we've proven in the remark is the following.

Proposition 3.9.26

Any solution to $\mathcal{N}_K(\alpha) = n$ for $n \in \mathbb{Z}$, $\alpha \in \mathcal{O}_d$ has the form $\alpha = \pm\eta^k(a + b\sqrt{d})$ where $a + b\sqrt{d} \in \mathcal{O}_d$ and

$$\mathcal{N}_K(a + b\sqrt{d}) = n, \quad |b| \leq \sqrt{\frac{|n|\eta}{d}},$$

with η a fundamental unit with $\eta > 1$ and $\mathcal{N}_K(\eta) = 1$.

Remark 3.9.27 — Such an η always exists. If the fundamental unit ϵ has norm 1 ($\mathcal{N}_K(\epsilon) = 1$), then $\eta = \epsilon$. If $\mathcal{N}_K(\epsilon) = -1$, then $\eta = \epsilon^2 > 1$ and $\mathcal{N}_K(\eta) = 1$.

Example 3.9.28

We compute the class group $\text{Cl}(\mathcal{O}_{79})$, where $K = \mathbb{Q}(\sqrt{79})$.

Solution. The Minkowski bound is $C_K = \frac{\sqrt{4 \cdot 79}}{2} < 9$. Then, we have some primes to look at. We have that $2 \cdot \mathcal{O}_K = \mathfrak{p}_2^2$ as $79 \equiv 3 \pmod{4}$. Next, we have that $3 \cdot \mathcal{O}_K = \mathfrak{p}_3 \cdot \overline{\mathfrak{p}}_3$ as $79 \equiv 1 \pmod{3}$. Next, $5 \cdot \mathcal{O}_K = \mathfrak{p}_5 \cdot \overline{\mathfrak{p}}_5$, as $79 \equiv 4 \pmod{5}$. Finally, $7 \cdot \mathcal{O}_K = \mathfrak{p}_7 \cdot \overline{\mathfrak{p}}_7$ as $79 \equiv 2 \equiv 3^2 \pmod{7}$. We now want to see if they are related in some way. As there are many primes involved, we need to be smart. The idea is to find elements with small norm to find relations between the ideals. Note that the norm of an arbitrary element $a + b\sqrt{79} \in \mathcal{O}_{79} = \mathbb{Z}[\sqrt{79}]$ has norm $a^2 - 79b^2$. The tactic will to fix b and then plug in a that *minimize* the quantity. For instance, we can start checking with $b = 1, b = 2$.

- $b = 1, a = 8$ gives -15 .
- $b = 1, a = 9$ gives 2 .
- $b = 1, a = 10$ gives 21 .
- $b = 2, a = 17$ gives -27 .

First, we get that $N(9 + \sqrt{79}) = 2$, so $\mathfrak{p}_2$ is principal. In particular, we have that $N(8 + \sqrt{79}) = -15$, and so

$$(8 + \sqrt{79}) = \begin{cases} \mathfrak{p}_3 \cdot \mathfrak{p}_5 \\ \mathfrak{p}_3 \cdot \overline{\mathfrak{p}}_5 \\ \overline{\mathfrak{p}}_3 \cdot \mathfrak{p}_5 \\ \overline{\mathfrak{p}}_3 \cdot \overline{\mathfrak{p}}_5 \end{cases}.$$

All of this imply that $[\mathfrak{p}_5] = [\mathfrak{p}_3]^{\pm 1}$. A similar analysis of $N(10 + \sqrt{79}) = 21$ tells us that $[\mathfrak{p}_7] = [\mathfrak{p}_3]^{\pm 1}$. Therefore, the class group is generated by $[\mathfrak{p}_3]$. We now want to determine the order of $\mathfrak{p}_3$. We know that $N(17 + 2\sqrt{79}) = 27$, so $(17 + 2\sqrt{79}) = \mathfrak{p}_3^3$ or $\bar{\mathfrak{p}}_3^3$. If this were not the case, then it would be a mix of the two. Thus, for the sake of contradiction, we assume that $\mathfrak{p}_3 \bar{\mathfrak{p}}_3 \mid (17 + 2\sqrt{79})$. This tells us that $17 + 2\sqrt{79} \in \mathfrak{p}_3 \cdot \bar{\mathfrak{p}}_3 = (3)$, but this is not true. Therefore, $[\mathfrak{p}_3]^3 = [1]$ and $[\bar{\mathfrak{p}}_3]^3 = [\mathfrak{p}_3]^{-3} = [1]$.

The final step is to ask whether $\mathfrak{p}_3$ is principal or not. That is, does there exists $\alpha \in \mathcal{O}_{79}$ such that $N(\alpha) = N(\mathfrak{p}_3) = \pm 3$? (the $\pm$ comes from taking the absolute value) Now, write $\alpha = a^2 - 79b^2$, so we need to see whether there are solutions $a^2 - 79b^2 = \pm 3$. We first search for $a^2 - 79b^2 = 3$. Taking the equation modulo 79, we have

$$a^2 \equiv 3 \pmod{79},$$

but $\left(\frac{3}{79}\right) = -1 \cdot \left(\frac{79}{3}\right) = -1 \cdot \left(\frac{1}{3}\right) = -1$ by quadratic reciprocity. Thus, this has no solution. Now, we check $a^2 - 79b^2 = -3$. In this case modulo 79 doesn't provide much (there is a solution). To solve this, we use the fundamental unit. In this case, the fundamental unit is $\epsilon = 80 + 9\sqrt{79}$ (solve for this!). We can check that $N(\epsilon) = 1$. By the previous proposition, we know that if $\mathcal{N}_K(\alpha) = -3$ has a solution, then it also has a solution with

$$|b| \leq \sqrt{\frac{|3|\epsilon}{79}} < 3.$$

We now want to check for when $\mathcal{N}_K(a + \sqrt{d}b) = -3$ for $0, \pm 1, \pm 2$.

- $b = 0$: $a^2 = -3$, so none.
- $b = \pm 1$: $a^2 = -3 + 79 = 76$, so none.
- $b = \pm 2$: $a^2 = 313$, so none.

Therefore, $\mathcal{N}_K(\alpha) = -3$ has no solution, and so $\mathfrak{p}_3$ is not principal, implying that $\text{Cl}(\mathcal{O}_{79}) \cong \mathbb{Z}/3\mathbb{Z}$. $\square$

§3.10 Dirichlet's Unit Theorem

We state, but won't prove the following theorem.

Theorem 3.10.1 (Dirichlet's Unit Theorem)

Let K be a number field and suppose that K has r real embeddings and s pairs of conjugate complex embeddings ($n = [K : \mathbb{Q}] = r + 2s$). Let $\mu(K) = \{x \in K \mid x^n = 1, \text{ for some } n\}$ be the set of roots of unity in K . Then,

$$\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}^{r+s-1}.$$

In other words there exists units $\epsilon_1, \dots, \epsilon_{r+s-1} \in \mathcal{O}_K^\times$ such that any unit $u \in \mathcal{O}_K^\times$ can be written uniquely in the form $u = \zeta \epsilon_1^{n_1} \dots \epsilon_{r+s-1}^{n_{r+s-1}}$ for $\zeta \in \mu(K)$ a root of unity and $n_1, \dots, n_{r+s-1} \in \mathbb{Z}$.

Remark 3.10.2 — Note that $\mu(K)$ is a cyclic and finite.

Example 3.10.3

Let $K = \mathbb{Q}$. Then, $r = 1$ and $s = 0$, so $r + s - 1 = 0$, and

$$\mathcal{O}_K^\times = \mathbb{Z}^\times = \{\pm 1\} = \mu(\mathbb{Q}).$$

Example 3.10.4

Let $K = \mathbb{Q}(\sqrt{-d})$. Then, $r = 0$ and $s = 1$, so $r + s - 1 = 0$. Then,

$$\mathcal{O}_d^\times = \mu(K) = \{\pm 1\},$$

unless $K = \mathbb{Q}(i)$ or $\mathbb{Q}(\zeta_3)$.

Example 3.10.5

Let $K = \mathbb{Q}(\sqrt{d})$. Then, $r = 2$, $s = 0$, so $r + s - 1 = 1$. Then,

$$\mathcal{O}_d^\times \cong \mu(K) \times \mathbb{Z} = \{\pm 1\} \times \mathbb{Z}$$

Example 3.10.6

Consider $K = \mathbb{Q}(\sqrt[3]{2})$. Then, $r = 1$, $s = 1$, so $r + s - 1 = 1$. Therefore,

$$\mathcal{O}_K^\times \cong \{\pm 1\} \times \mathbb{Z}.$$

We can conclude that $\mathcal{N}(1 + \sqrt[3]{2} + \sqrt[3]{4}) = 1$. It turns out that

$$\mathbb{Z}[\sqrt[3]{2}]^\times \cong \{\pm(1 + \sqrt[3]{2} + \sqrt[3]{4})^n \mid n \in \mathbb{Z}\}.$$

Example 3.10.7

Let $K = \mathbb{Q}(\zeta_5)$. Then, $r = 0$, $s = 2$, so $r + s - 1 = 1$. However, $\mathbb{Q}(\sqrt{5}) \subseteq \mathbb{Q}(\zeta_5)$, so

$$\mathcal{O}_5^\times = \left\{ \pm \left(\frac{1 + \sqrt{5}}{2} \right)^n \mid n \in \mathbb{Z} \right\} \subseteq \mathbb{Z}[\zeta_5]^\times.$$

In fact,

$$\mathbb{Z}[\zeta_5]^\times = \{ \zeta_{10^k} \left(\frac{1 + \sqrt{5}}{2} \right)^n \mid n \in \mathbb{Z} \}$$